Metabolite Reorganization Results

Christian Anderson

24 February 2026

Contents

Node/Edge Overlaps

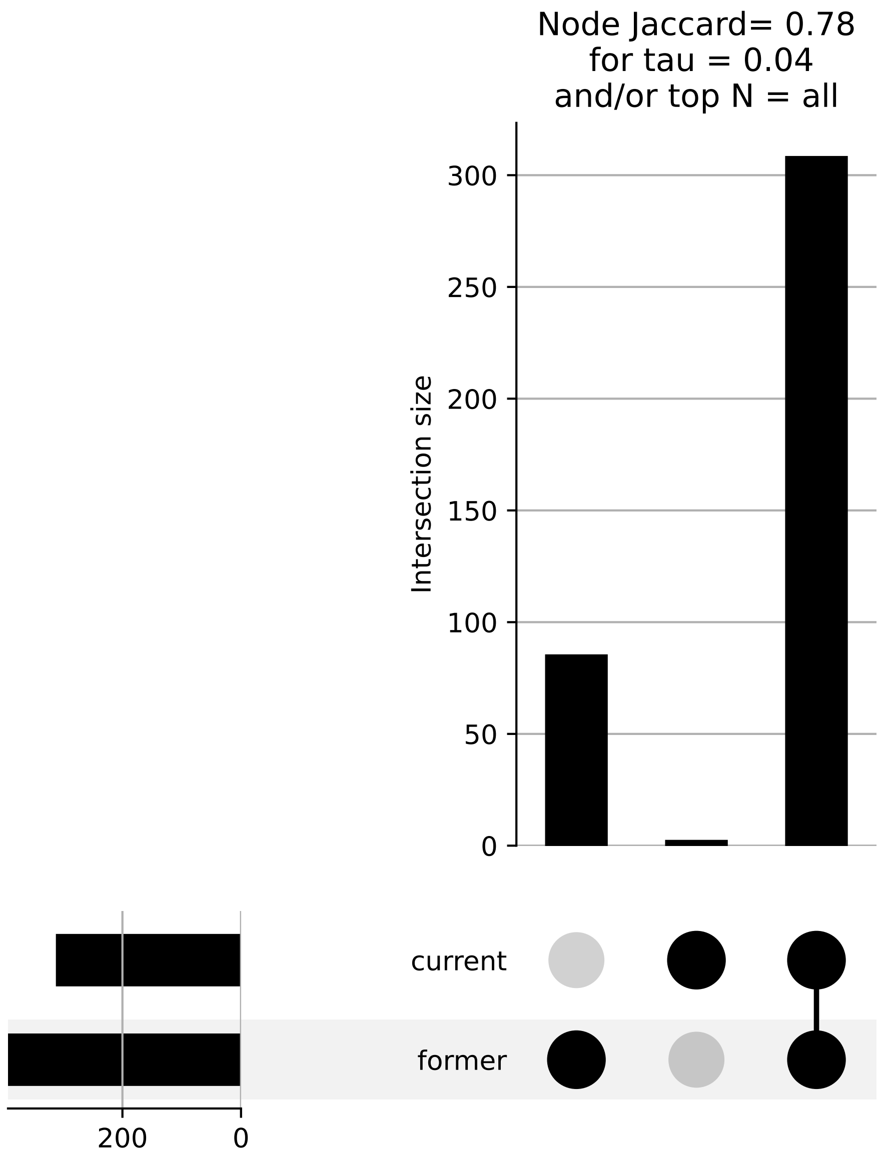


Figure 1: Node intersection and difference

results/005/node-upset.pdf

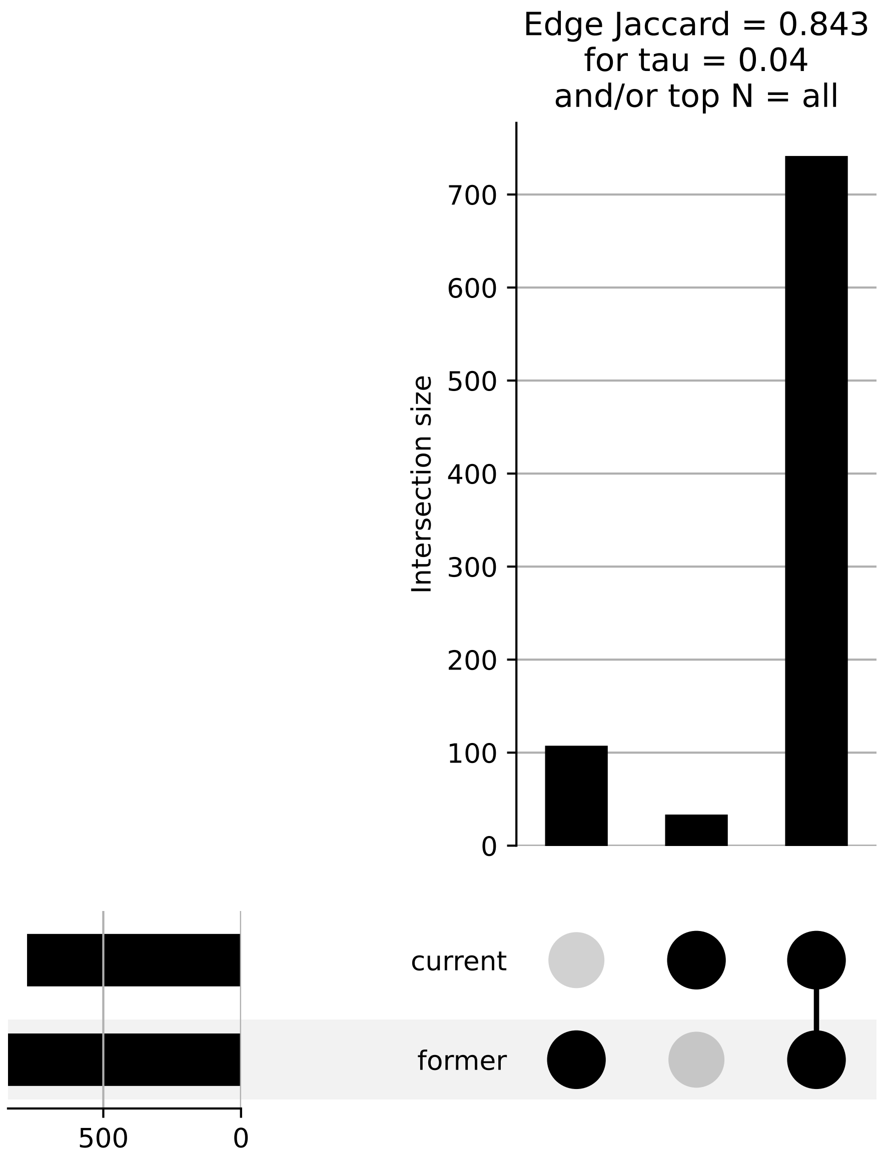


Figure 2: Edge intersection and difference.

results/005/edge-upset.pdf

Current Smokers

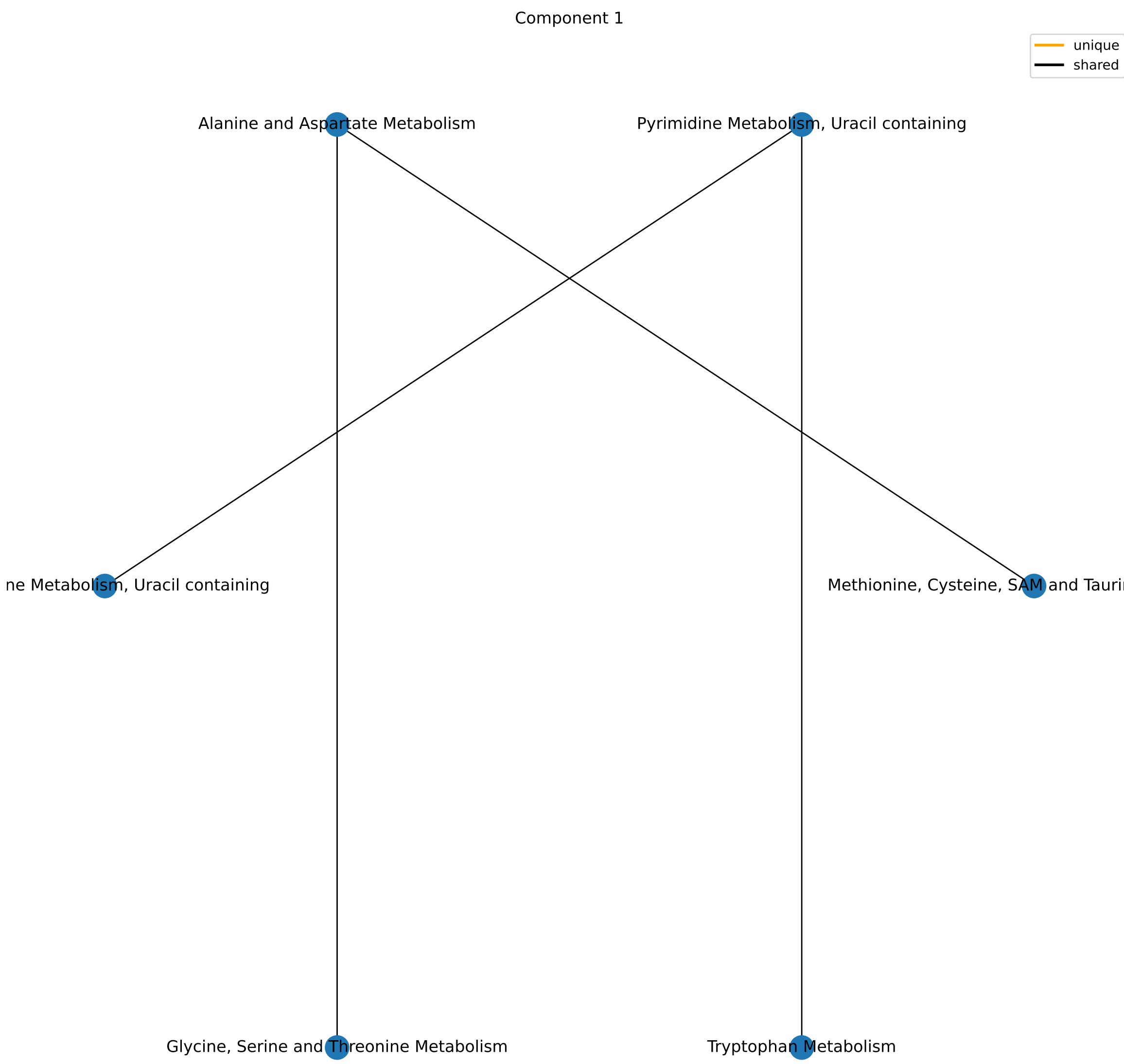
Component 1

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Amino Acid	180	5	0.00989176323329891	0.0890258690996902
Nucleotide	32	2	0.0317712984883834	0.142970843197725
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Lipid	363	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/curr-components/1-superpathway.csv

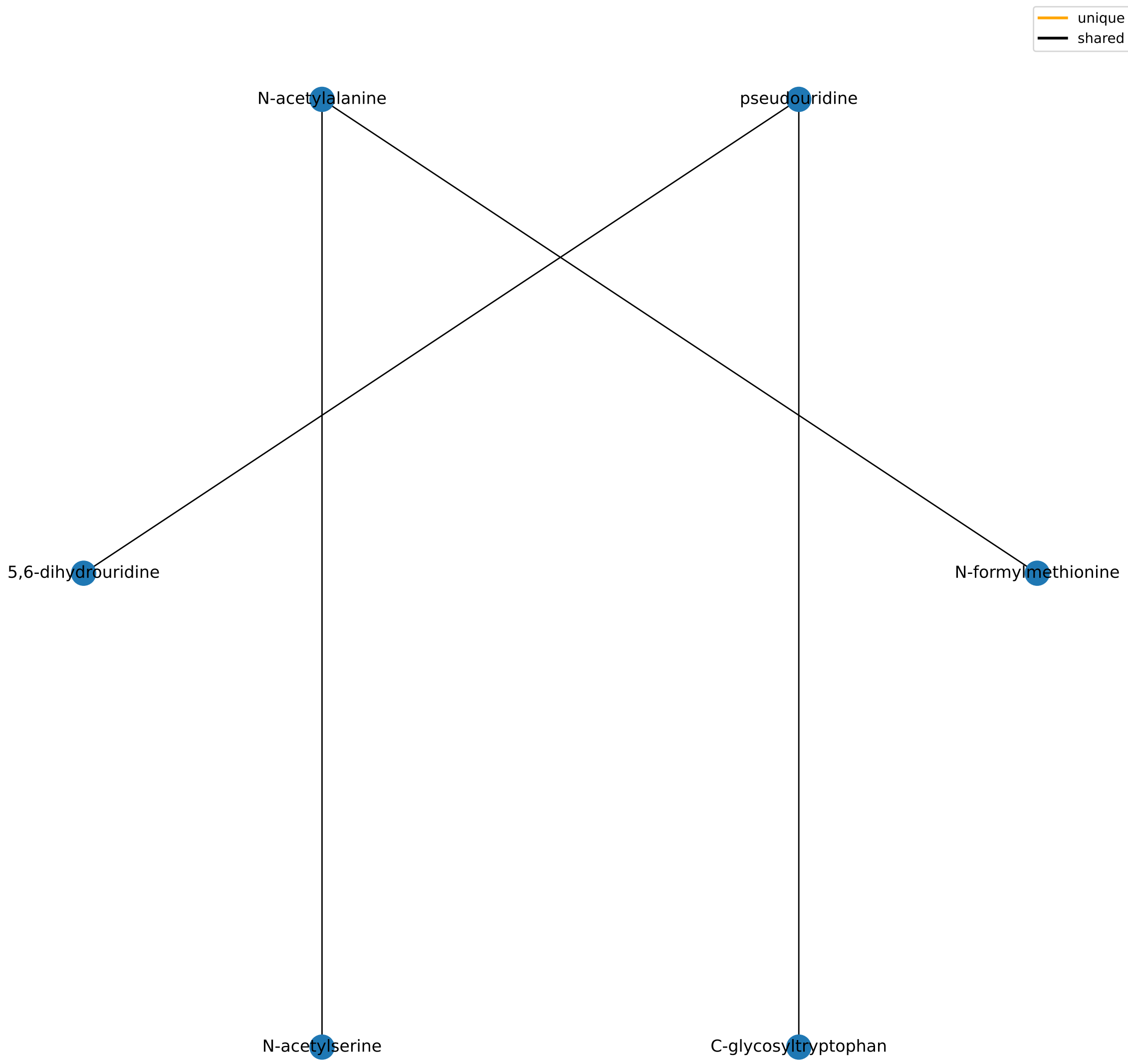
Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Alanine and Aspartate Metabolism	8	2	0.00199579061602098	0.165319357383607
Pyrimidine Metabolism, Uracil containing	10	2	0.00317921841122322	0.165319357383607
Glycine, Serine and Threonine Metabolism	10	1	0.0891120444102552	1
Tryptophan Metabolism	16	1	0.139240270348925	1
Methionine, Cysteine, SAM and Taurine Metabolism	17	1	0.147360645156954	1
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1

results/006/enrichment/curr-components/1-subpathway.csv

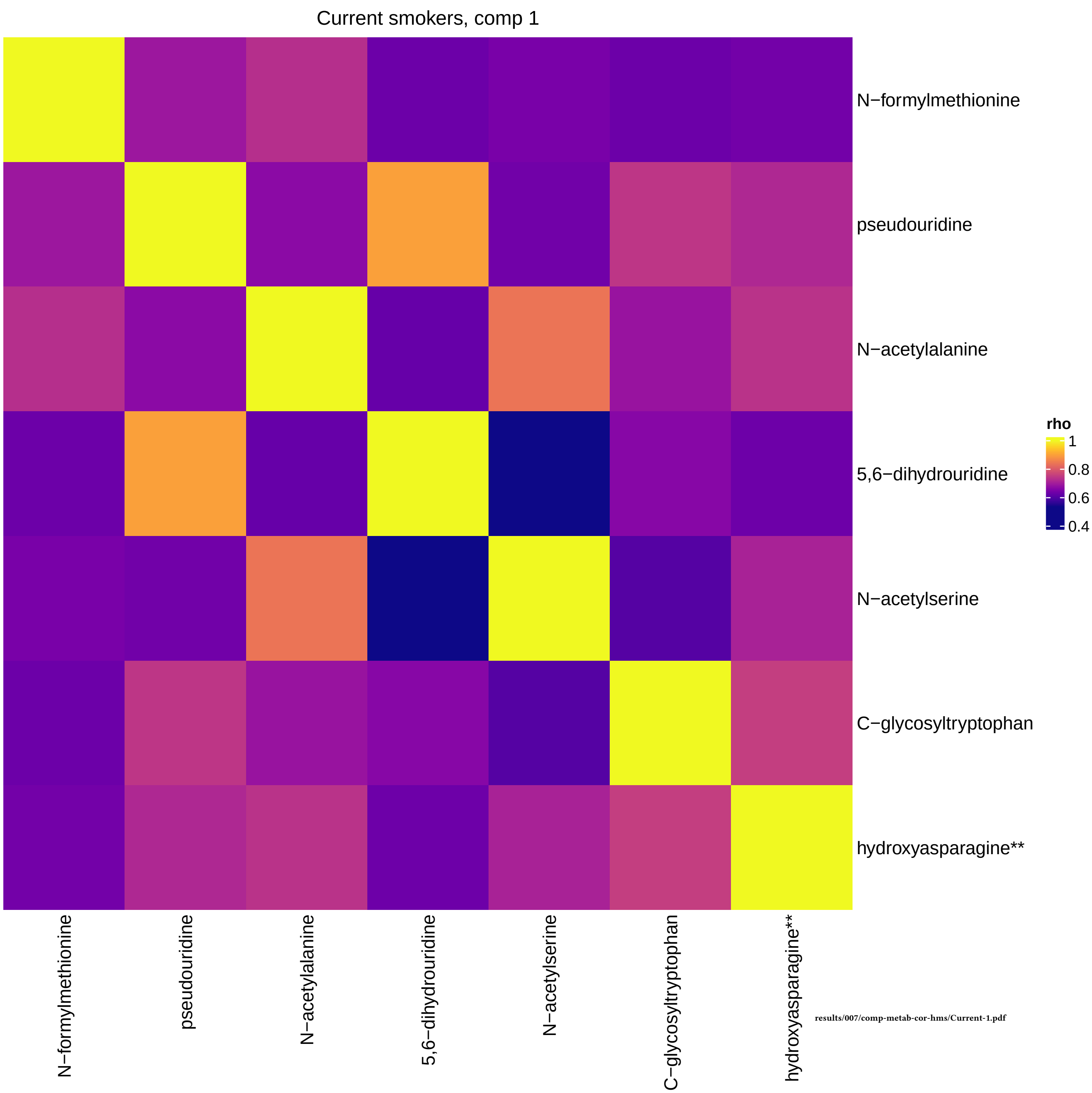


results/005_1/component-graphs/current/sub_pathway-0.pdf

Component 1



results/005_1/component-graphs/current/chemical_name-0.pdf


Current smokers, comp 1
N−formylmethionine
pseudouridine
N−acetylalanine
5,6−dihydrouridine
N−acetylserine
C−glycosyltryptophan
hydroxyasparagine**
rho
1
0.8
0.6
0.4
results/007/comp-metab-cor-hms/Current-1.pdf

Component 2

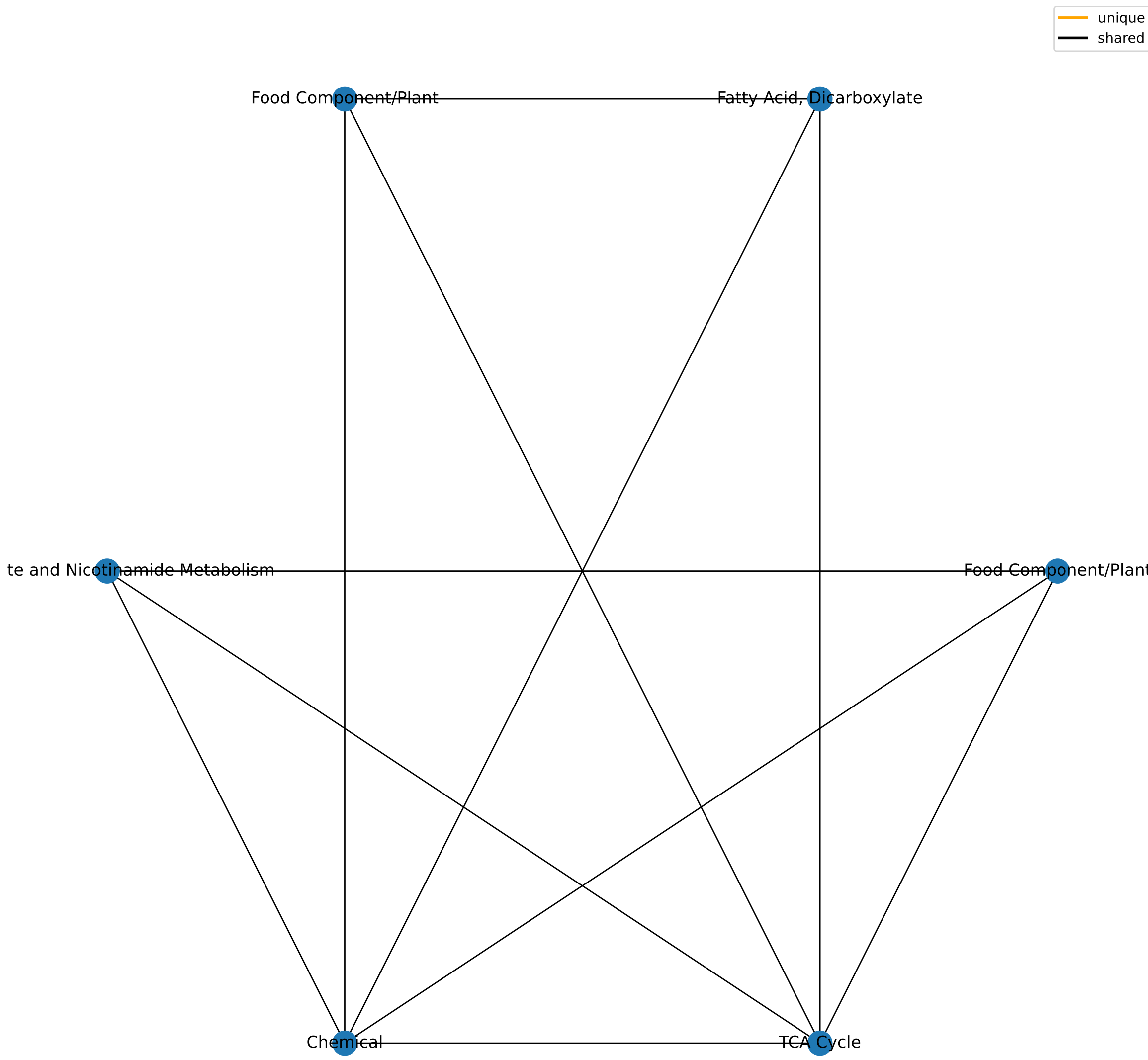
Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Xenobiotics	96	4	0.00626769567392165	0.0564092610652948
Energy	10	1	0.0891120444102552	0.401004199846148
Cofactors and Vitamins	25	1	0.209993587754082	0.629980763262246
Lipid	363	1	0.989829439076062	1
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1

results/006/enrichment/curr-components/2-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Food Component/Plant	38	3	0.00354588213275155	0.368771741806161
Nicotinate and Nicotinamide Metabolism	5	1	0.0454470159344476	1
TCA Cycle	9	1	0.0805187618103524	1
Chemical	18	1	0.155415267942246	1
Fatty Acid, Dicarboxylate	23	1	0.194717567032683	1
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1

results/006/enrichment/curr-components/2-subpathway.csv

Component 2



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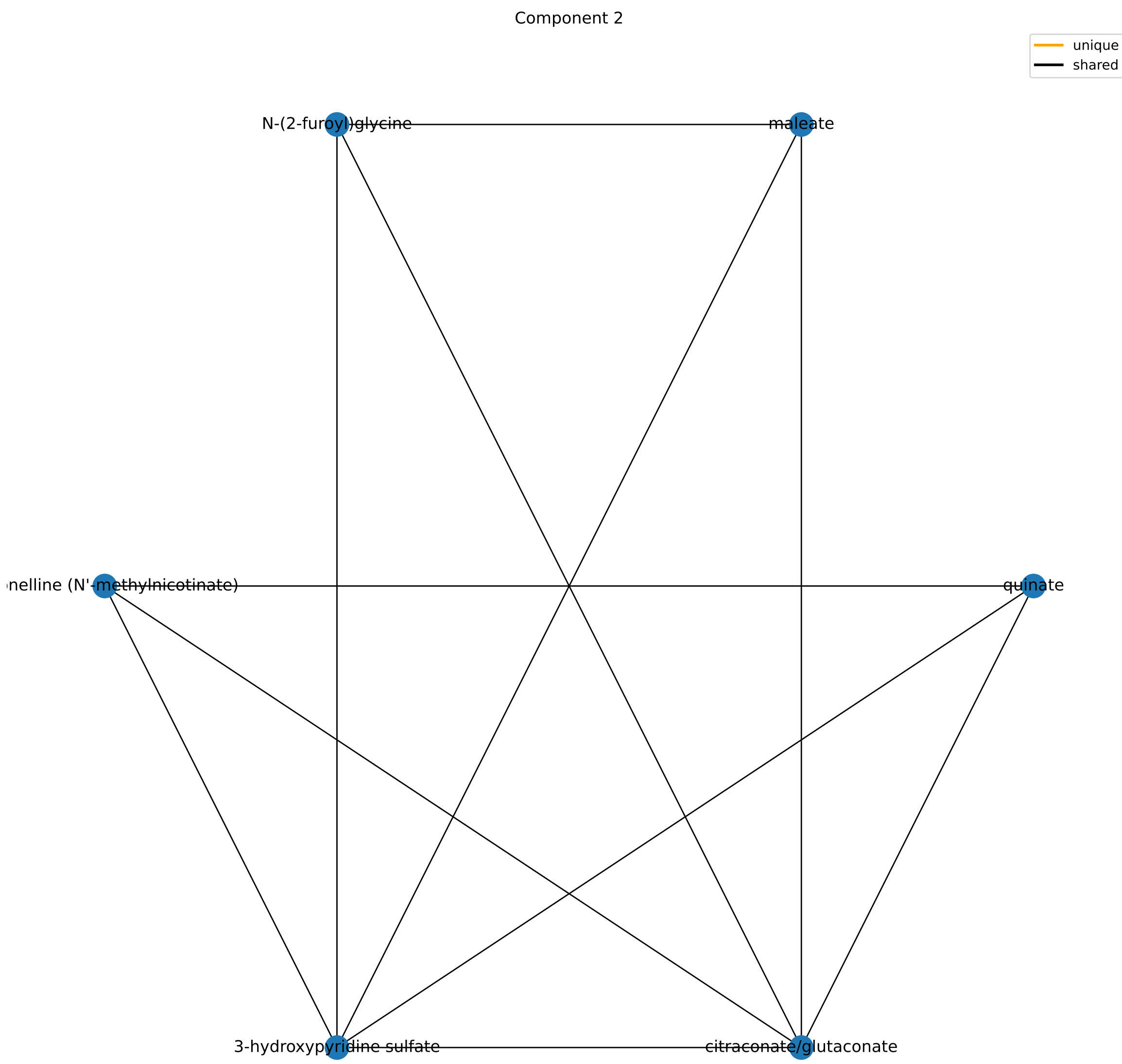

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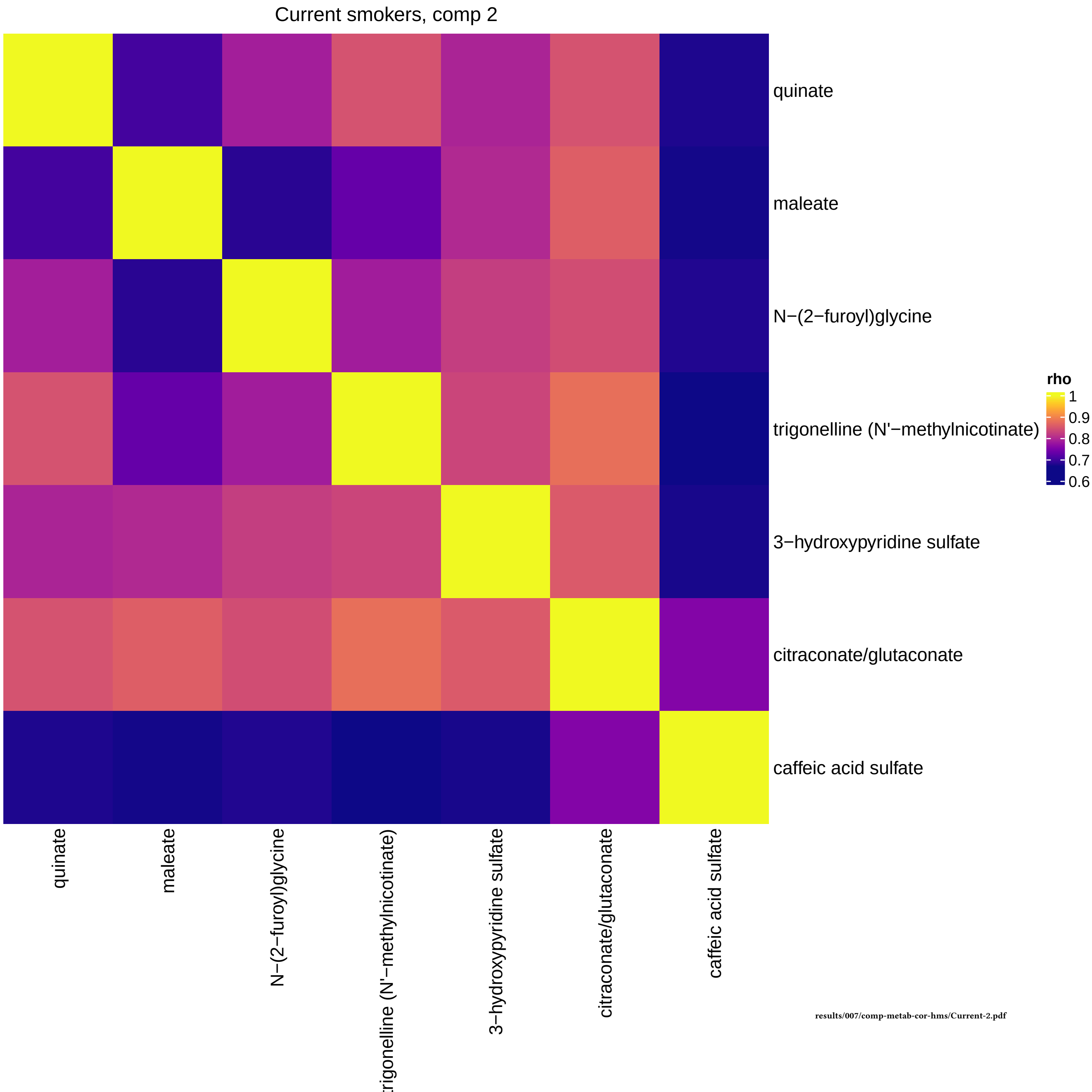

Current smokers, comp 2
quinate
maleate
N−(2−furoyl)glycine
trigonelline (N'−methylnicotinate)
3−hydroxypyridine sulfate
citraconate/glutaconate
caffeic acid sulfate
rho
1
0.9
0.8
0.7
0.6
results/007/comp-metab-cor-hms/Current-2.pdf

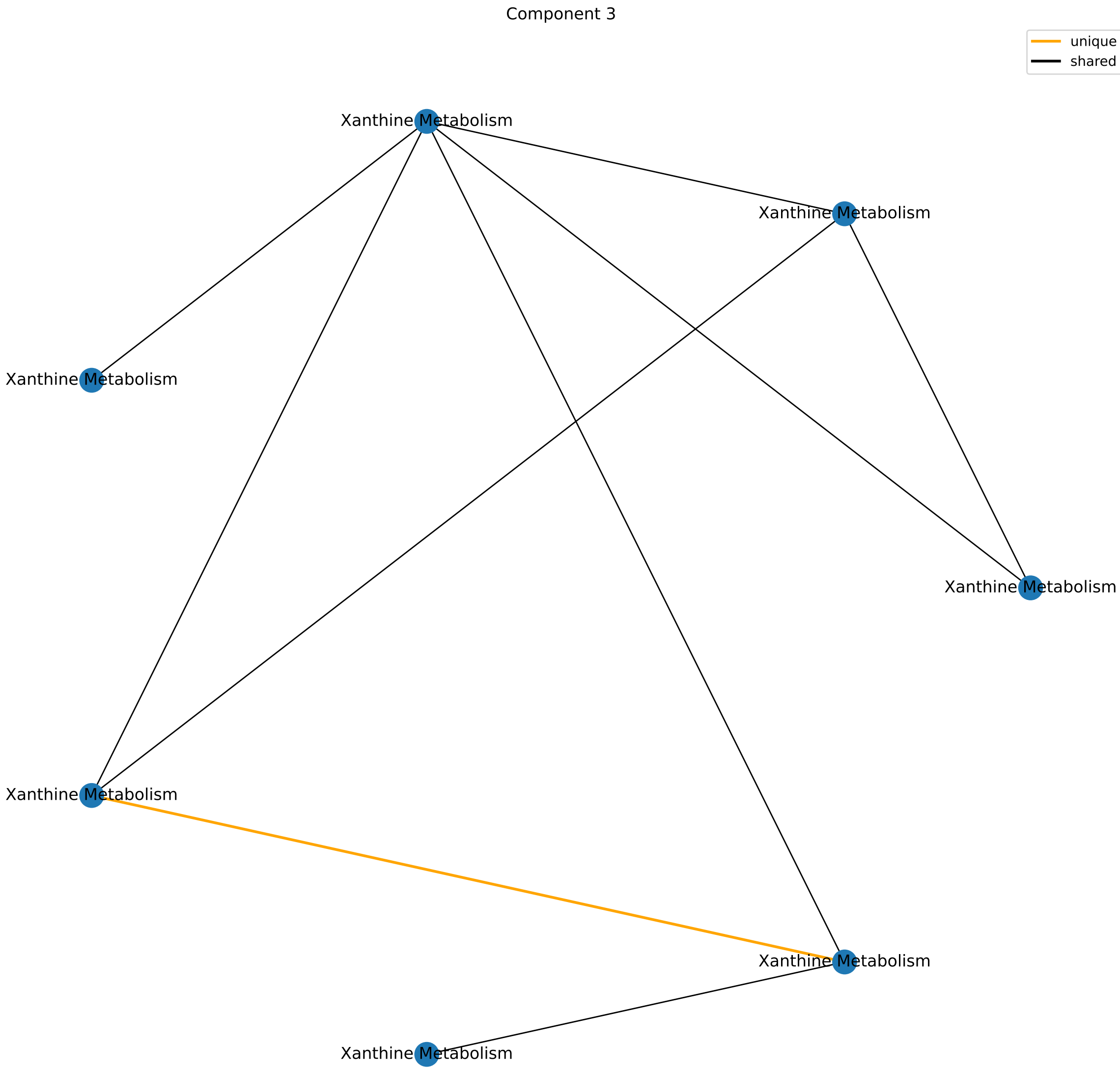
Component 3

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Xenobiotics	96	8	5.09107440612931e-08	4.58196696551638e-07
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Lipid	363	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1

results/006/enrichment/curr-components/3-superpathway.csv

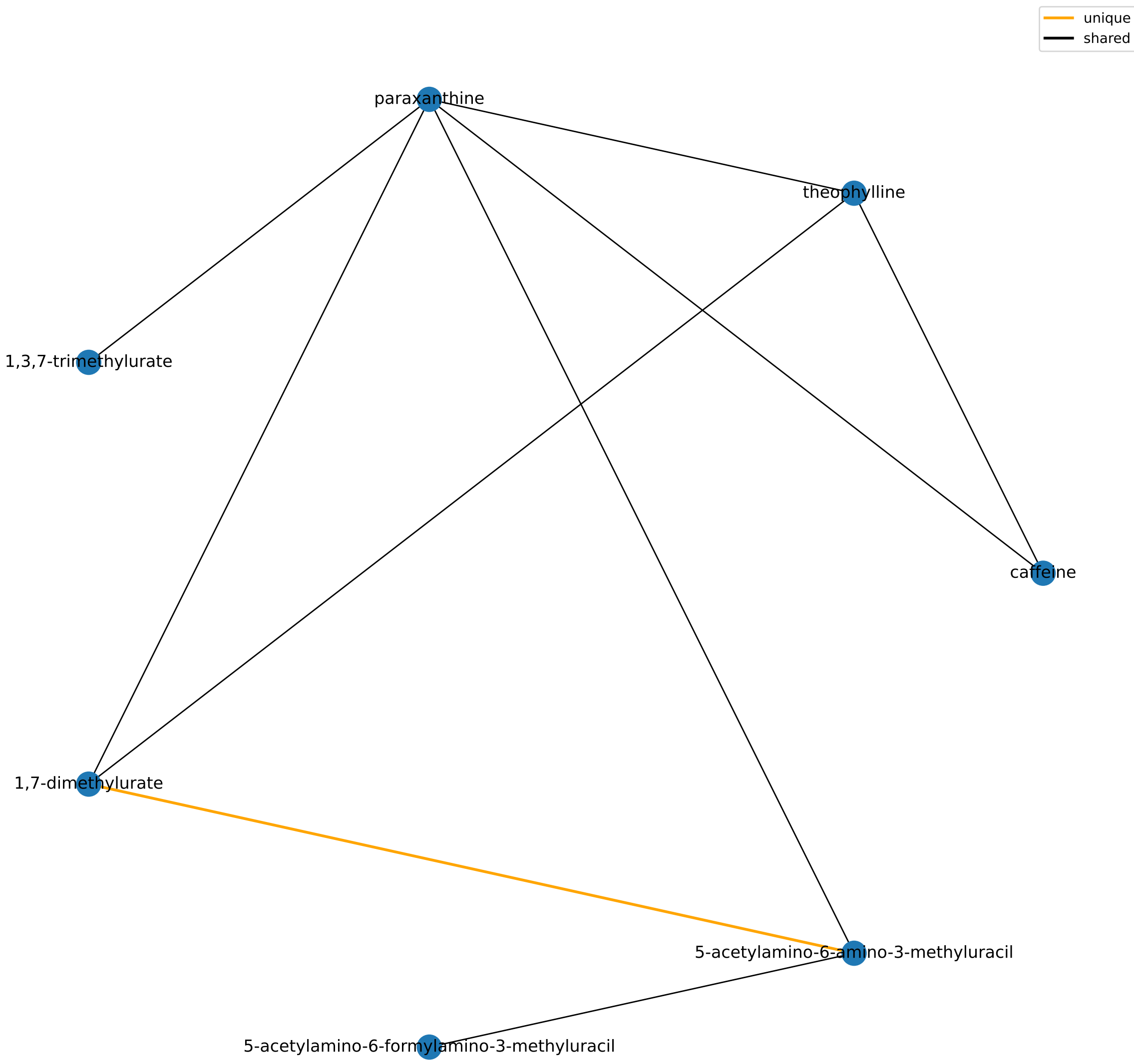
Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Xanthine Metabolism	14	8	1.15296977833151e-15	1.19908856946477e-13
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1
Benzoate Metabolism	21	0	1	1

results/006/enrichment/curr-components/3-subpathway.csv



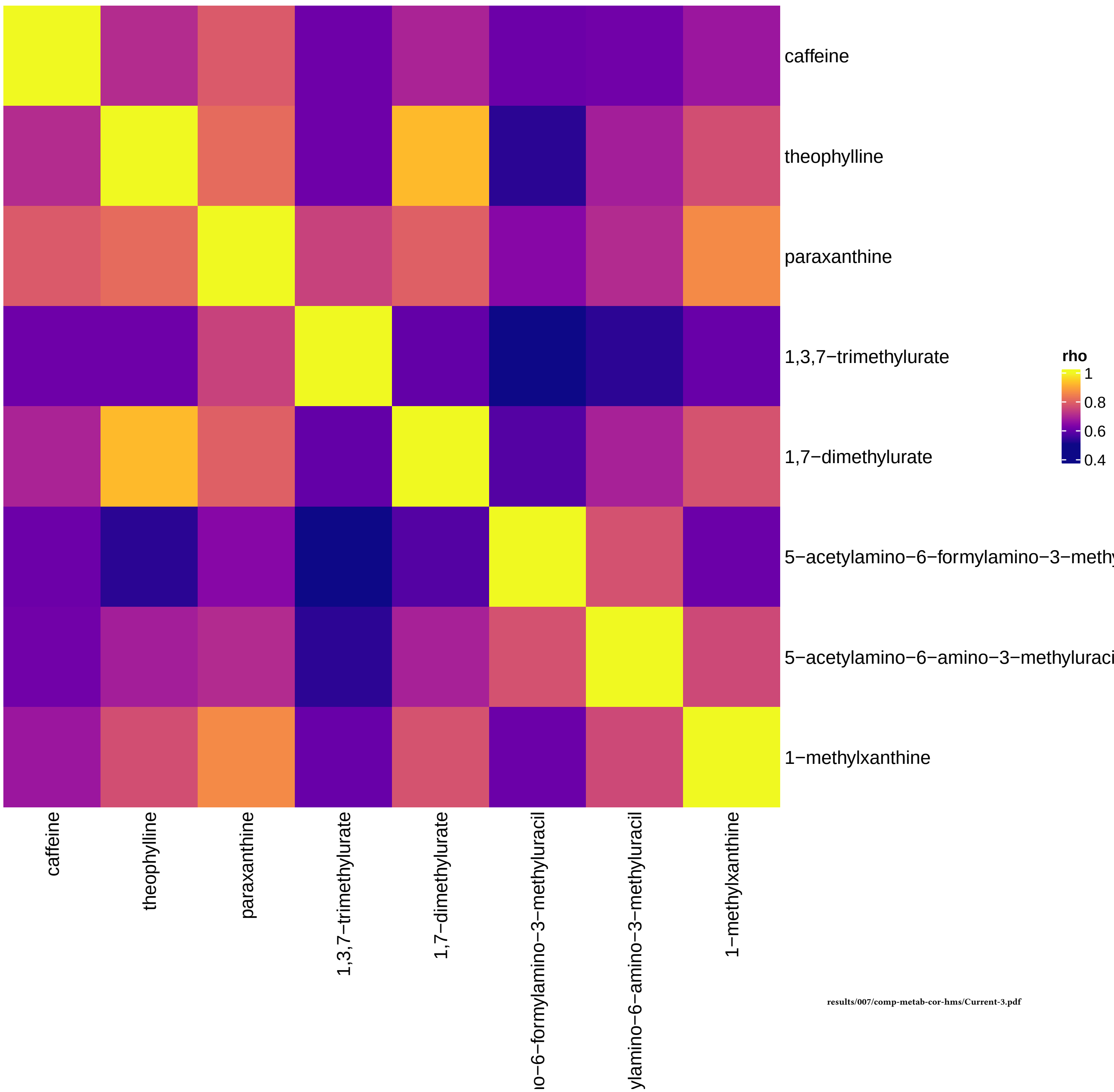
results/005_1/component-graphs/current/sub_pathway-2.pdf

Component 3



results/005_1/component-graphs/current/chemical_name-2.pdf

Current smokers, comp 3



results/007/comp-metab-cor-hms/Current-3.pdf

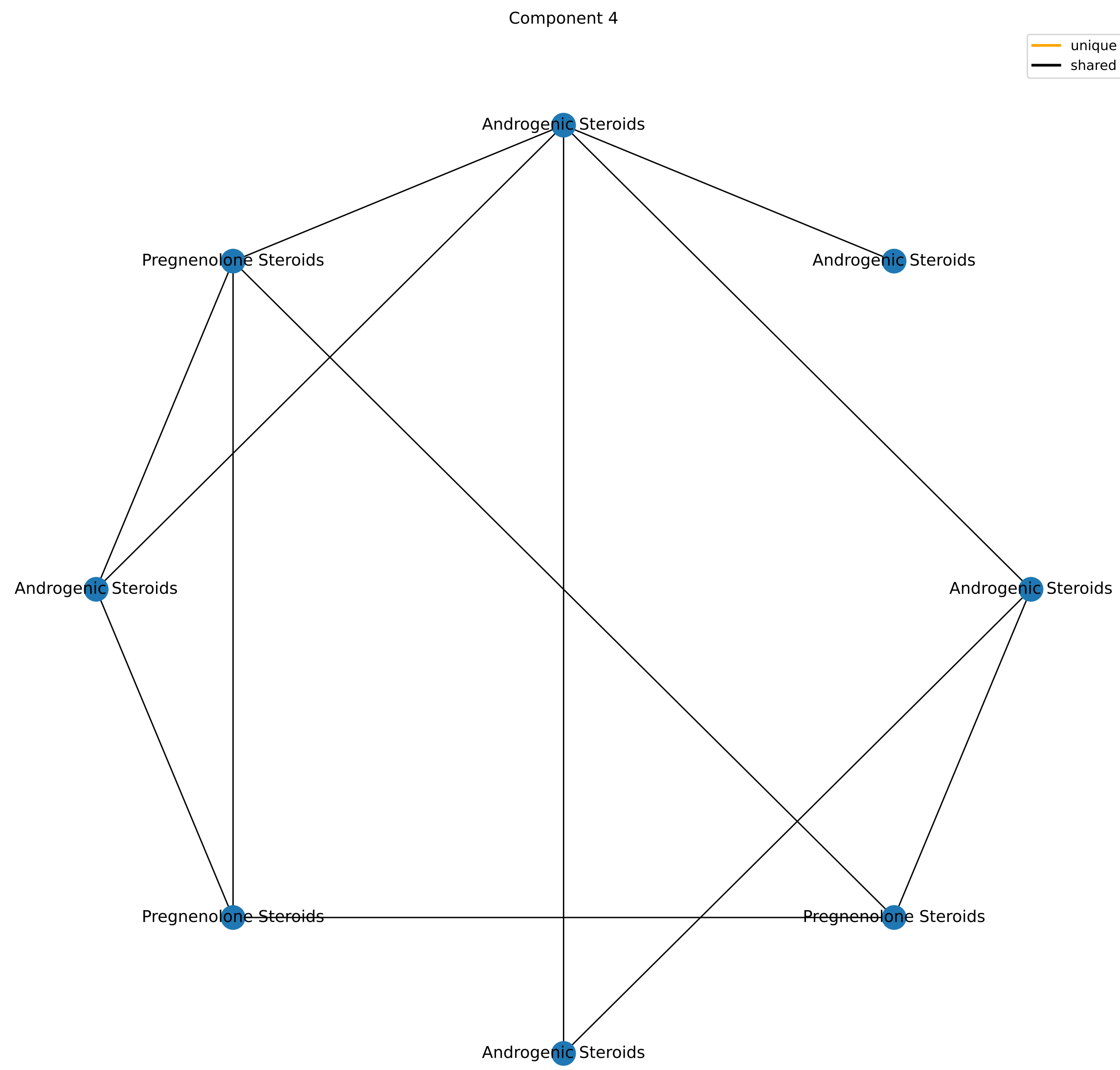
Component 4

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	9	0.00125727345198129	0.0113154610678316
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/curr-components/4-superpathway.csv

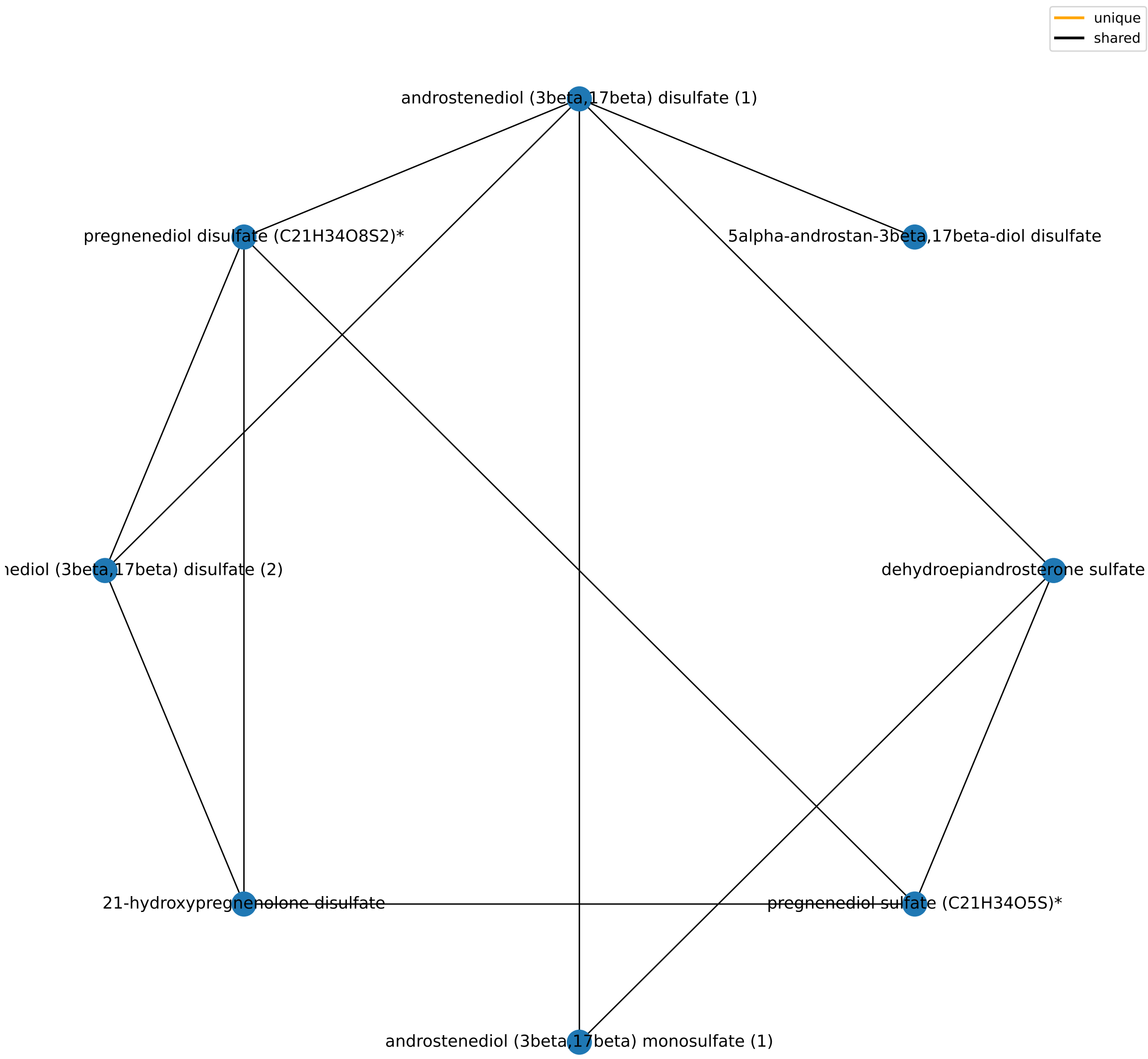
Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Pregnenolone Steroids	4	4	9.23310882952272e-09	9.60243318270362e-07
Androgenic Steroids	14	5	1.1776034831734e-07	6.12353811250167e-06
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1
Benzoate Metabolism	21	0	1	1

results/006/enrichment/curr-components/4-subpathway.csv



results/005_1/component-graphs/current/sub_pathway-3.pdf

Component 4



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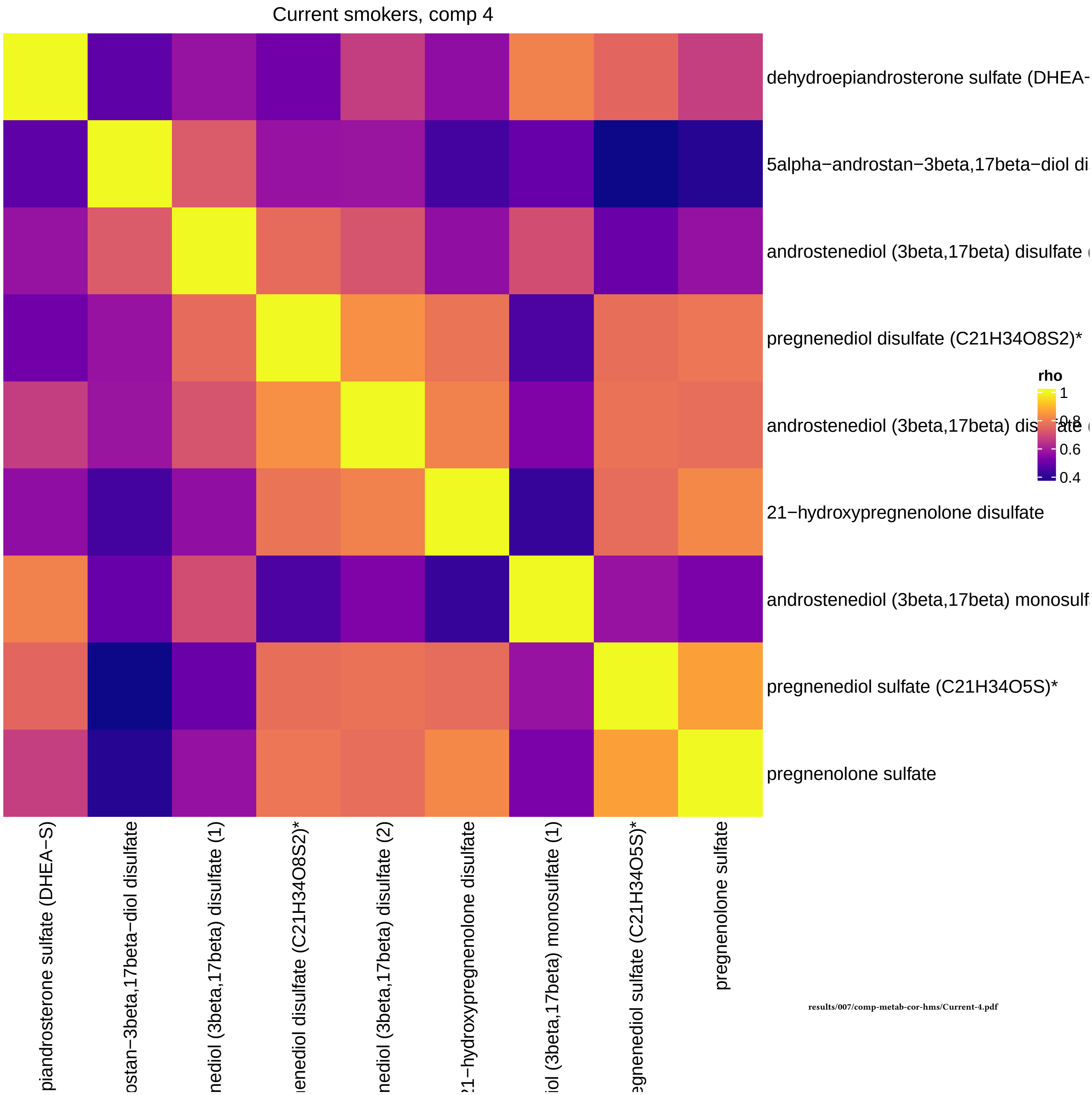


results/007/comp-metab-cor-hms/Current-4.pdf

Component 5

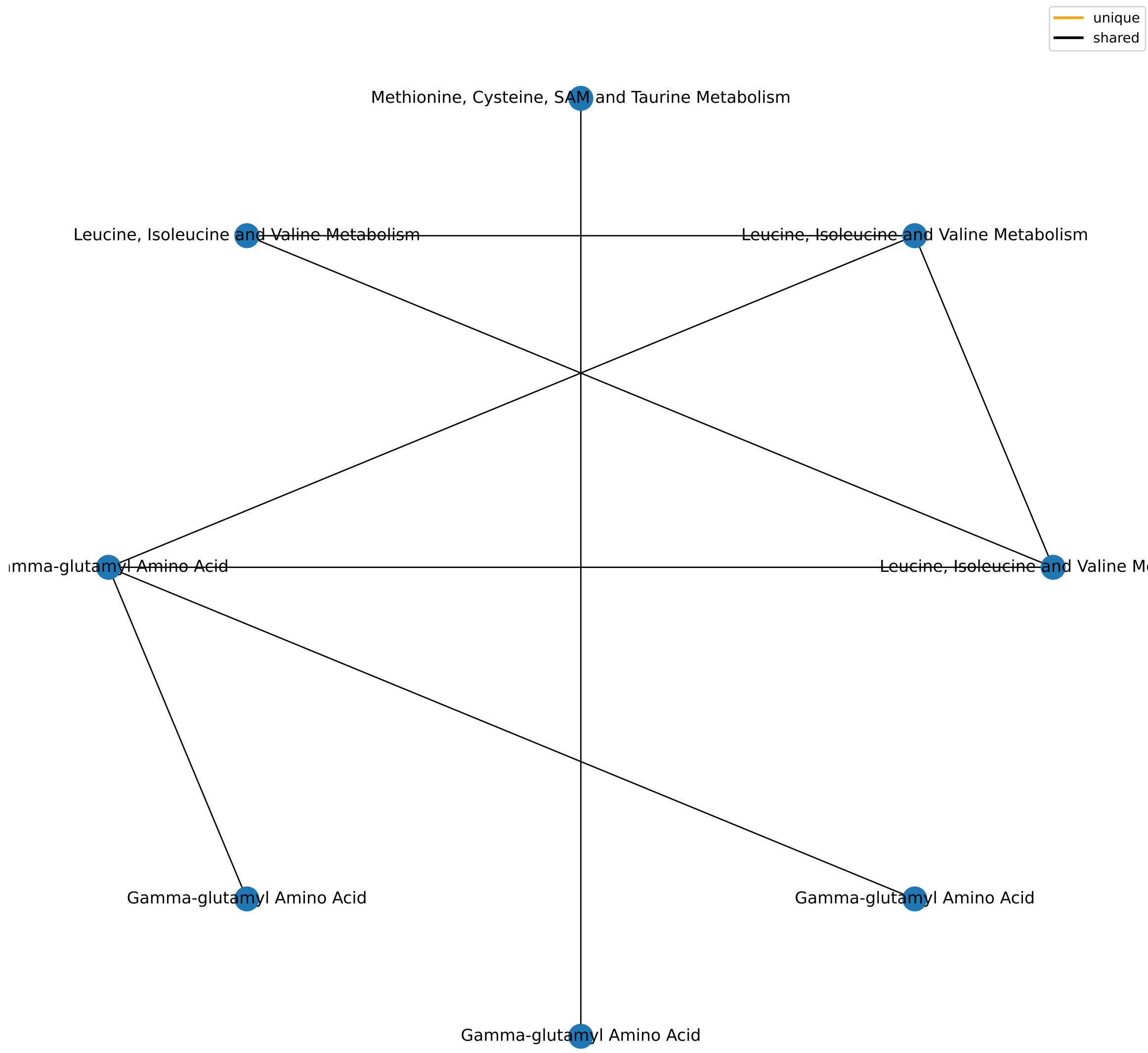
Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Peptide	25	6	2.83911221223453e-07	2.55520099101108e-06
Amino Acid	180	5	0.0939525030391726	0.422786263676277
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Lipid	363	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Xenobiotics	96	0	1	1

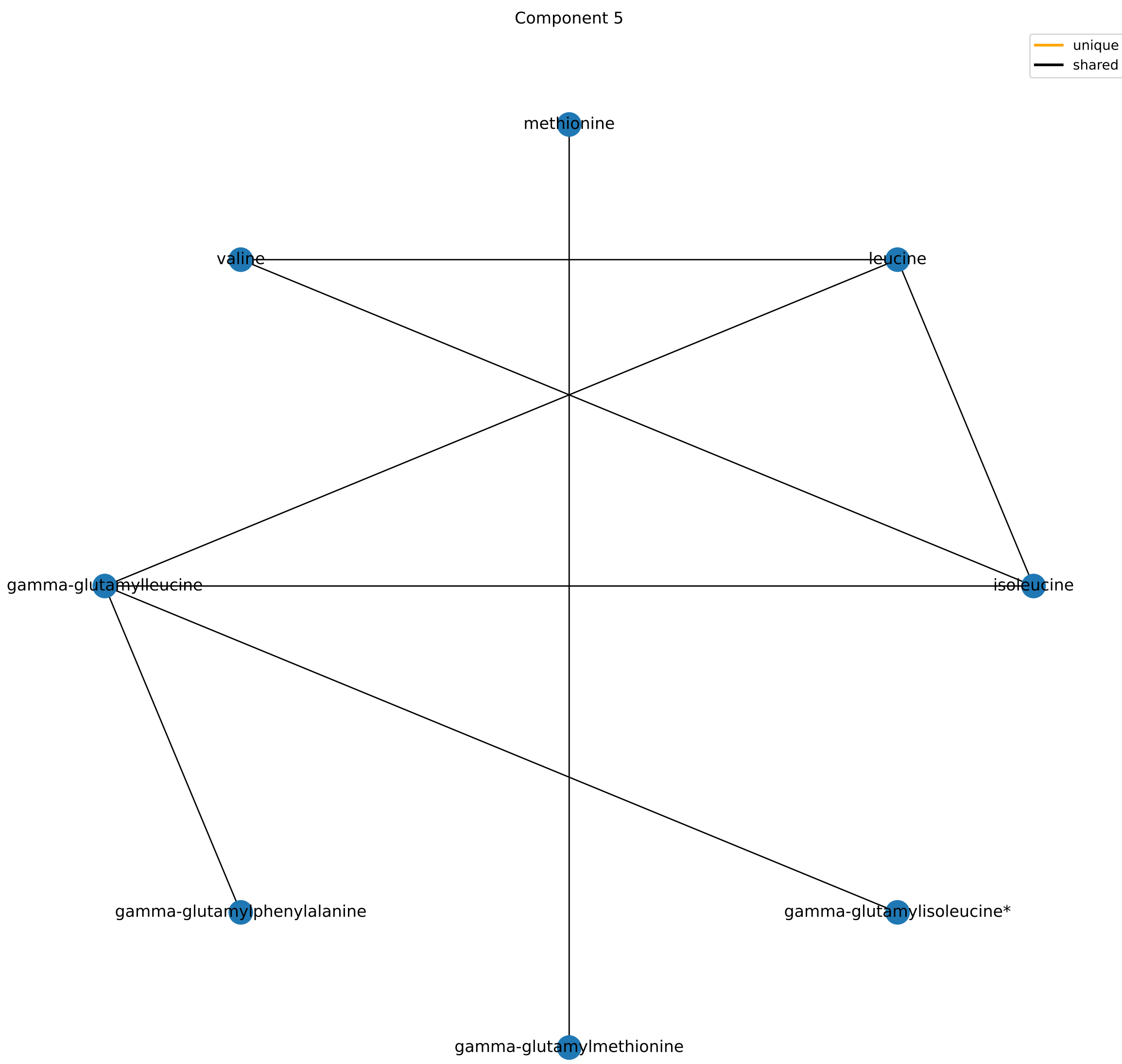
results/006/enrichment/curr-components/5-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Gamma-glutamyl Amino Acid	17	6	2.07829426601731e-08	2.161426036658e-06
Leucine, Isoleucine and Valine Metabolism	31	3	0.00819186524766017	0.425976992878329
Lysine Metabolism	12	1	0.161980404221457	1
Methionine, Cysteine, SAM and Taurine Metabolism	17	1	0.222126008571237	1
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1

results/006/enrichment/curr-components/5-subpathway.csv

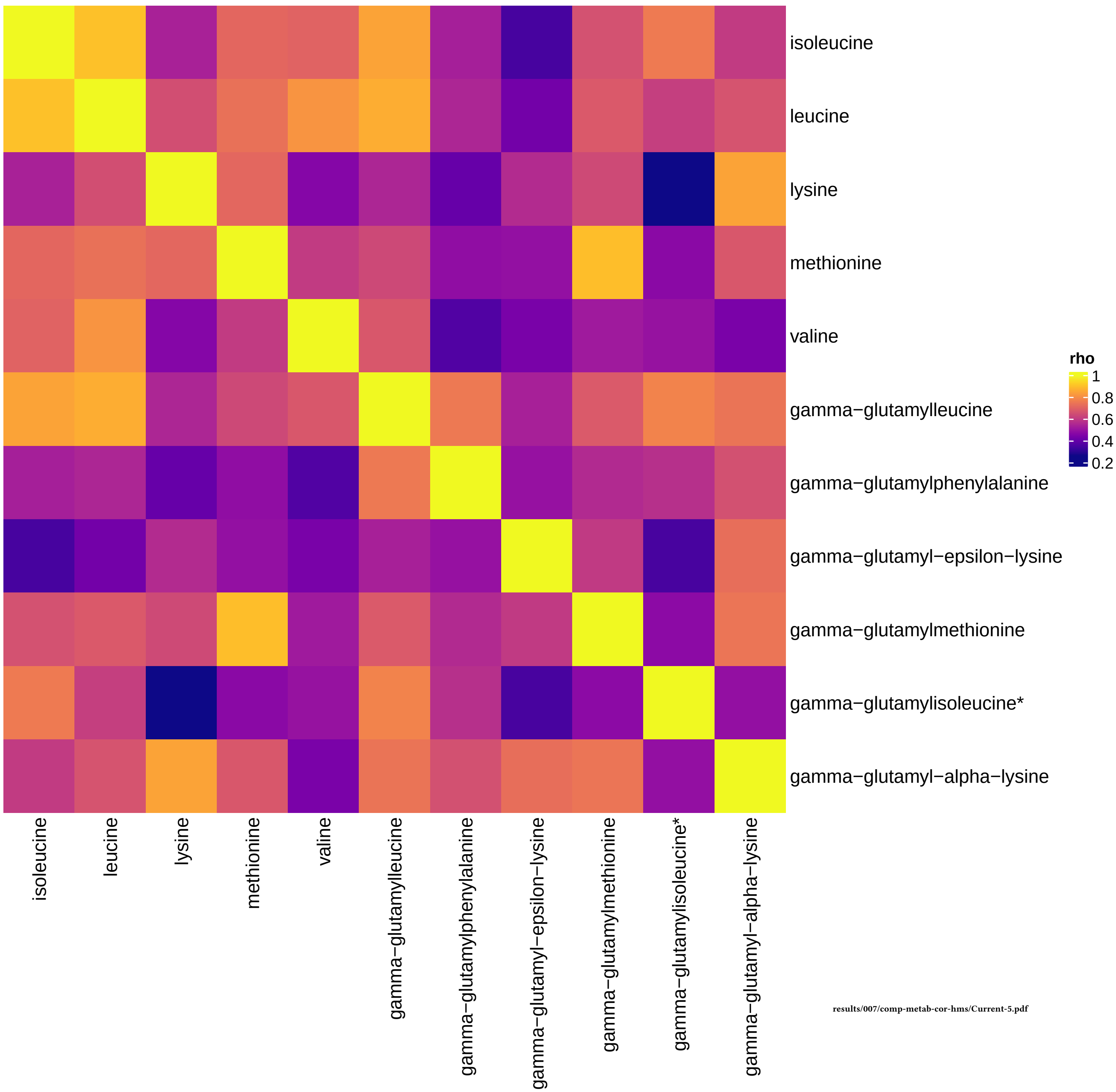
Component 5


unique
shared
Methionine, Cysteine, SAM and Taurine Metabolism
Leucine, Isoleucine and Valine Metabolism
Leucine, Isoleucine and Valine Metabolism
mma-glutamyl Amino Acid
Leucine, Isoleucine and Valine Me
Gamma-glutamyl Amino Acid
Gamma-glutamyl Amino Acid
Gamma-glutamyl Amino Acid



results/005_1/component-graphs/current/chemical_name-4.pdf

Current smokers, comp 5



results/007/comp-metab-cor-hms/Current-5.pdf

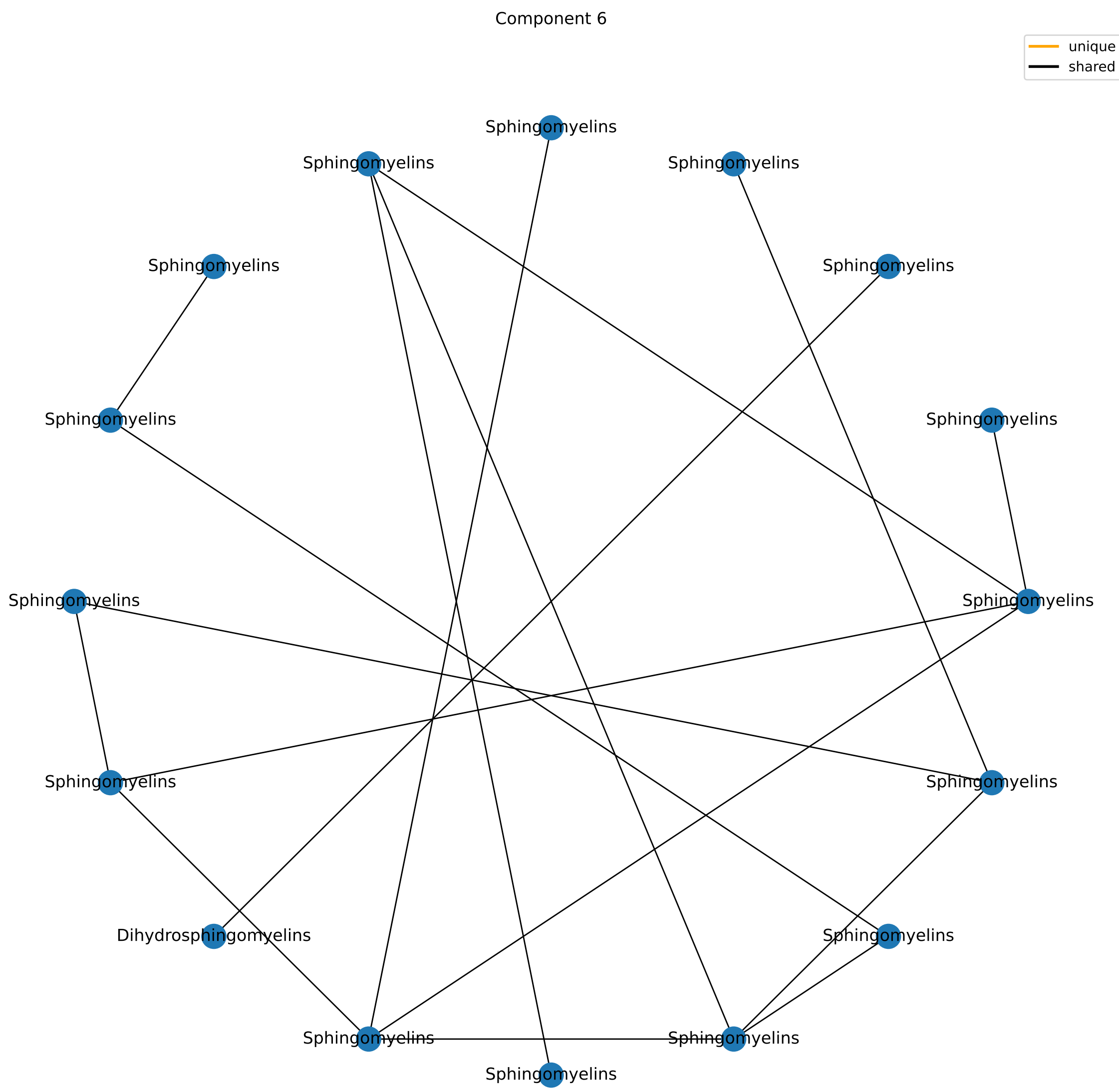
Component 6

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	17	3.00119981151321e-06	2.70107983036189e-05
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/curr-components/6-superpathway.csv

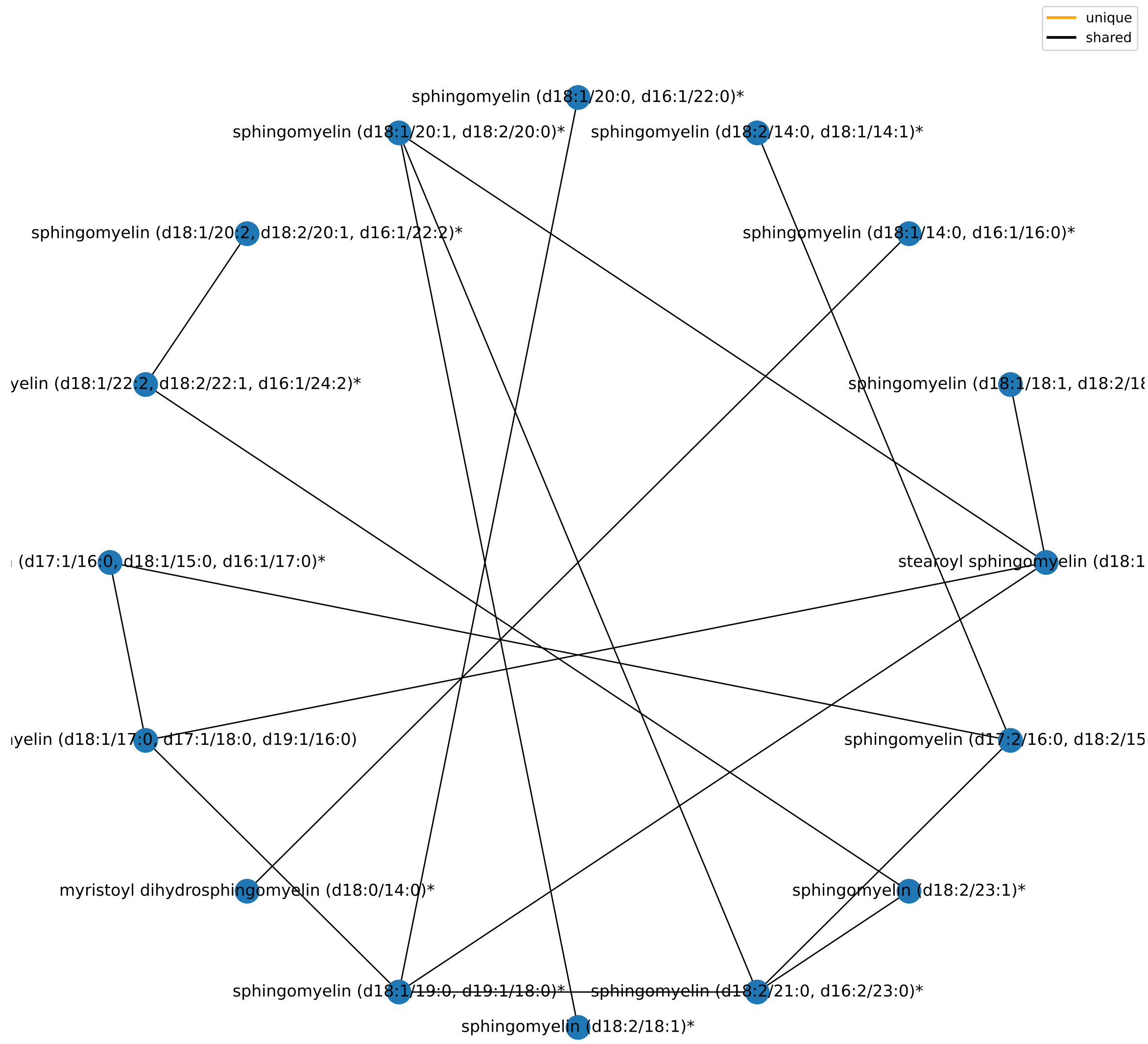
Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Sphingomyelins	28	16	1.05223178291208e-24	1.09432105422856e-22
Dihydrosphingomyelins	5	1	0.107490109220638	1
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1

results/006/enrichment/curr-components/6-subpathway.csv



results/005_1/component-graphs/current/sub_pathway-5.pdf

Component 6



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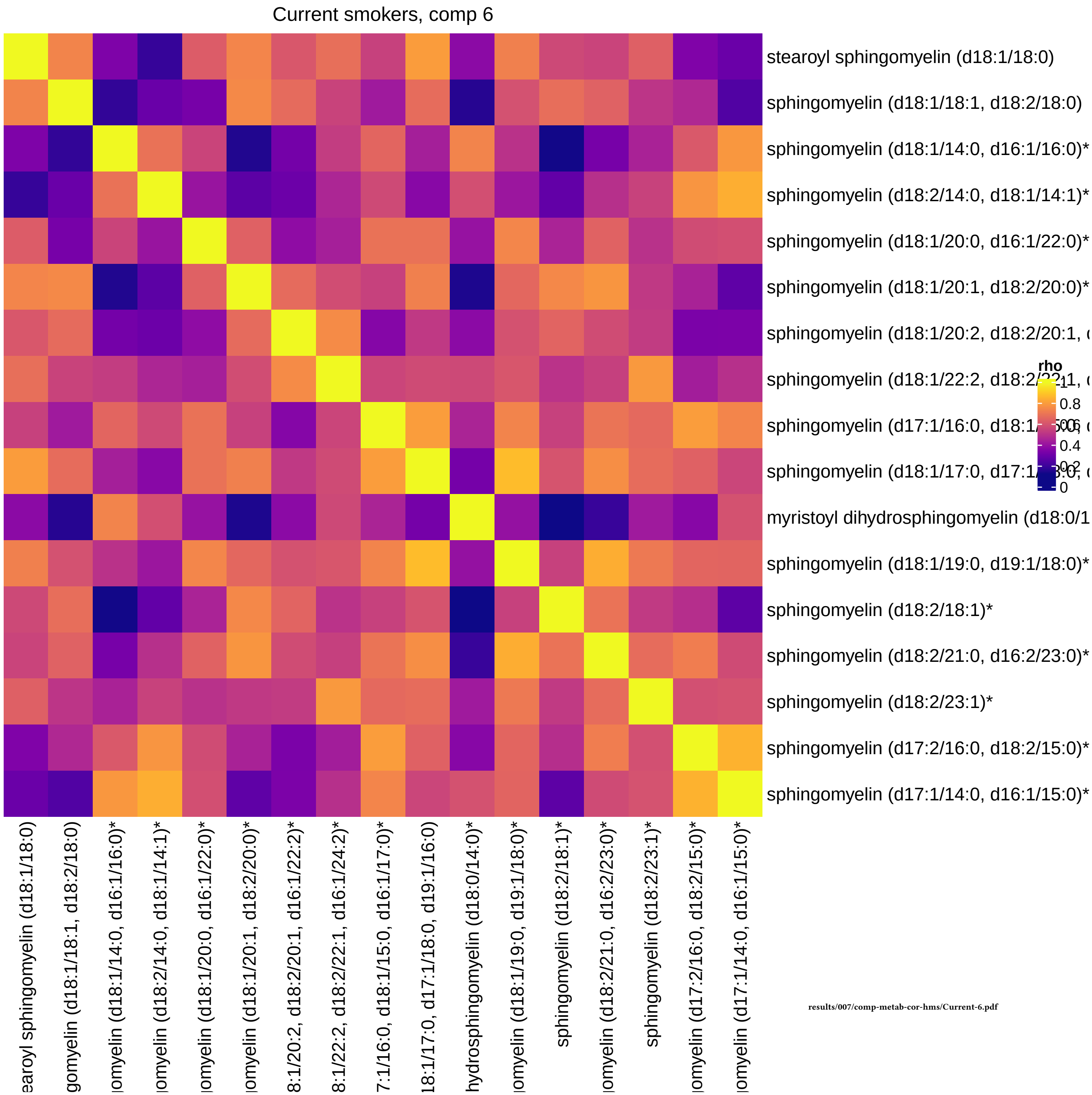

Current smokers, comp 6
stearoyl sphingomyelin (d18:1/18:0)
sphingomyelin (d18:1/18:1, d18:2/18:0)
sphingomyelin (d18:1/14:0, d16:1/16:0)*
sphingomyelin (d18:2/14:0, d18:1/14:1)*
sphingomyelin (d18:1/20:0, d16:1/22:0)*
sphingomyelin (d18:1/20:1, d18:2/20:0)*
sphingomyelin (d18:1/20:2, d18:2/20:1, d
sphingomyelin (d18:1/22:2, d18:2/22:1, d
sphingomyelin (d17:1/16:0, d18:1/15:0, d
sphingomyelin (d18:1/17:0, d17:1/18:0, d
myristoyl dihydrosphingomyelin (d18:0/1
sphingomyelin (d18:1/19:0, d19:1/18:0)*
sphingomyelin (d18:2/18:1)*
sphingomyelin (d18:2/21:0, d16:2/23:0)*
sphingomyelin (d18:2/23:1)*
sphingomyelin (d17:2/16:0, d18:2/15:0)*
sphingomyelin (d17:1/14:0, d16:1/15:0)*
rho
1
0.8
0.6
0.4
0.2
0
earoyl sphingomyelin (d18:1/18:0)
gomyelin (d18:1/18:1, d18:2/18:0)
omyelin (d18:1/14:0, d16:1/16:0)*
omyelin (d18:2/14:0, d18:1/14:1)*
omyelin (d18:1/20:0, d16:1/22:0)*
omyelin (d18:1/20:1, d18:2/20:0)*
8:1/20:2, d18:2/20:1, d16:1/22:2)*
8:1/22:2, d18:2/22:1, d16:1/24:2)*
7:1/16:0, d18:1/15:0, d16:1/17:0)*
18:1/17:0, d17:1/18:0, d19:1/16:0)
hydrosphingomyelin (d18:0/14:0)*
omyelin (d18:1/19:0, d19:1/18:0)*
sphingomyelin (d18:2/18:1)*
omyelin (d18:2/21:0, d16:2/23:0)*
sphingomyelin (d18:2/23:1)*
omyelin (d17:2/16:0, d18:2/15:0)*
omyelin (d17:1/14:0, d16:1/15:0)*
results/007/comp-metab-cor-hms/Current-6.pdf

Component 7

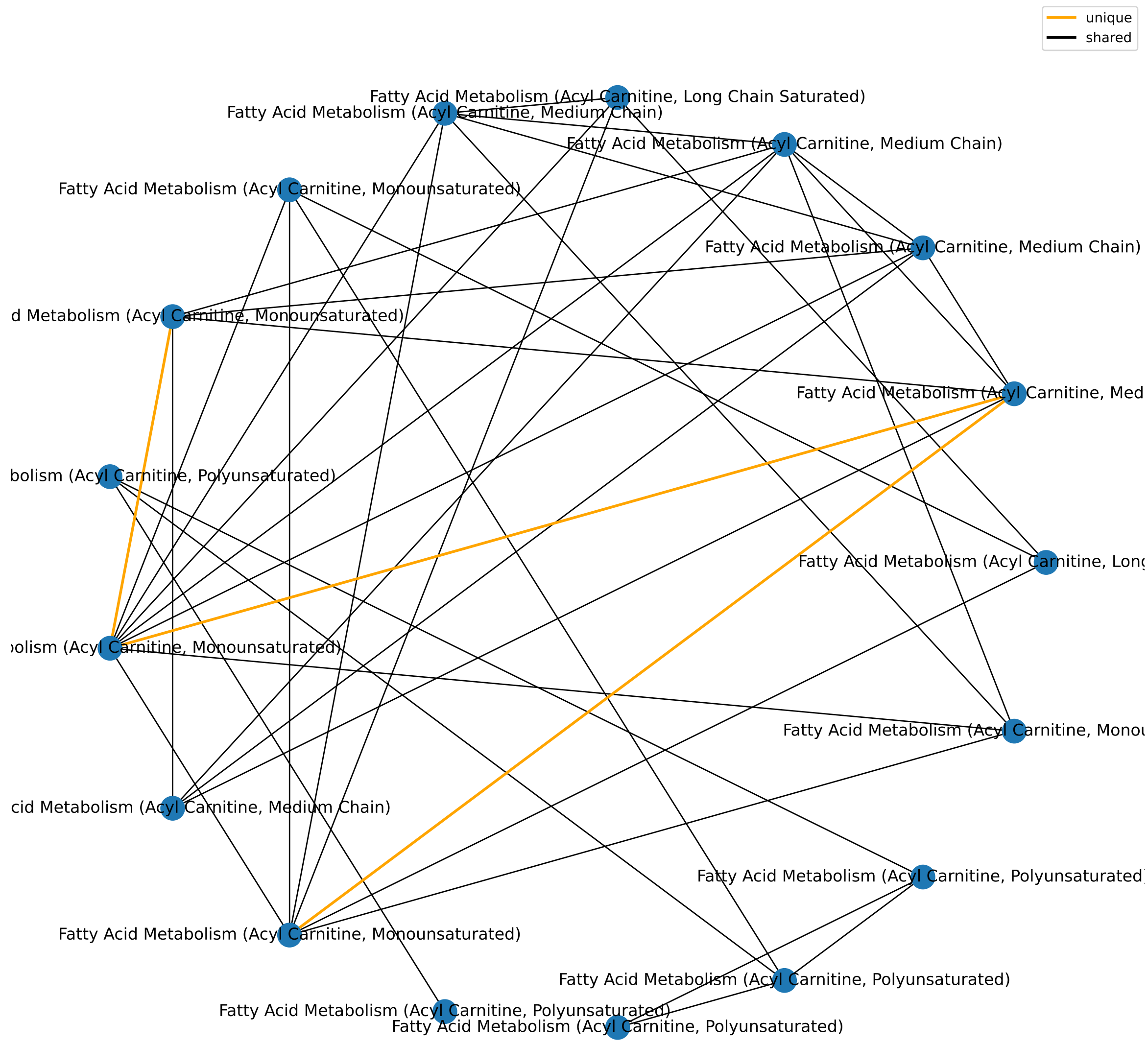
Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	18	1.40136995247445e-06	1.26123295722701e-05
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/curr-components/7-superpathway.csv

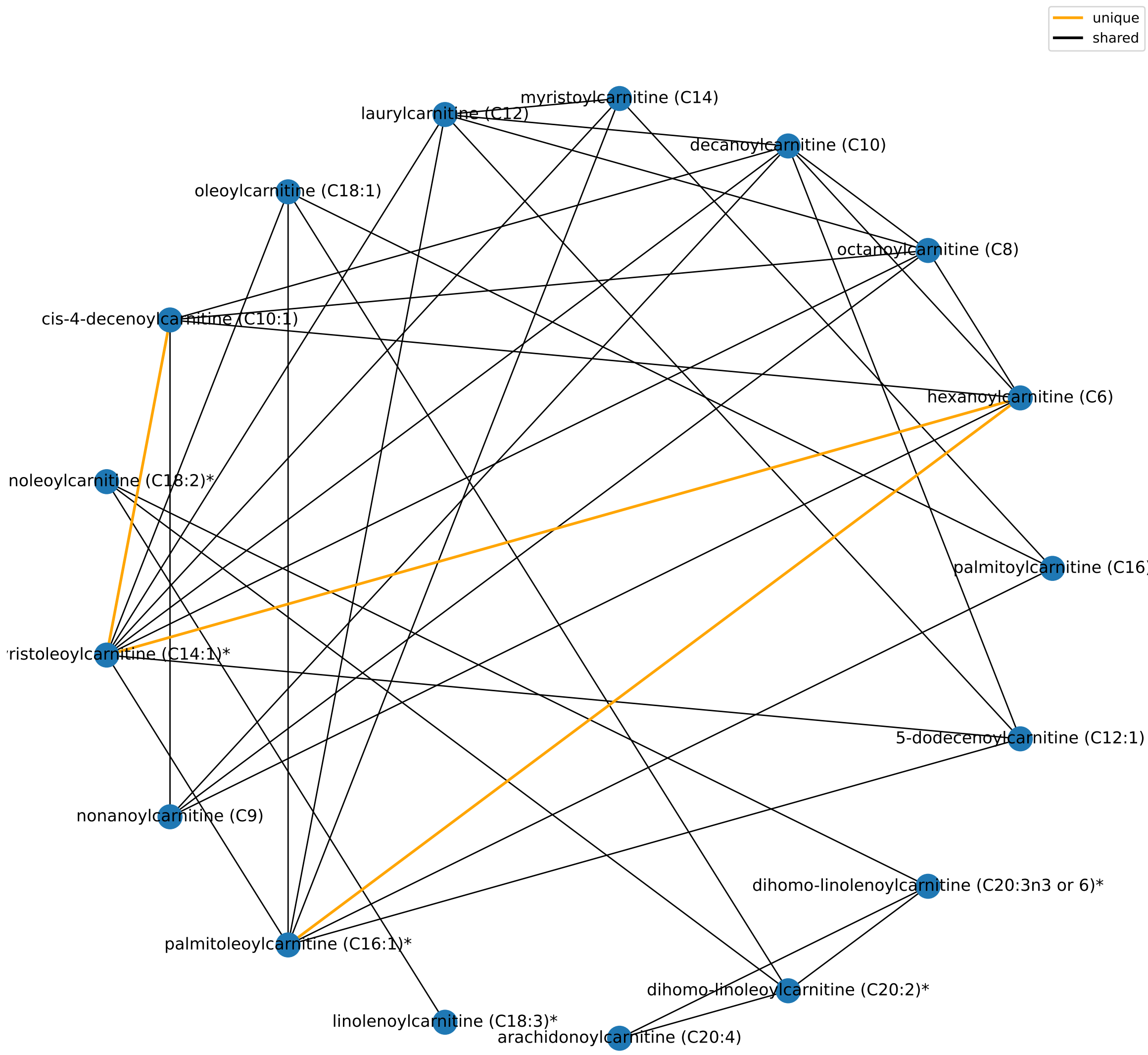
Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	4.16347082498372e-09	2.16500482899153e-07
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	4.16347082498372e-09	2.16500482899153e-07
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	5	2.23228315962357e-07	7.73858162002839e-06
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	2	0.0137139680231022	0.356563168600658
Fatty Acid Metabolism (Acyl Carnitine, Hydroxy)	3	1	0.0696515480675982	1
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1

results/006/enrichment/curr-components/7-subpathway.csv

Component 7

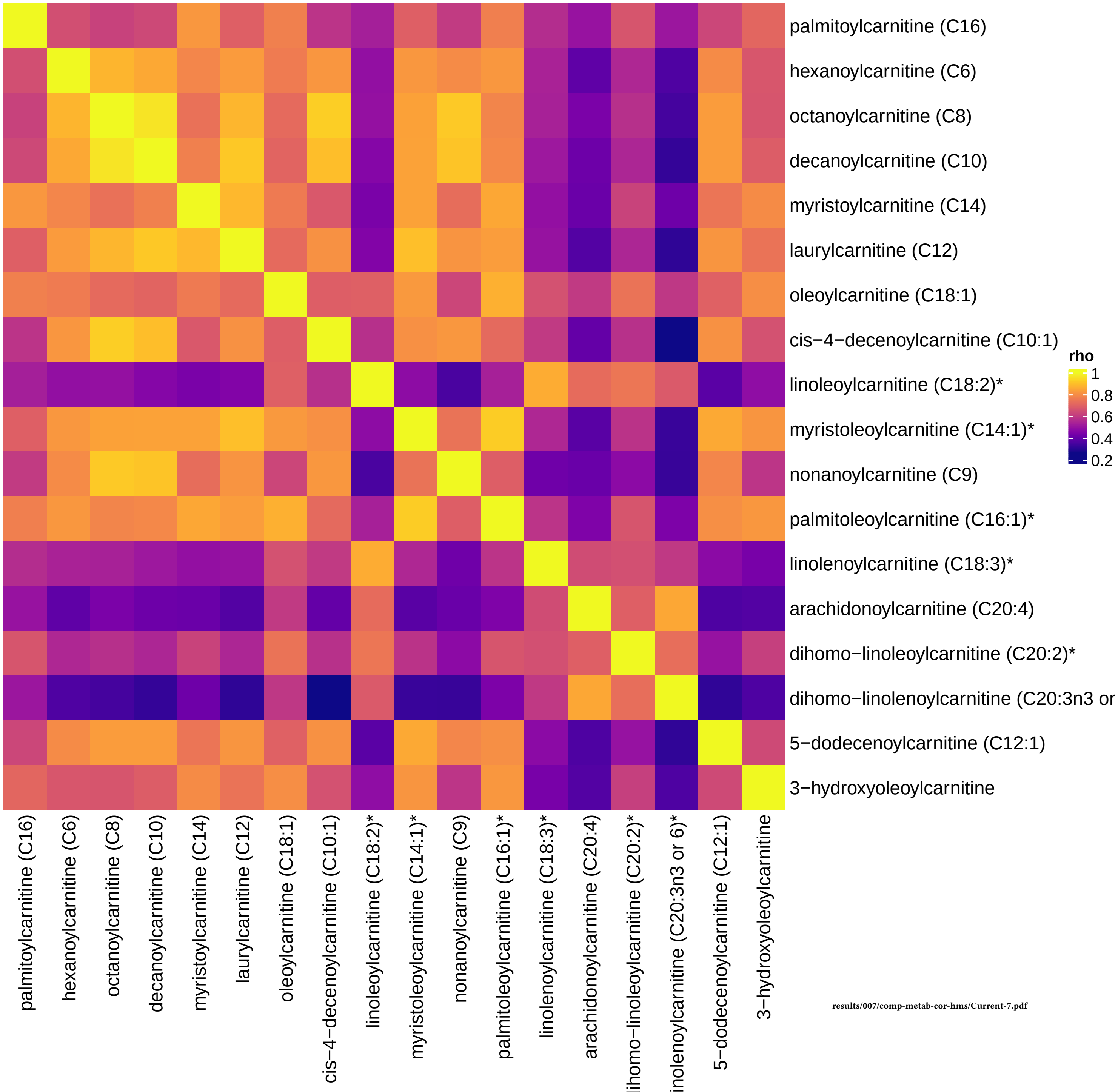

unique
shared
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)
d Metabolism (Acyl Carnitine, Monounsaturated)
Fatty Acid Metabolism (Acyl Carnitine, Med
bolism (Acyl Carnitine, Polyunsaturated)
Fatty Acid Metabolism (Acyl Carnitine, Long
bolism (Acyl Carnitine, Monounsaturated)
Fatty Acid Metabolism (Acyl Carnitine, Monou
cid Metabolism (Acyl Carnitine, Medium Chain)
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)

Component 7



results/005_1/component-graphs/current/chemical_name-6.pdf

Current smokers, comp 7



results/007/comp-metab-cor-hms/Current-7.pdf

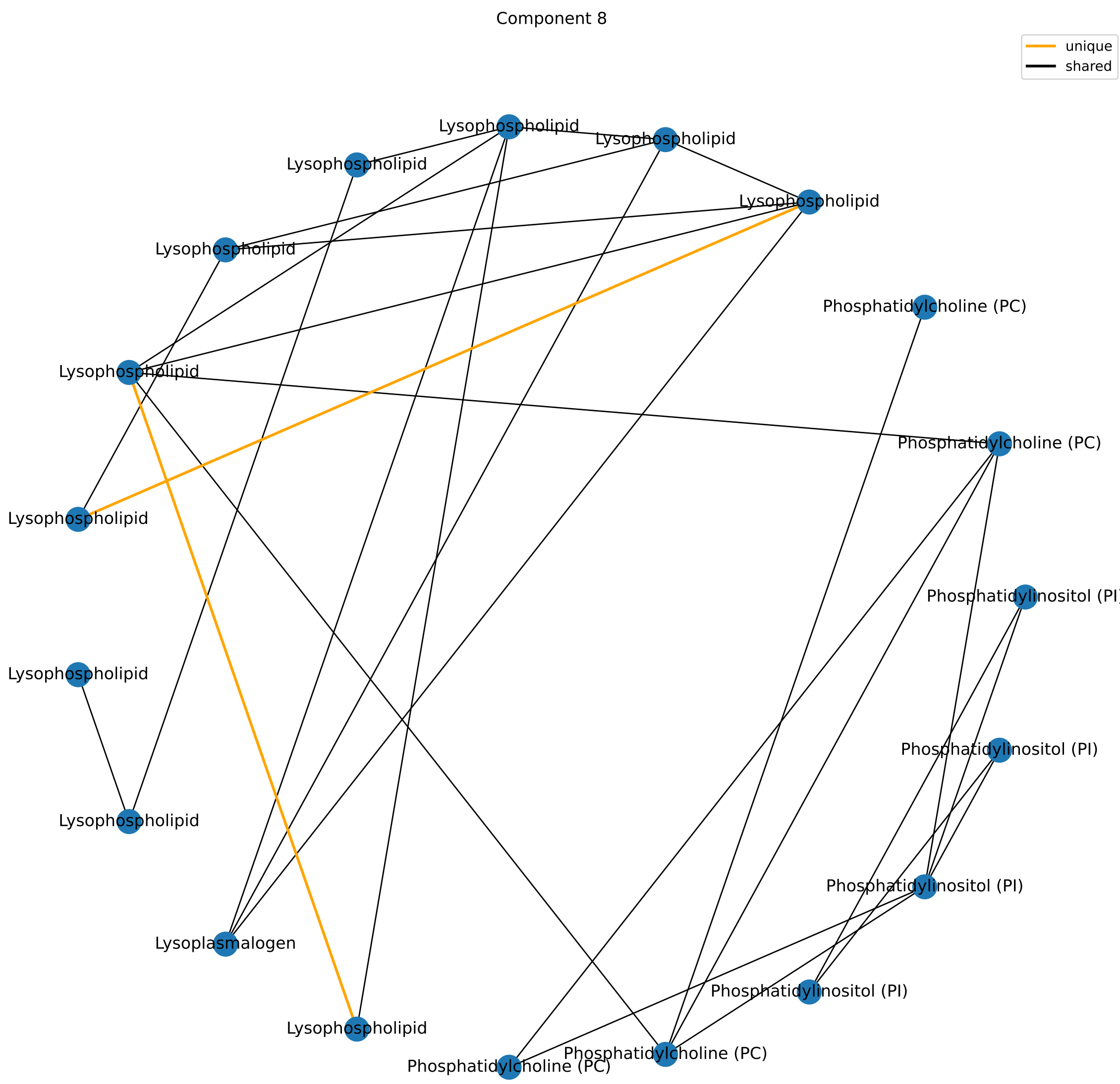
Component 8

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	20	3.04126441794368e-07	2.73713797614932e-06
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/curr-components/8-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lysophospholipid	24	10	1.87584221908952e-11	1.9508759078531e-09
Phosphatidylinositol (PI)	6	4	5.1462124580108e-06	0.000267603047816562
Phosphatidylcholine (PC)	19	5	6.92333380142202e-05	0.00240008905115963
Lysoplasmalogen	3	1	0.0771847063625122	1
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1

results/006/enrichment/curr-components/8-subpathway.csv



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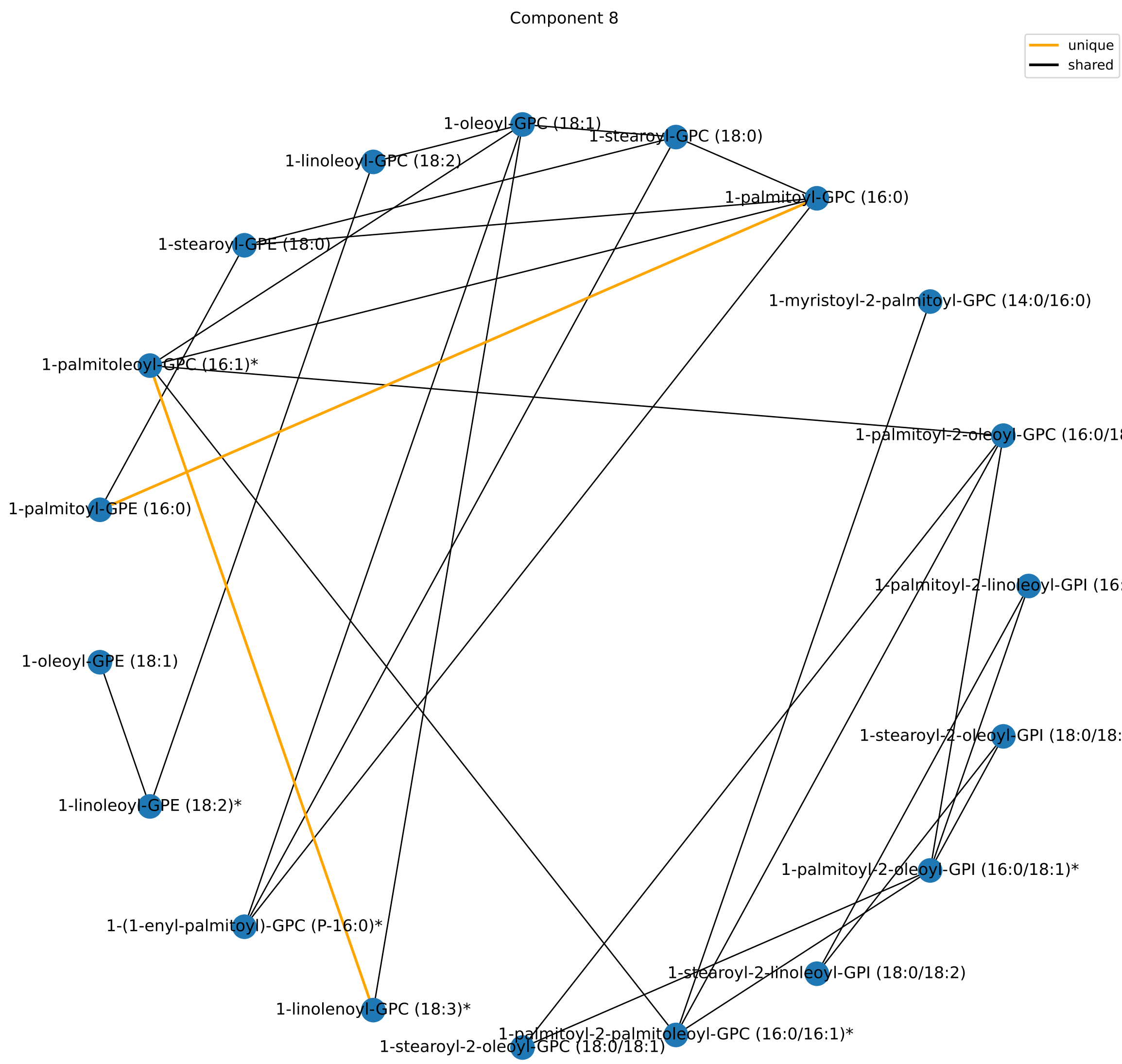
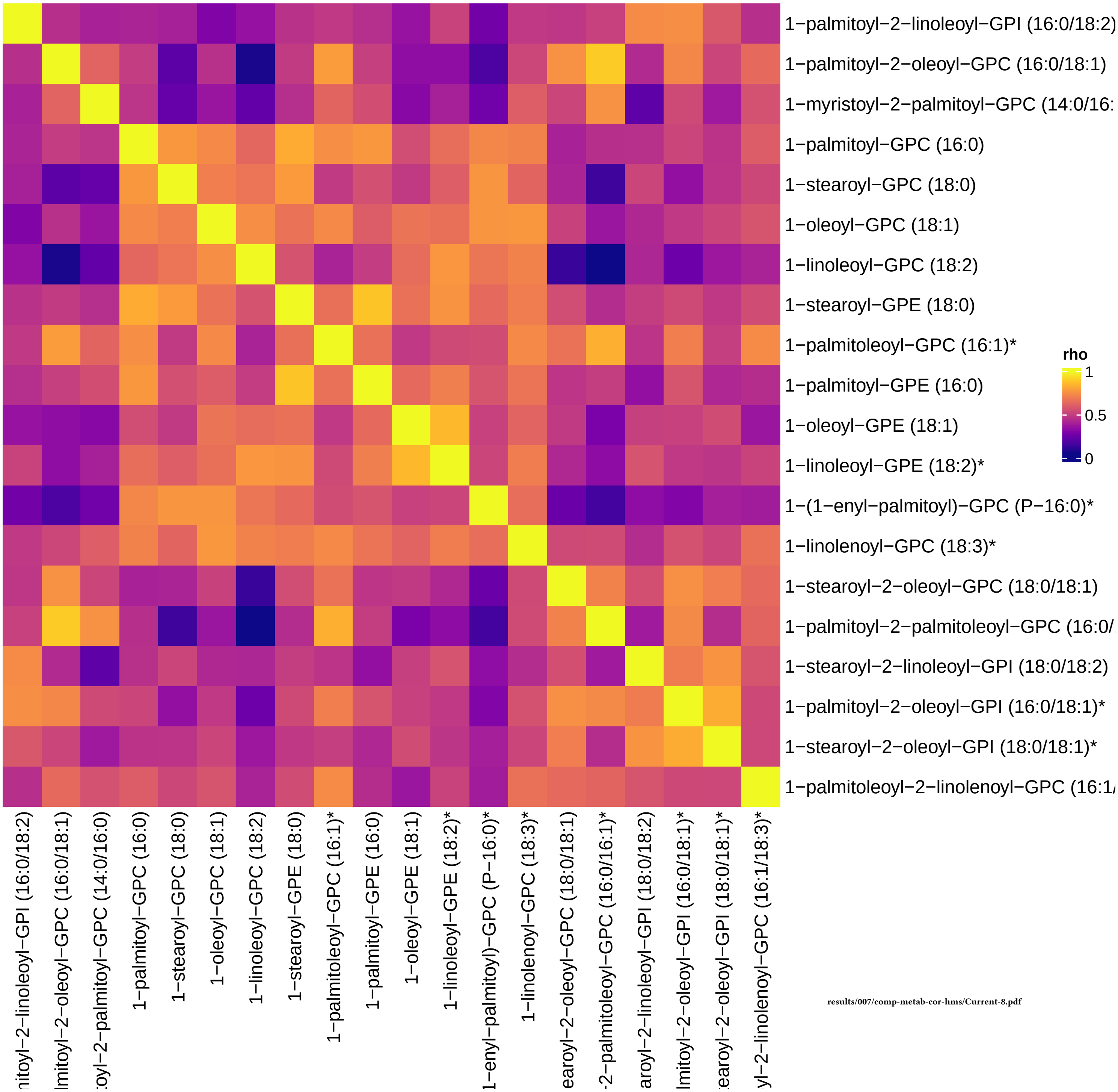


results/005_1/component-graphs/current/chemical_name-7.pdf

Current smokers, comp 8



results/007/comp-metab-cor-hms/Current-8.pdf

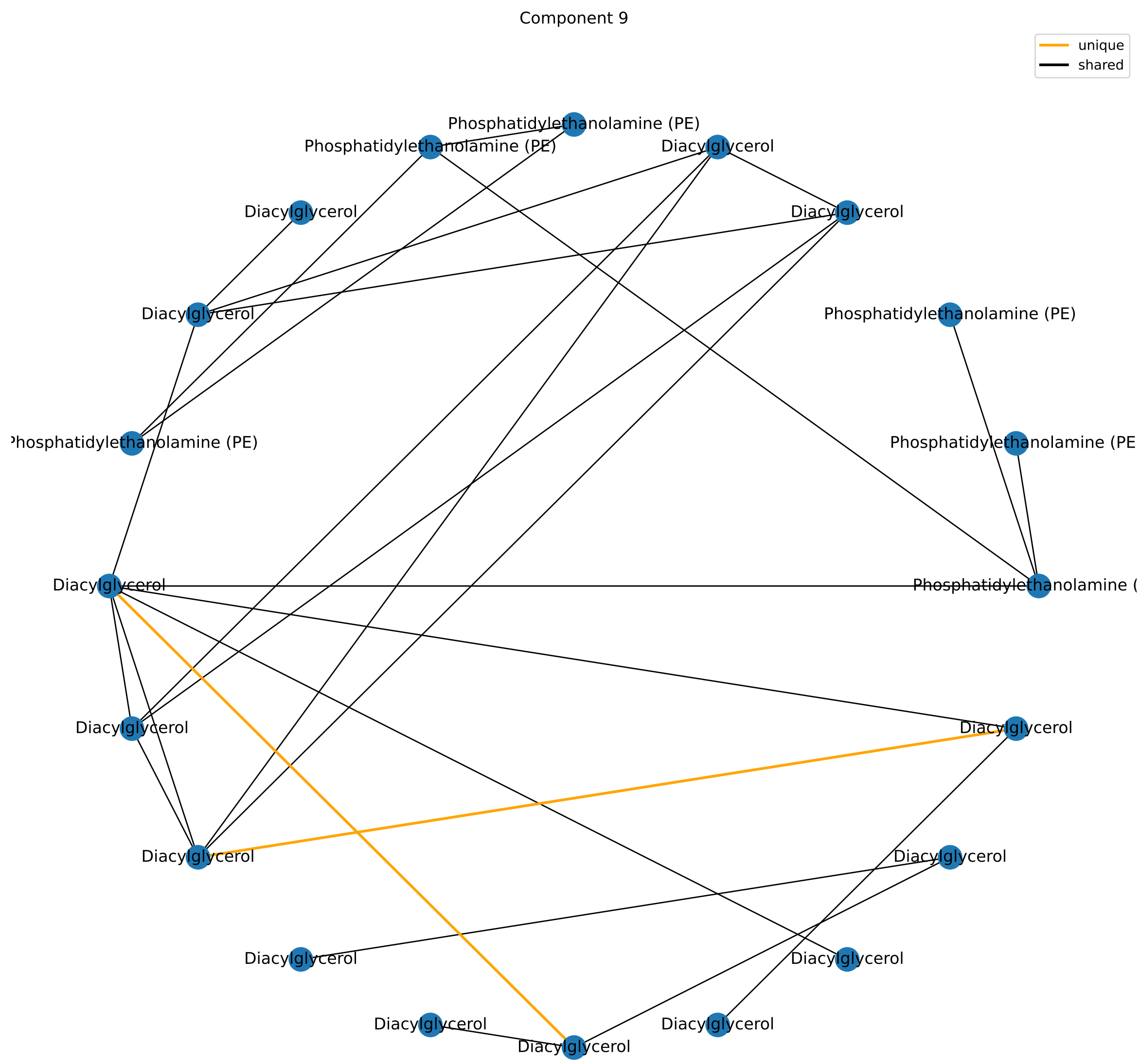
Component 9

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	21	1.41348739207951e-07	1.27213865287156e-06
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/curr-components/9-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Diacylglycerol	19	15	1.95739137628689e-23	2.03568703133837e-21
Phosphatidylethanolamine (PE)	11	6	8.90386378099915e-08	4.63000916611956e-06
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1

results/006/enrichment/curr-components/9-subpathway.csv



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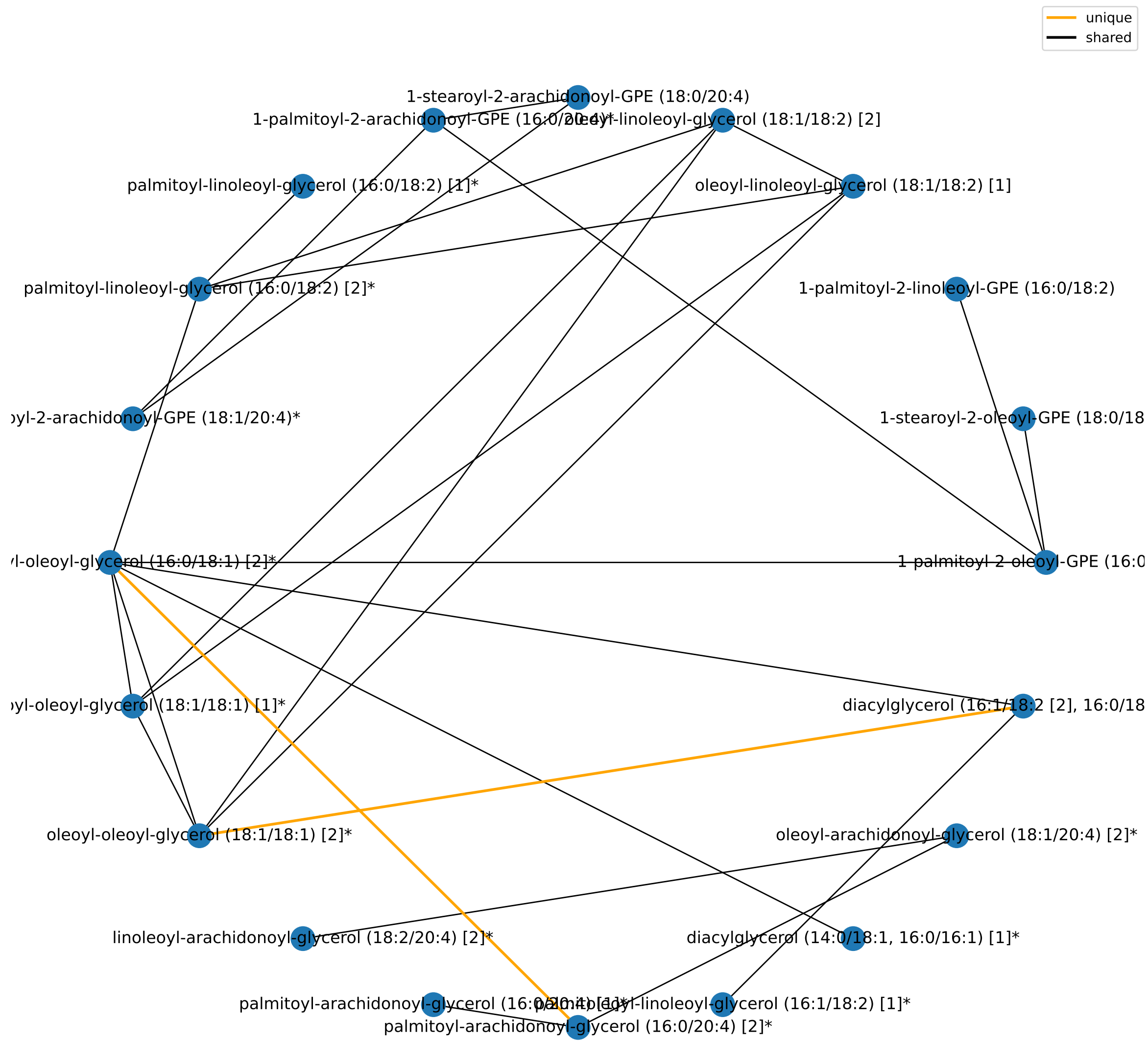
Component 9



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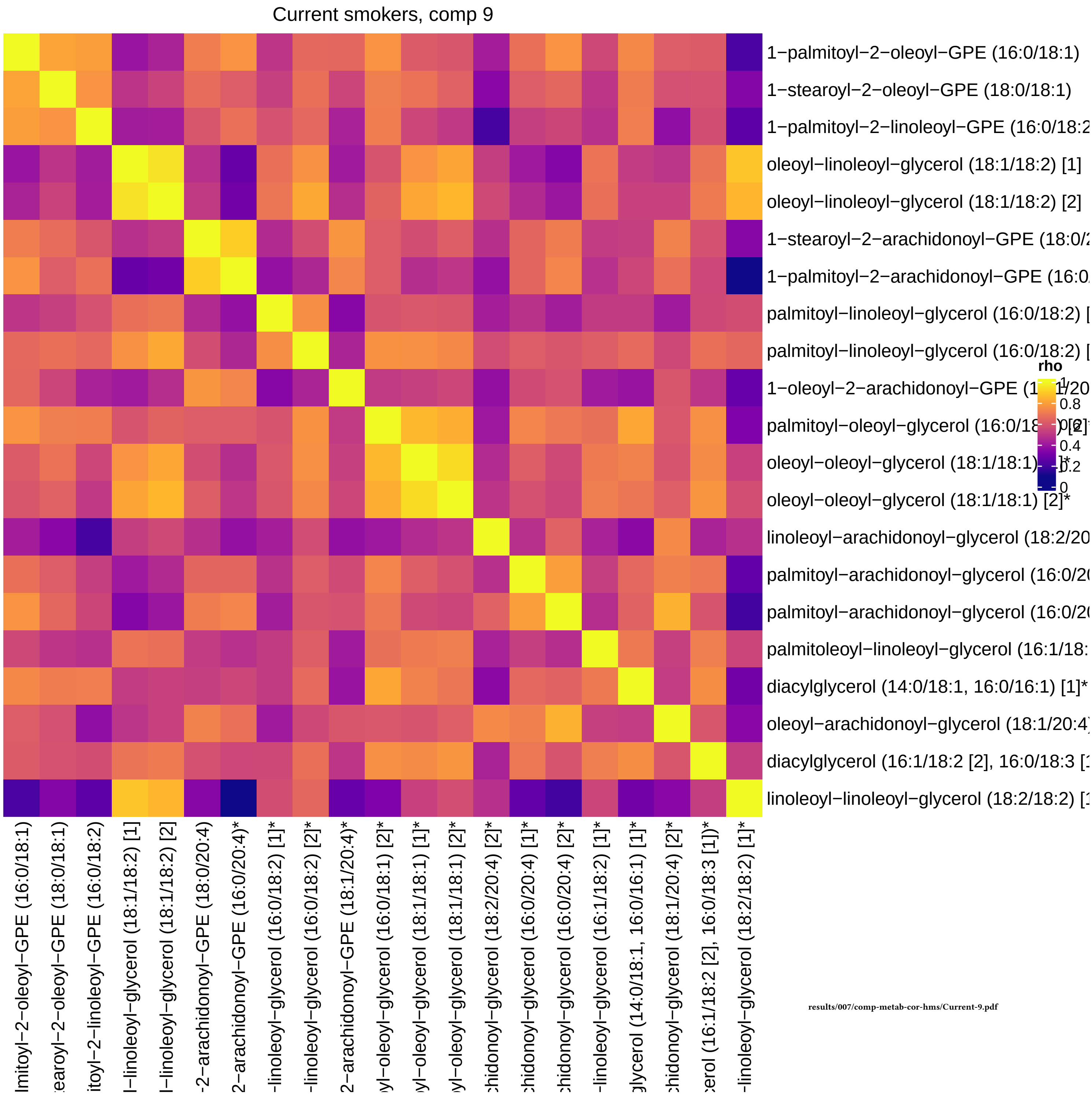


results/007/comp-metab-cor-hms/Current-9.pdf

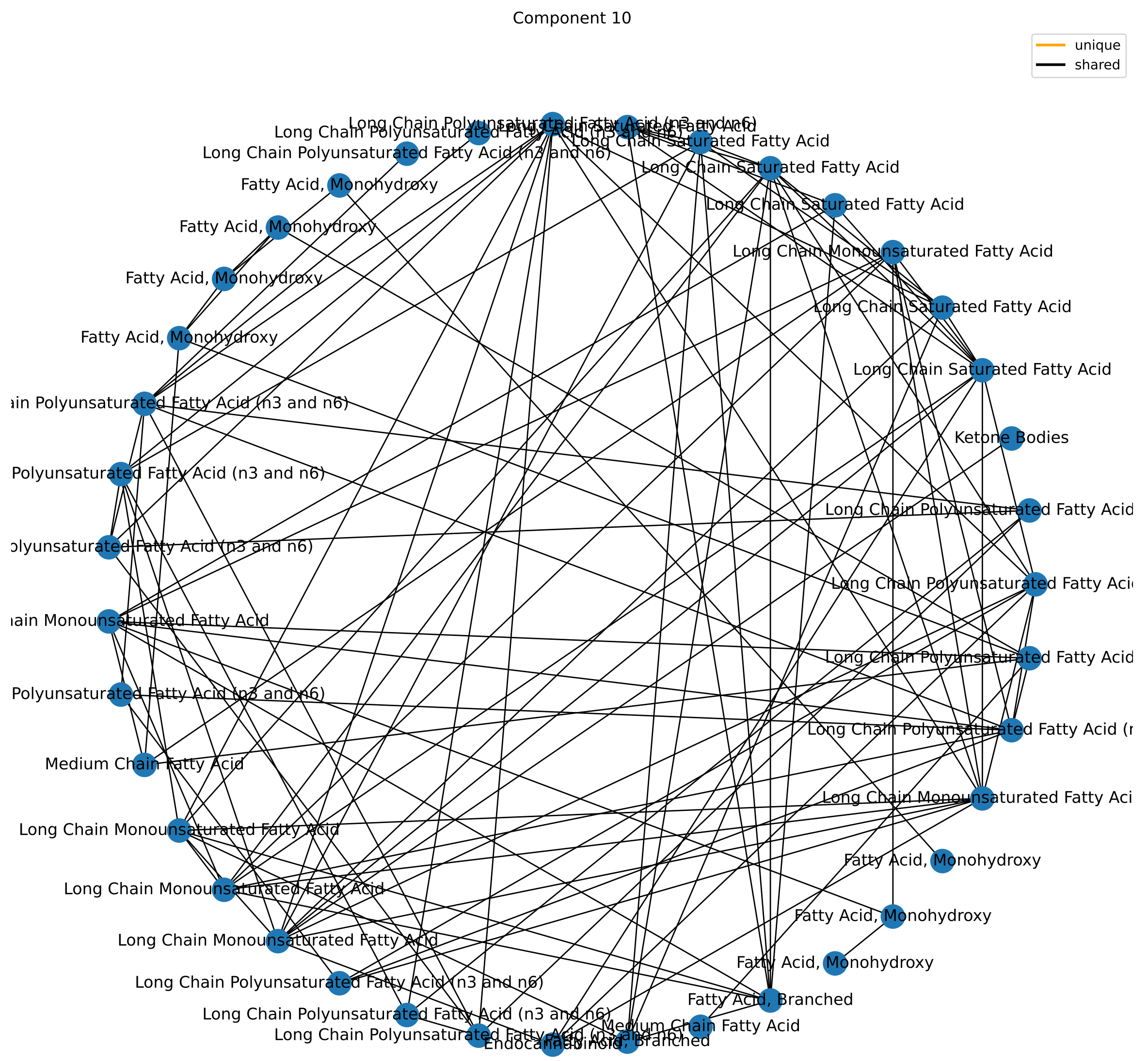
Component 10

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	40	4.95853354748542e-14	4.46268019273687e-13
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/curr-components/10-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	1.10435252485237e-19	1.14852662584646e-17
Long Chain Monounsaturated Fatty Acid	7	6	1.00002990198054e-07	3.46677032686586e-06
Long Chain Saturated Fatty Acid	7	6	1.00002990198054e-07	3.46677032686586e-06
Fatty Acid, Monohydroxy	20	7	3.12384764927581e-05	0.000812200388811711
Fatty Acid, Branched	2	2	0.00271868889485297	0.0565487290129417
Ketone Bodies	1	1	0.0527704485488129	0.914687774846089
Medium Chain Fatty Acid	8	2	0.0621548590530695	0.923443620217033
Endocannabinoid	5	1	0.237998725613951	1
Fatty Acid, Dicarboxylate	23	1	0.718014455091994	1

results/006/enrichment/curr-components/10-subpathway.csv



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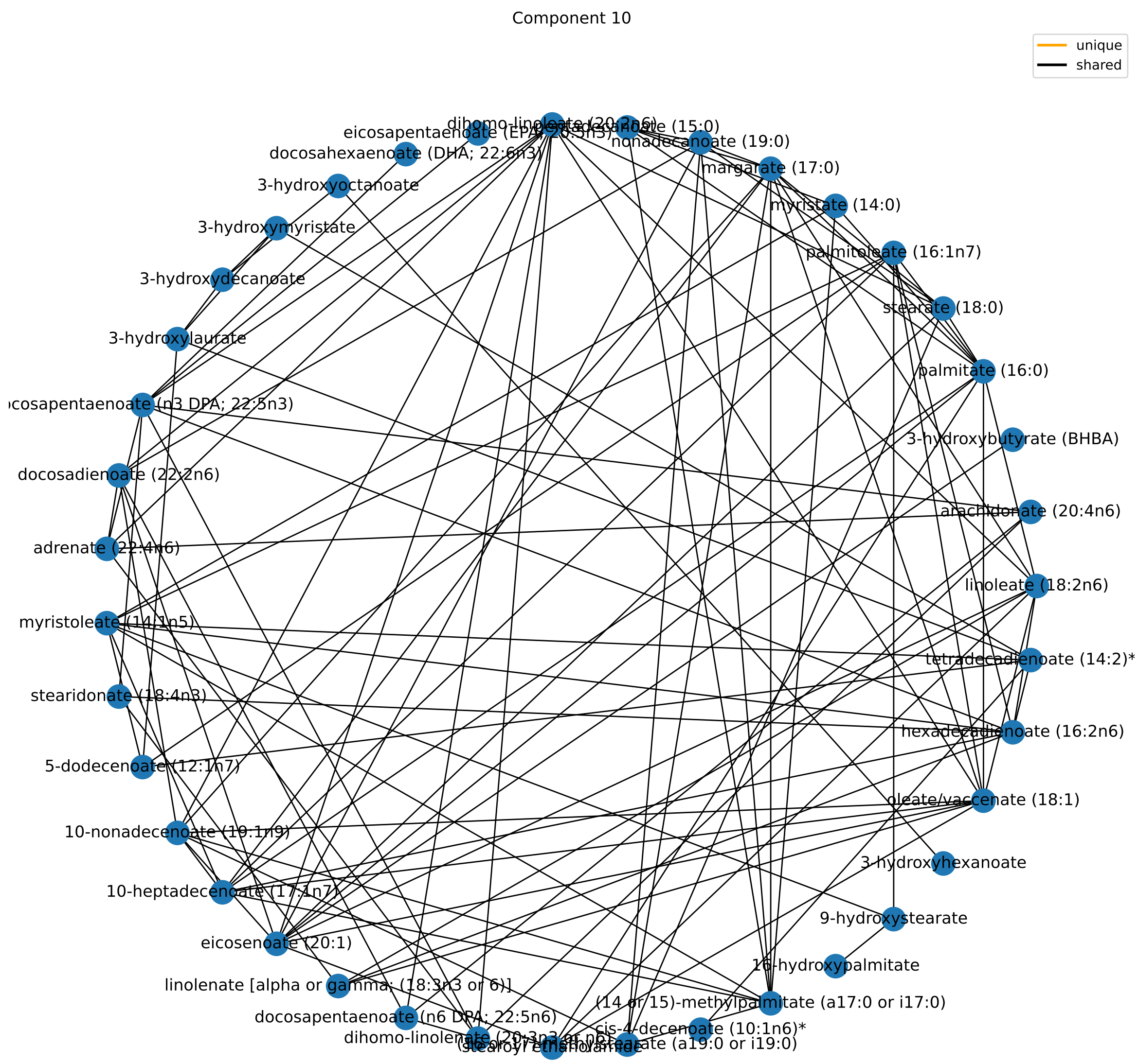

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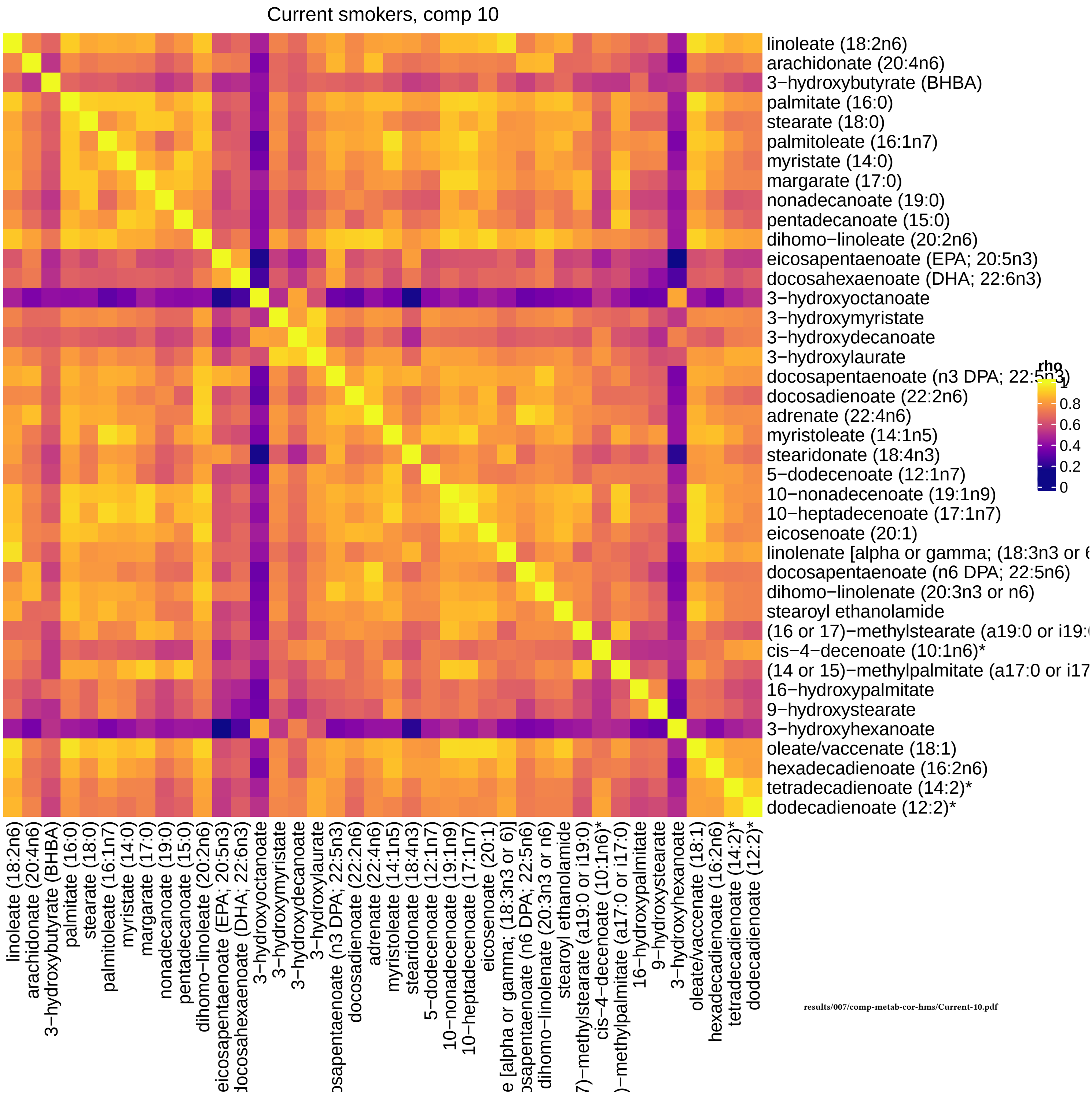


results/007/comp-metab-cor-hms/Current-10.pdf

Unique To Current Smokers

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	2	0.229007713408365	1
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

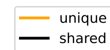
results/006/enrichment/curr-unique-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Ketone Bodies	1	1	0.00263852242744056	0.274406332453818
Sphingomyelins	28	1	0.0725611095039085	1
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1

results/006/enrichment/curr-unique-subpathway.csv

results/005_1/sub_pathway-node-only-curr-graph.pdf

Nodes Only in Current Smokers



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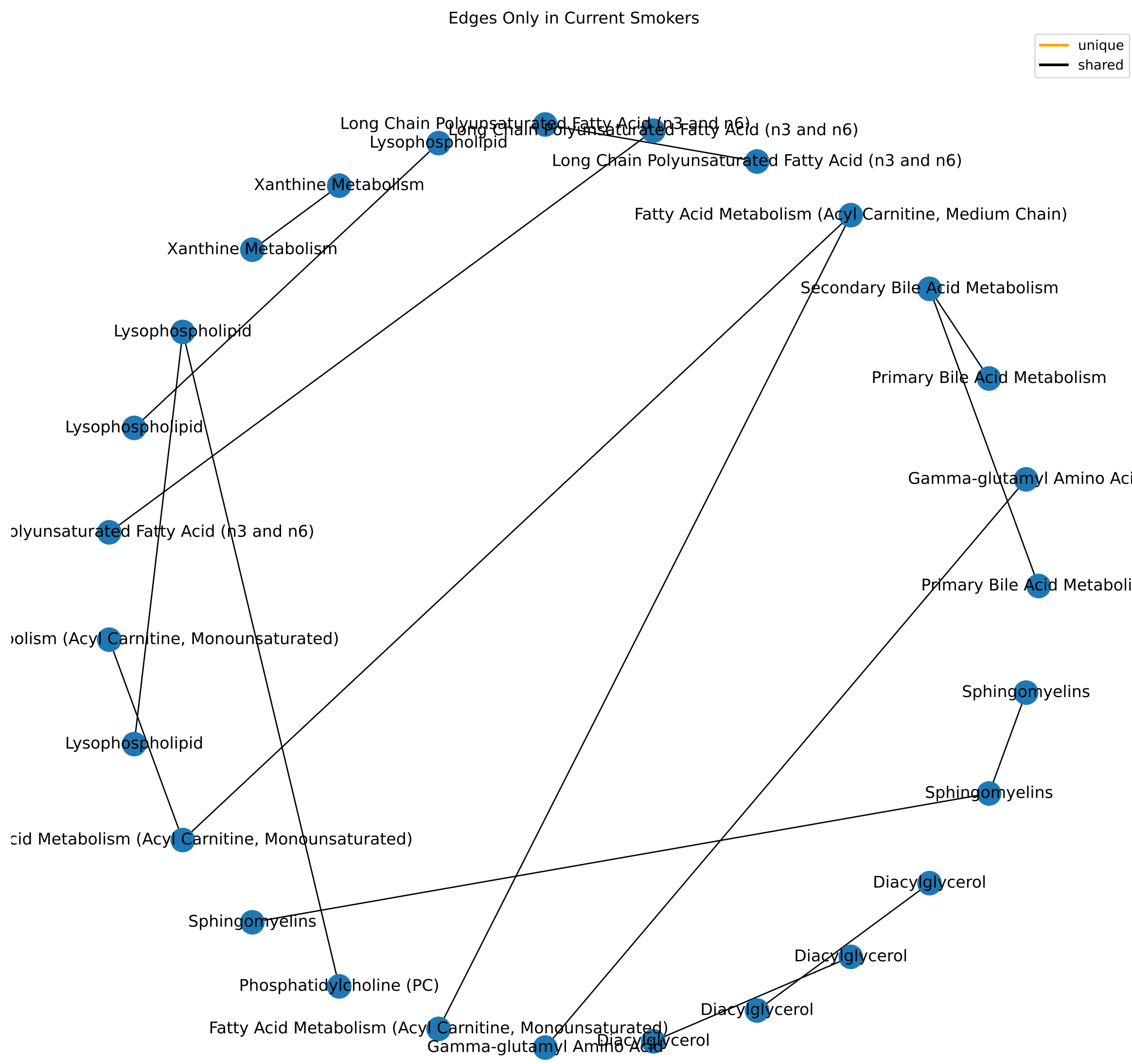
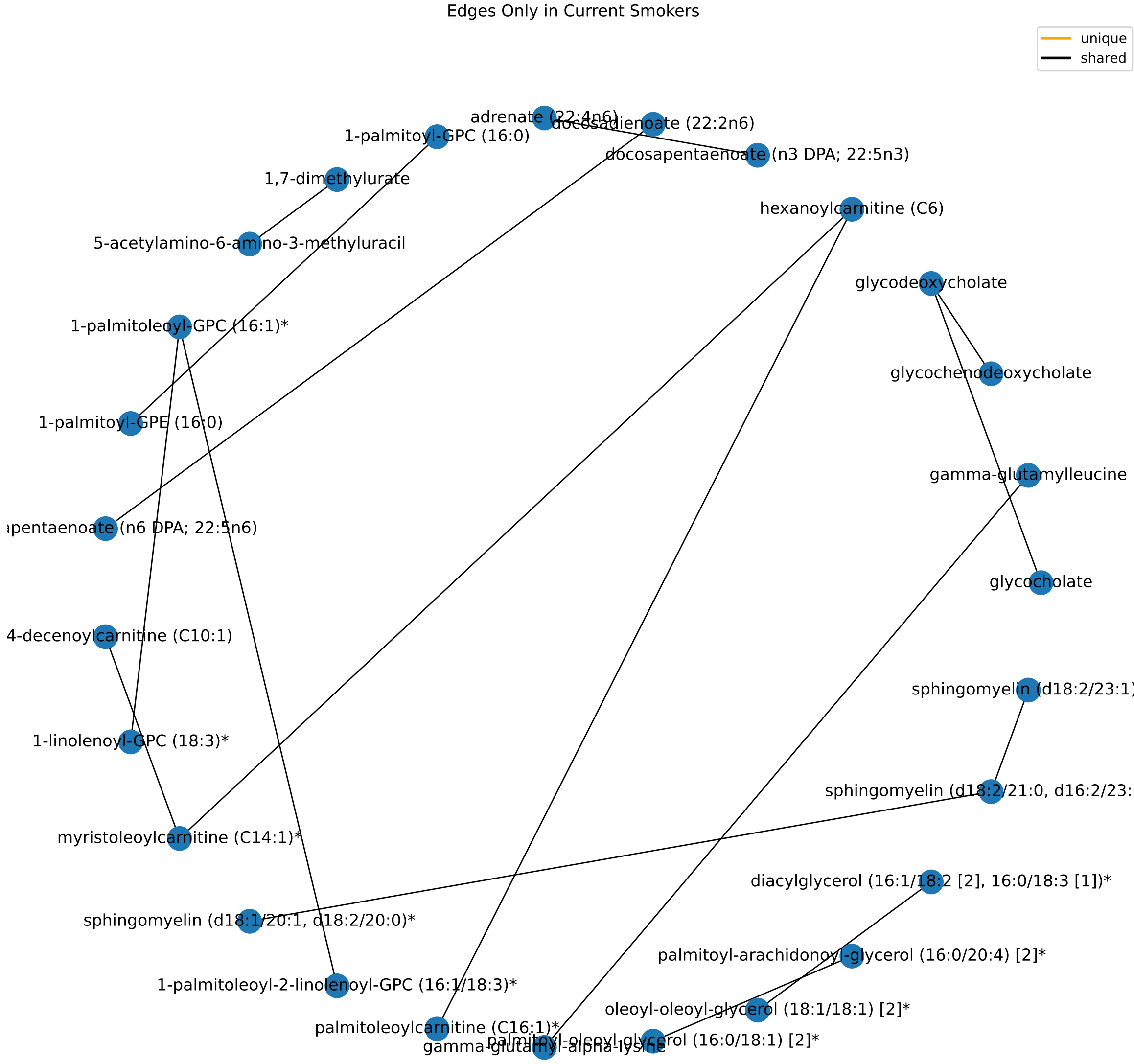


results/005_1/sub_pathway-edge-only-curr-graph.pdf

Edges Only in Current Smokers



results/005_1/chemical_name-edge-only-curr-graph.pdf

Former Smokers

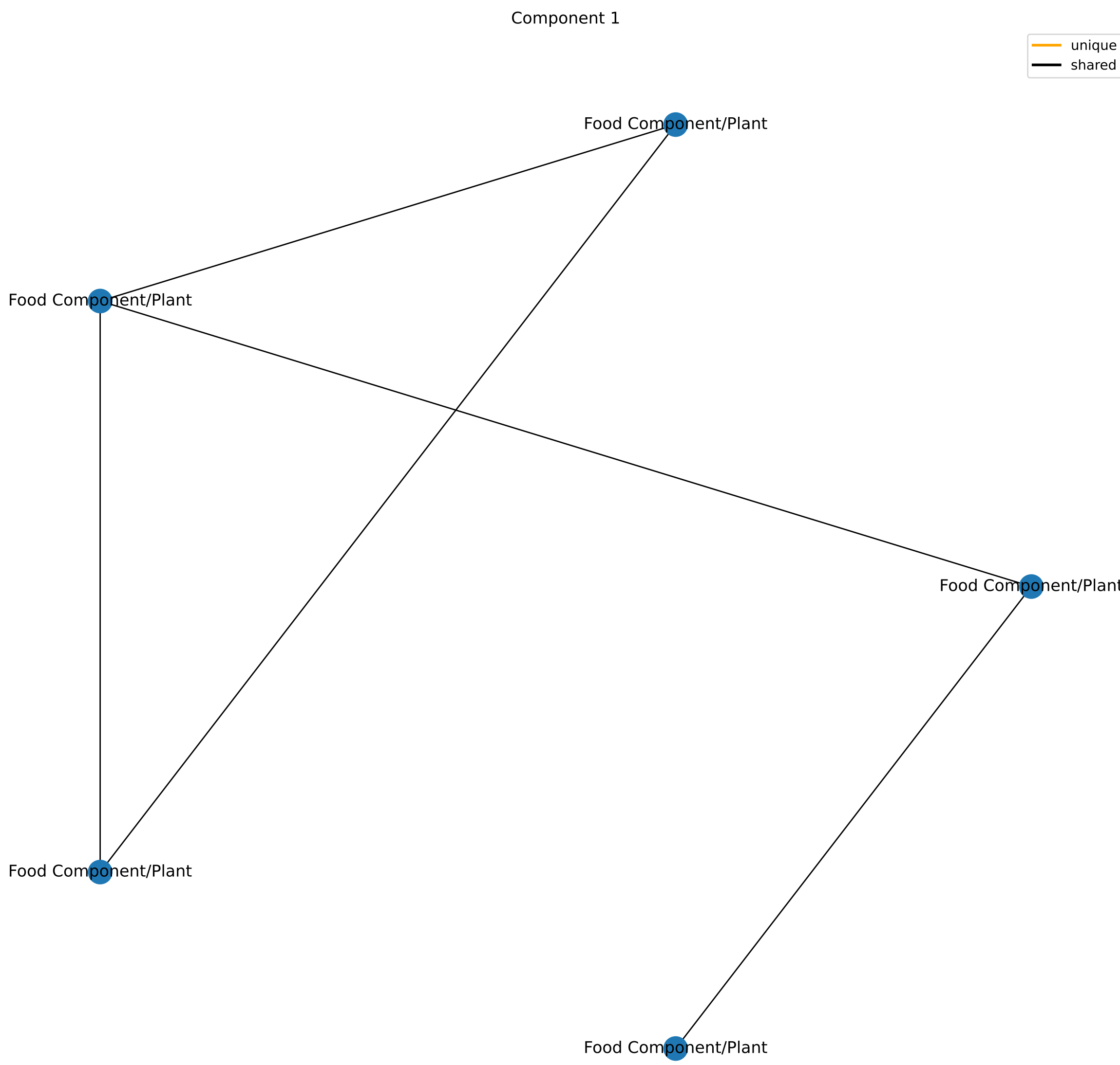
Component 1

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Xenobiotics	96	6	3.5895061835335e-06	3.23055556518015e-05
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Lipid	363	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1

results/006/enrichment/form-components/1-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Food Component/Plant	38	6	1.06892828318722e-08	1.11168541451471e-06
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1
Benzoate Metabolism	21	0	1	1

results/006/enrichment/form-components/1-subpathway.csv



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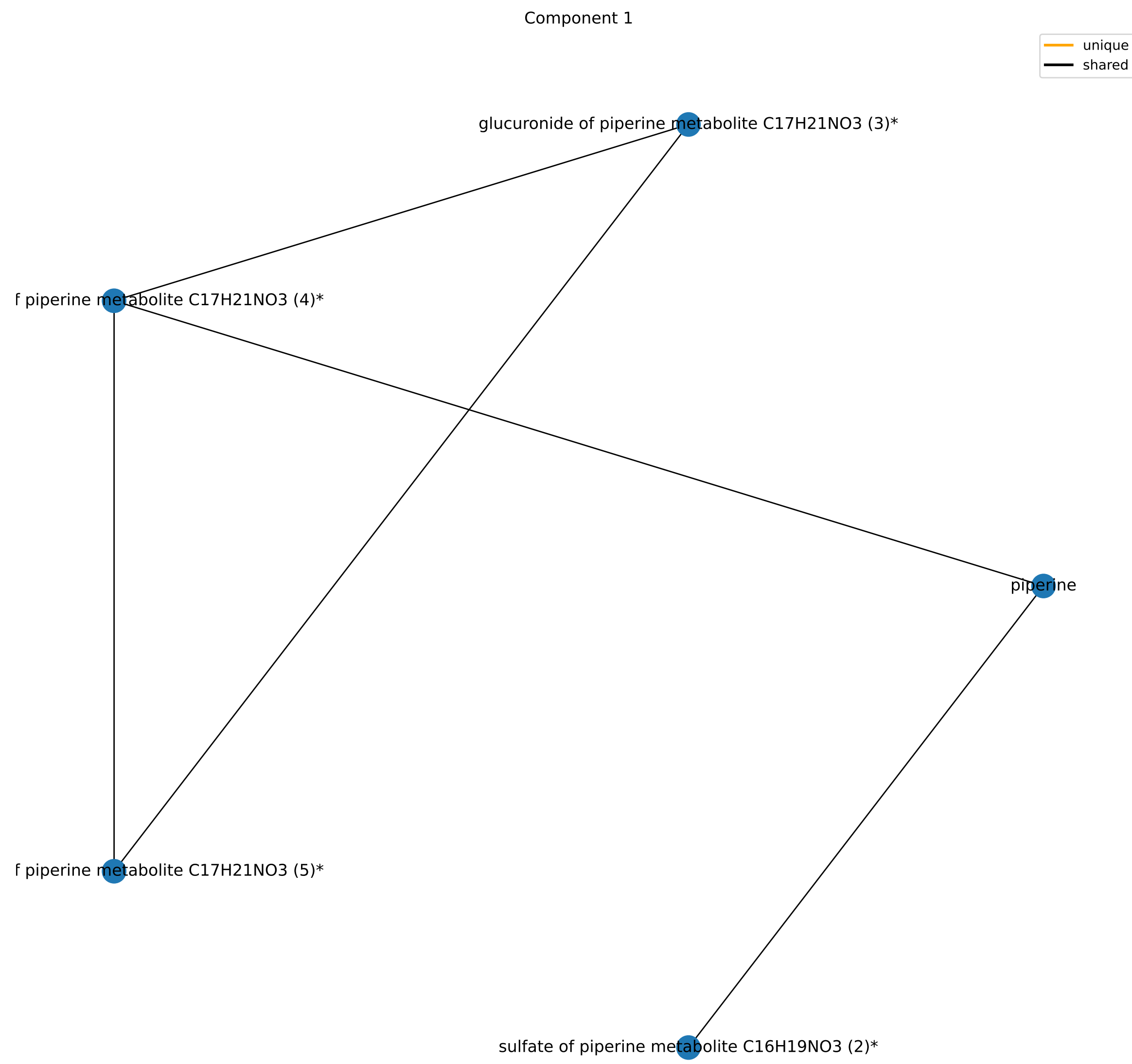

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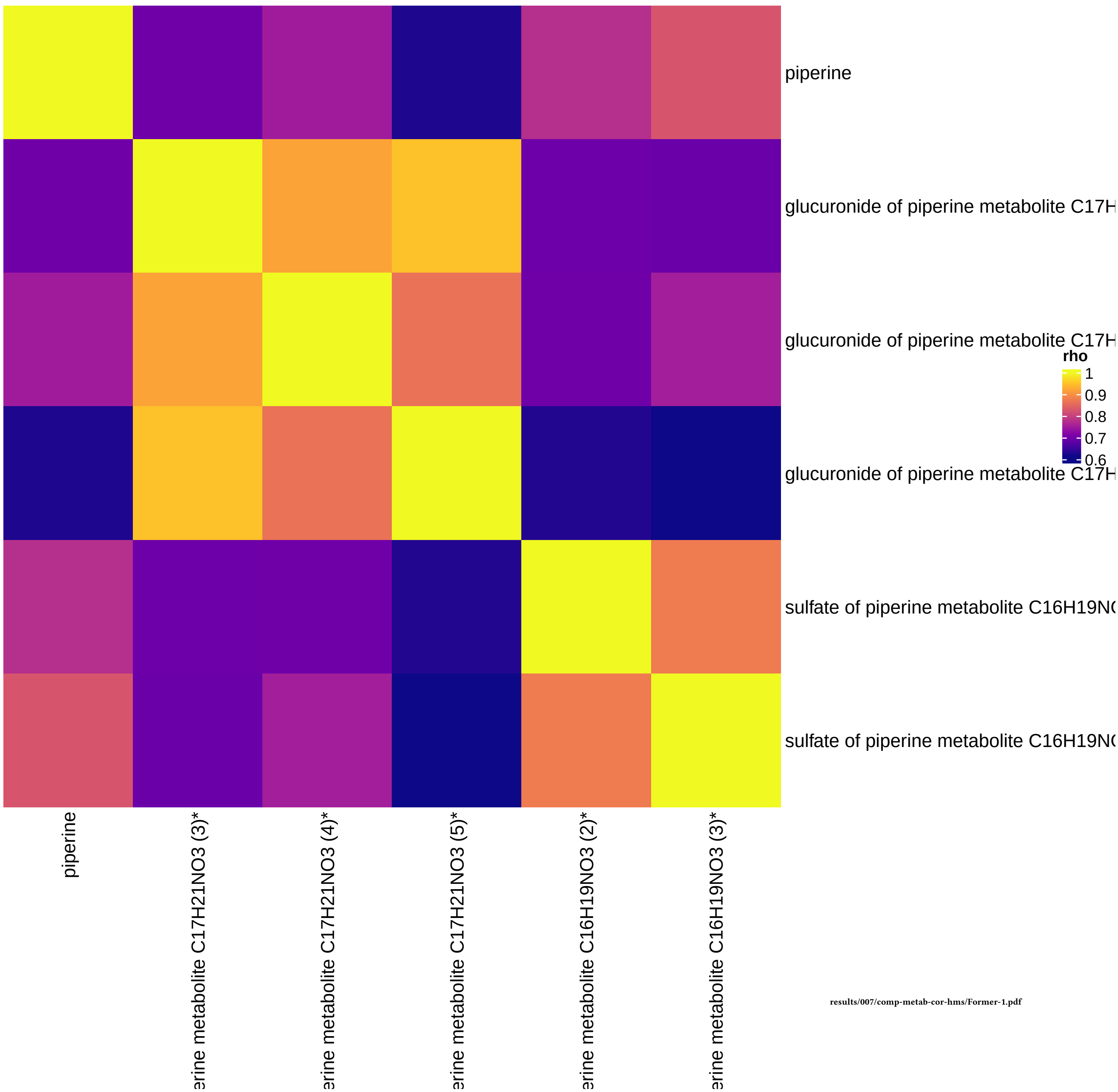
Former smokers, comp 1



results/007/comp-metab-cor-hms/Former-1.pdf

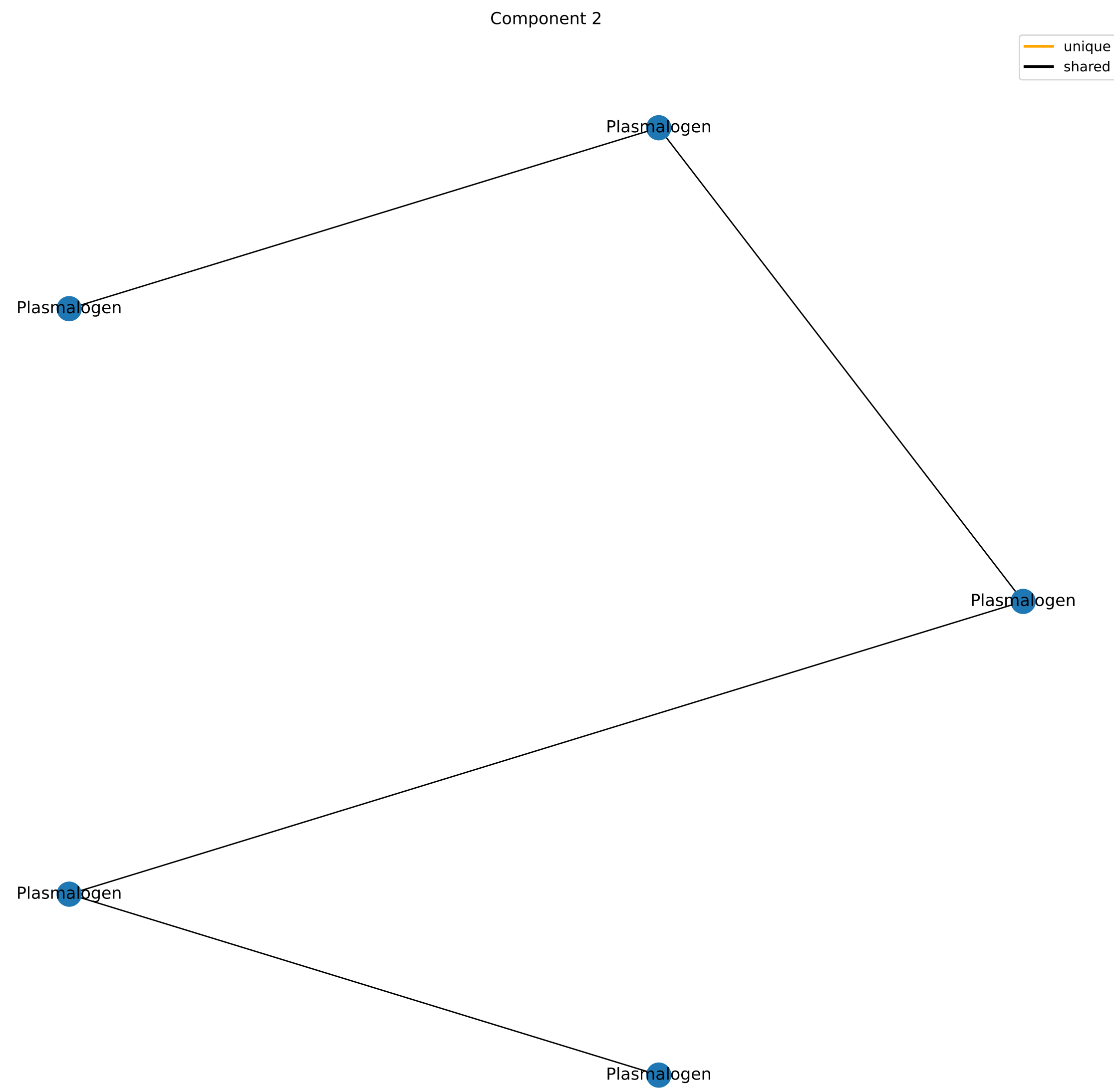
Component 2

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	6	0.0118032666679744	0.10622940001177
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/form-components/2-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Plasmalogen	11	6	1.78885161607769e-12	1.8604056807208e-10
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1
Benzoate Metabolism	21	0	1	1

results/006/enrichment/form-components/2-subpathway.csv



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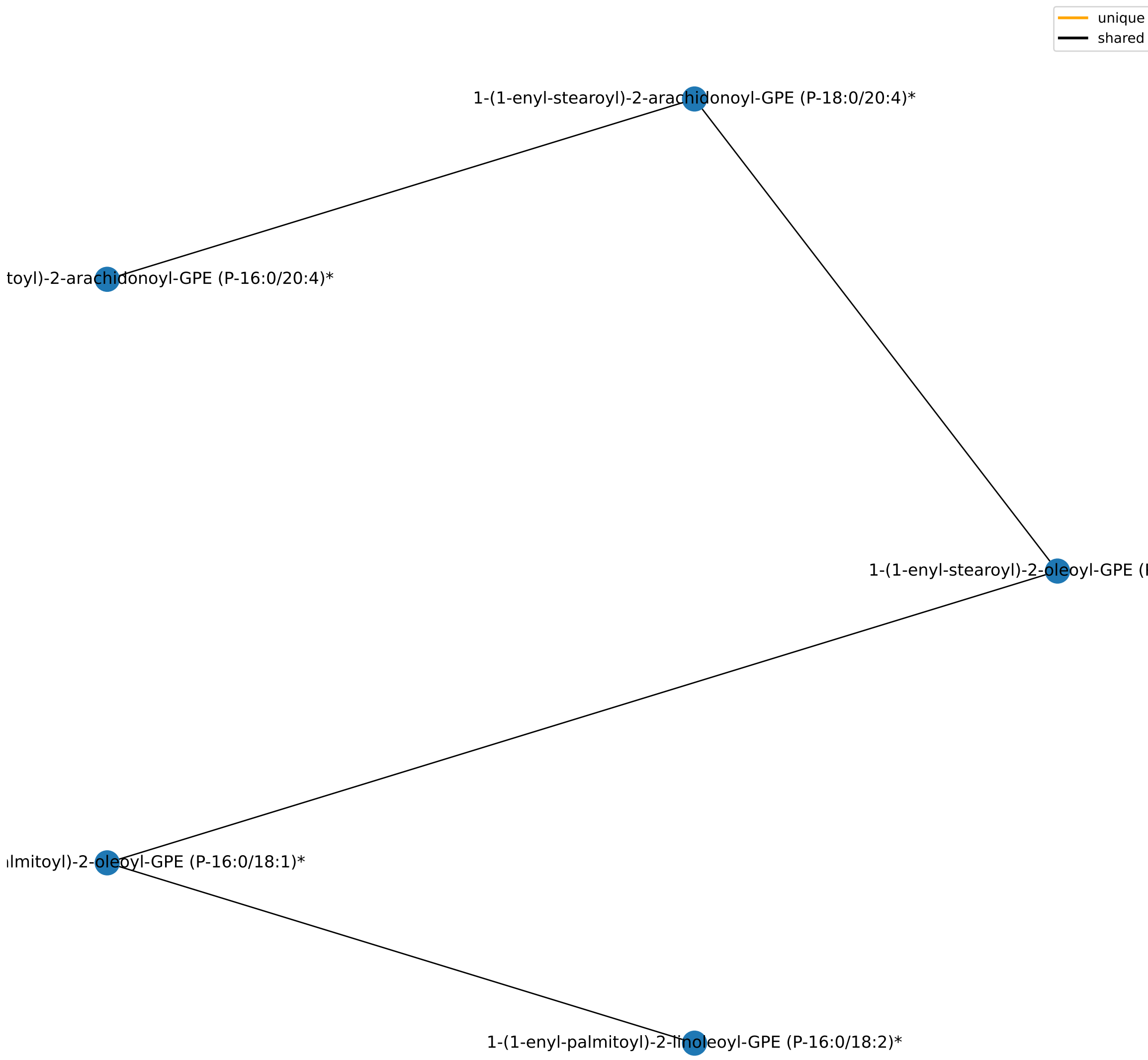
Component 2



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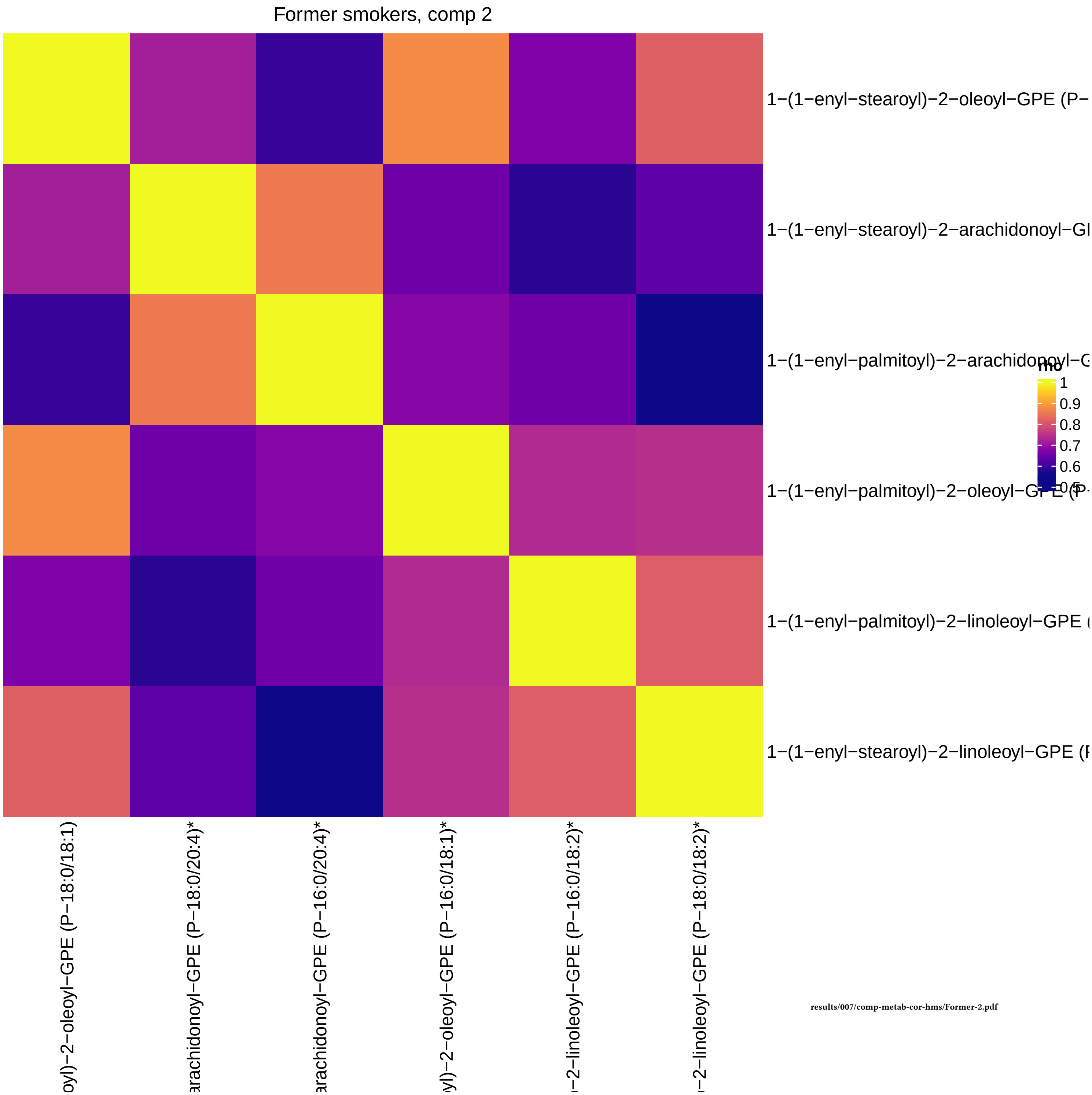


results/007/comp-metab-cor-hms/Former-2.pdf

Component 3

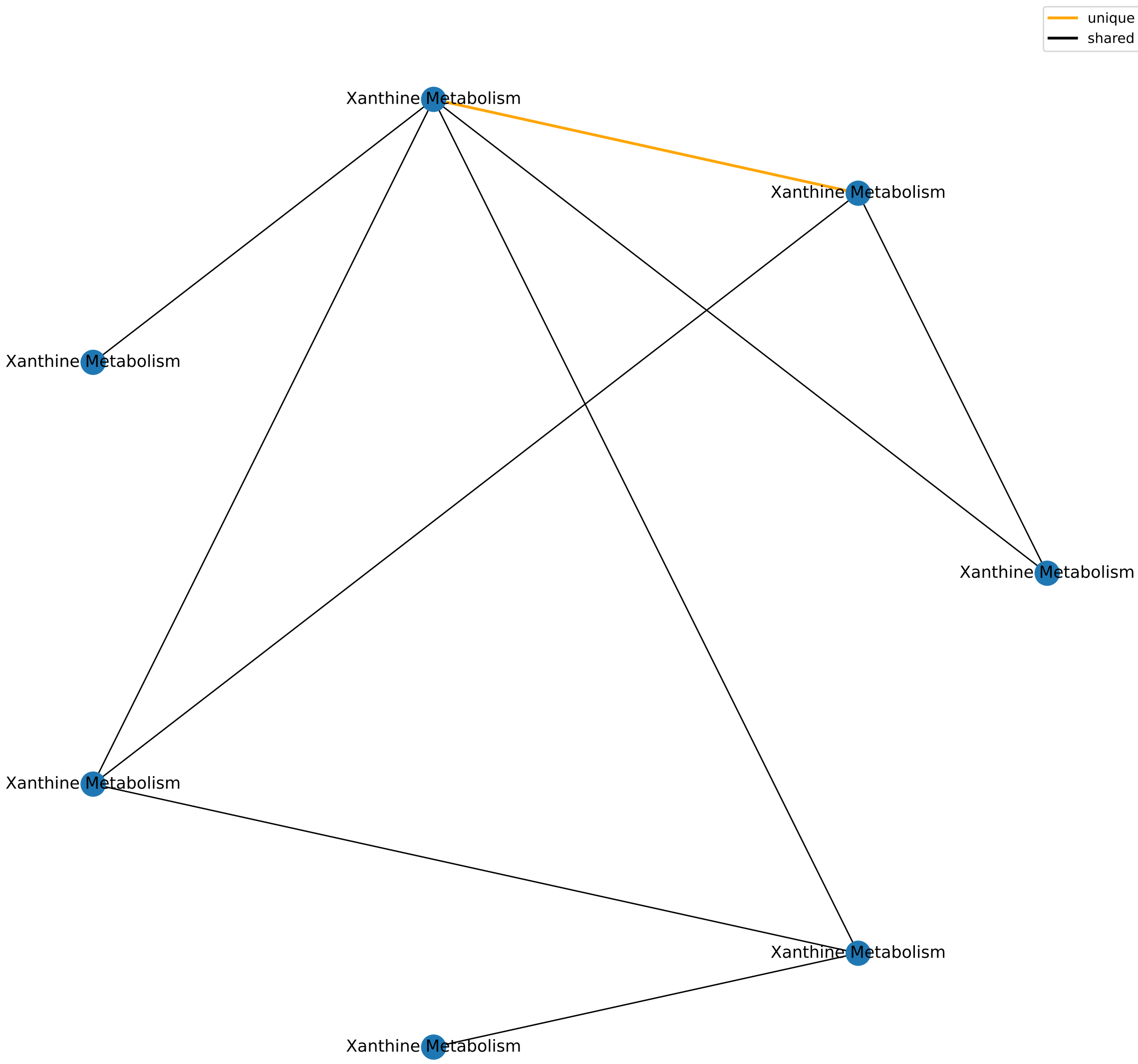
Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Xenobiotics	96	8	5.09107440612931e-08	4.58196696551638e-07
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Lipid	363	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1

results/006/enrichment/form-components/3-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Xanthine Metabolism	14	8	1.15296977833151e-15	1.19908856946477e-13
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1
Benzoate Metabolism	21	0	1	1

results/006/enrichment/form-components/3-subpathway.csv

Component 3



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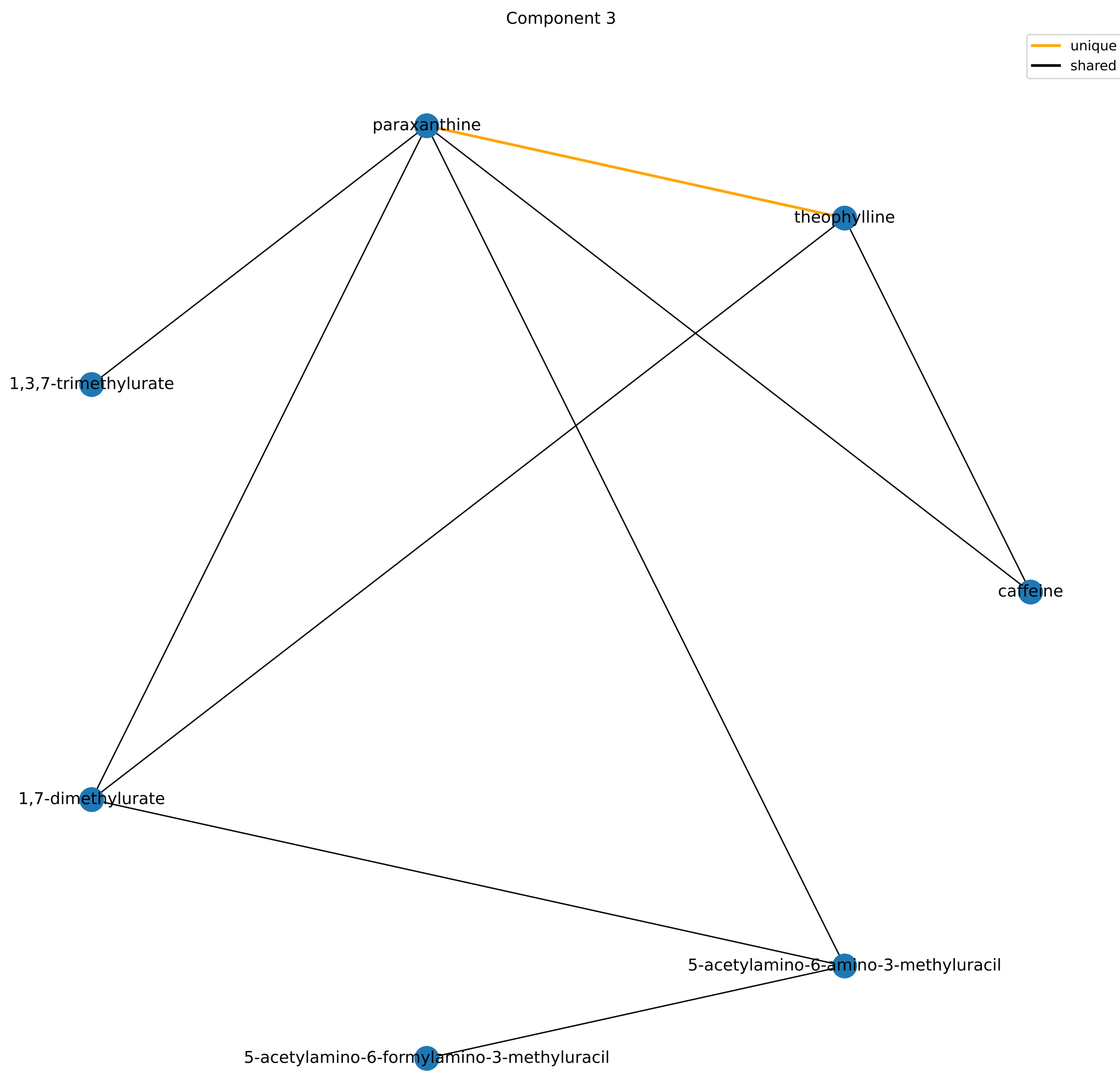

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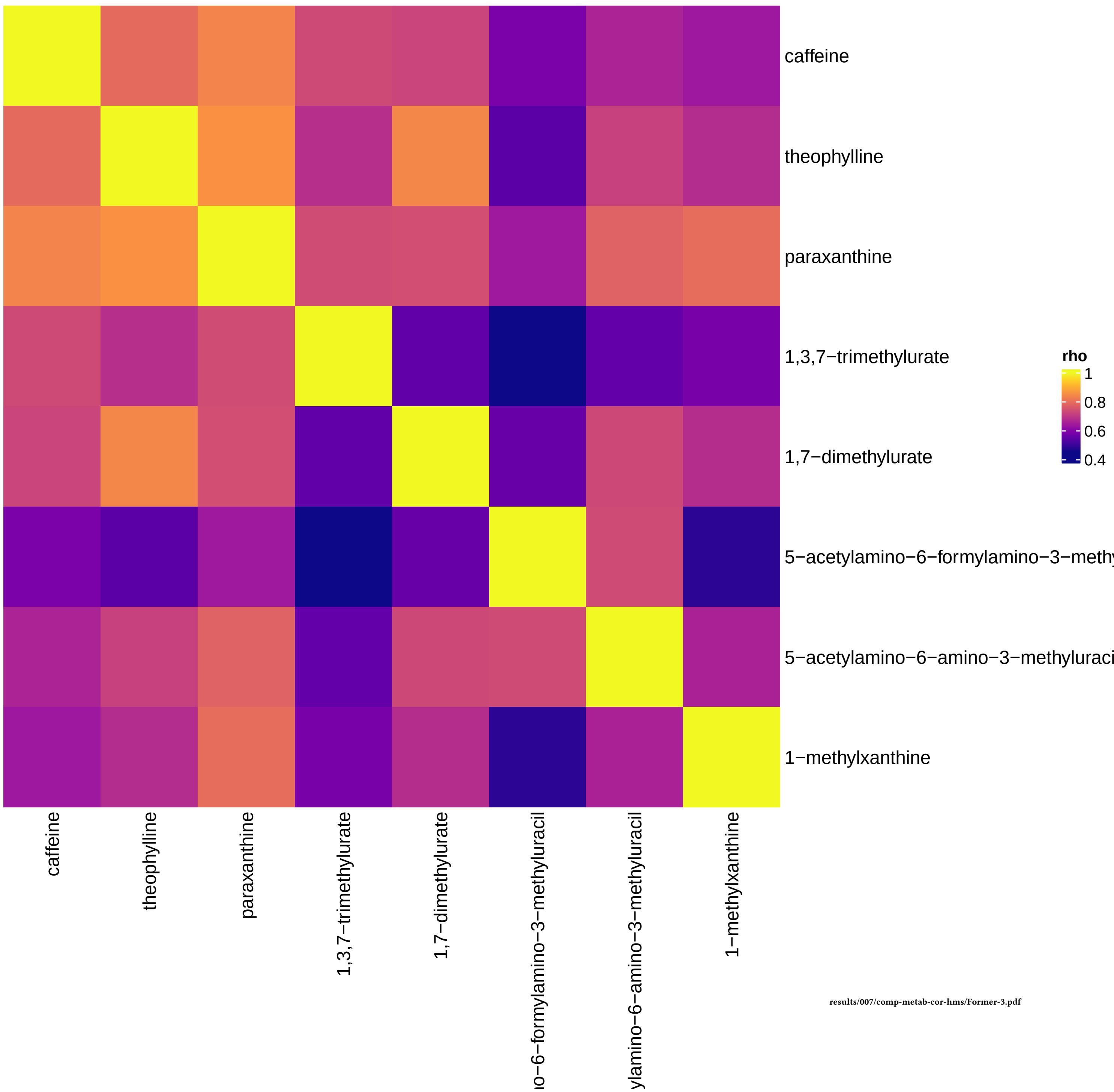
Former smokers, comp 3



results/007/comp-metab-cor-hms/Former-3.pdf

Component 4

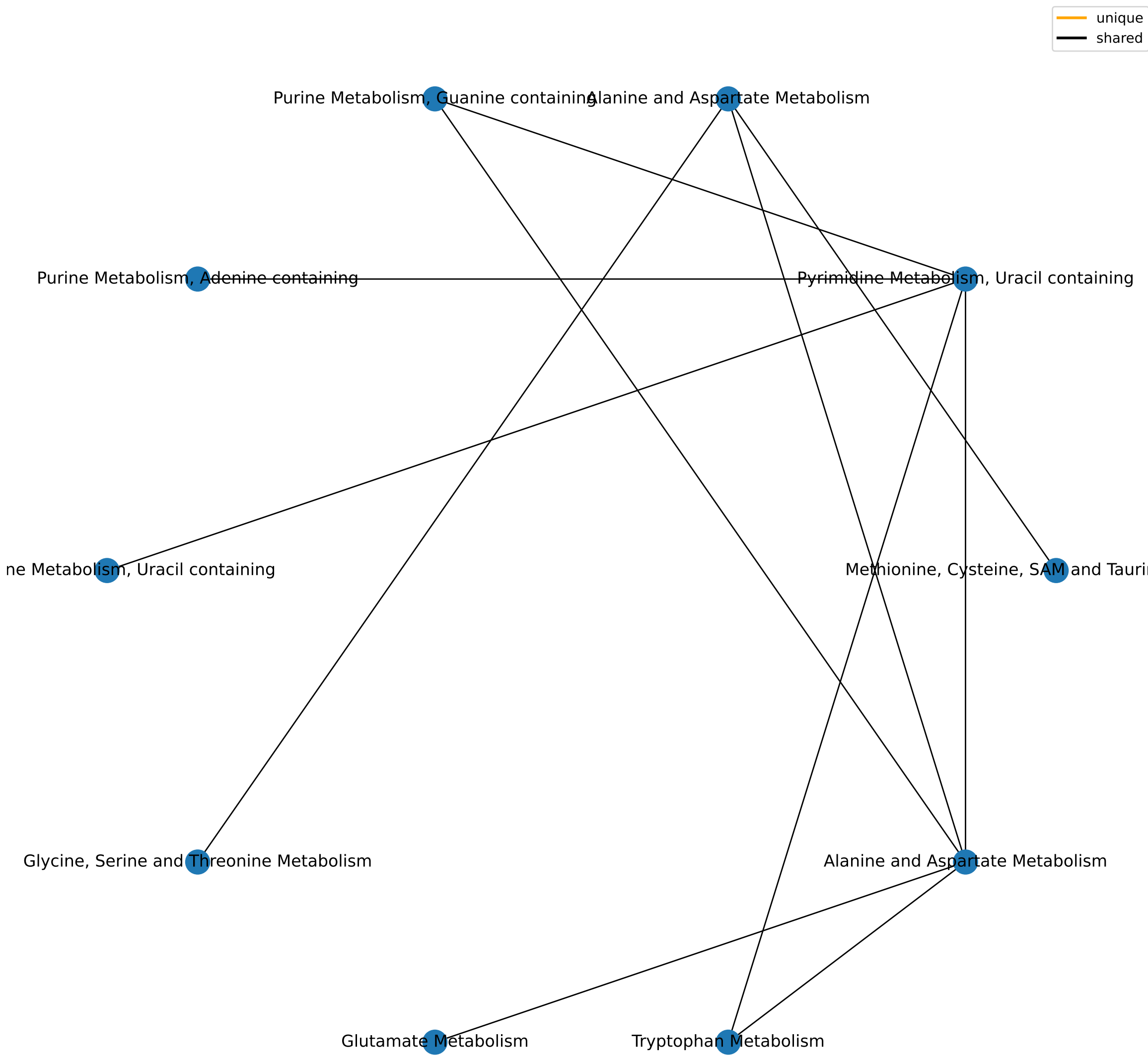
Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Nucleotide	32	4	0.000704179318330813	0.00633761386497732
Amino Acid	180	7	0.00524133302800007	0.0235859986260003
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Lipid	363	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/form-components/4-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Alanine and Aspartate Metabolism	8	2	0.00511709276014077	0.420882695469342
Pyrimidine Metabolism, Uracil containing	10	2	0.00809389798979504	0.420882695469342
Methionine, Cysteine, SAM and Taurine Metabolism	17	2	0.0231349875080654	0.749633151274128
Purine Metabolism, Guanine containing	2	1	0.0288320442797741	0.749633151274128
Purine Metabolism, Adenine containing	4	1	0.0569063833760697	1
Glycine, Serine and Threonine Metabolism	10	1	0.136760963109043	1
Glutamate Metabolism	11	1	0.149455654828028	1
Tryptophan Metabolism	16	1	0.210420654391051	1
Acetylated Peptides	3	0	1	1

results/006/enrichment/form-components/4-subpathway.csv

Component 4



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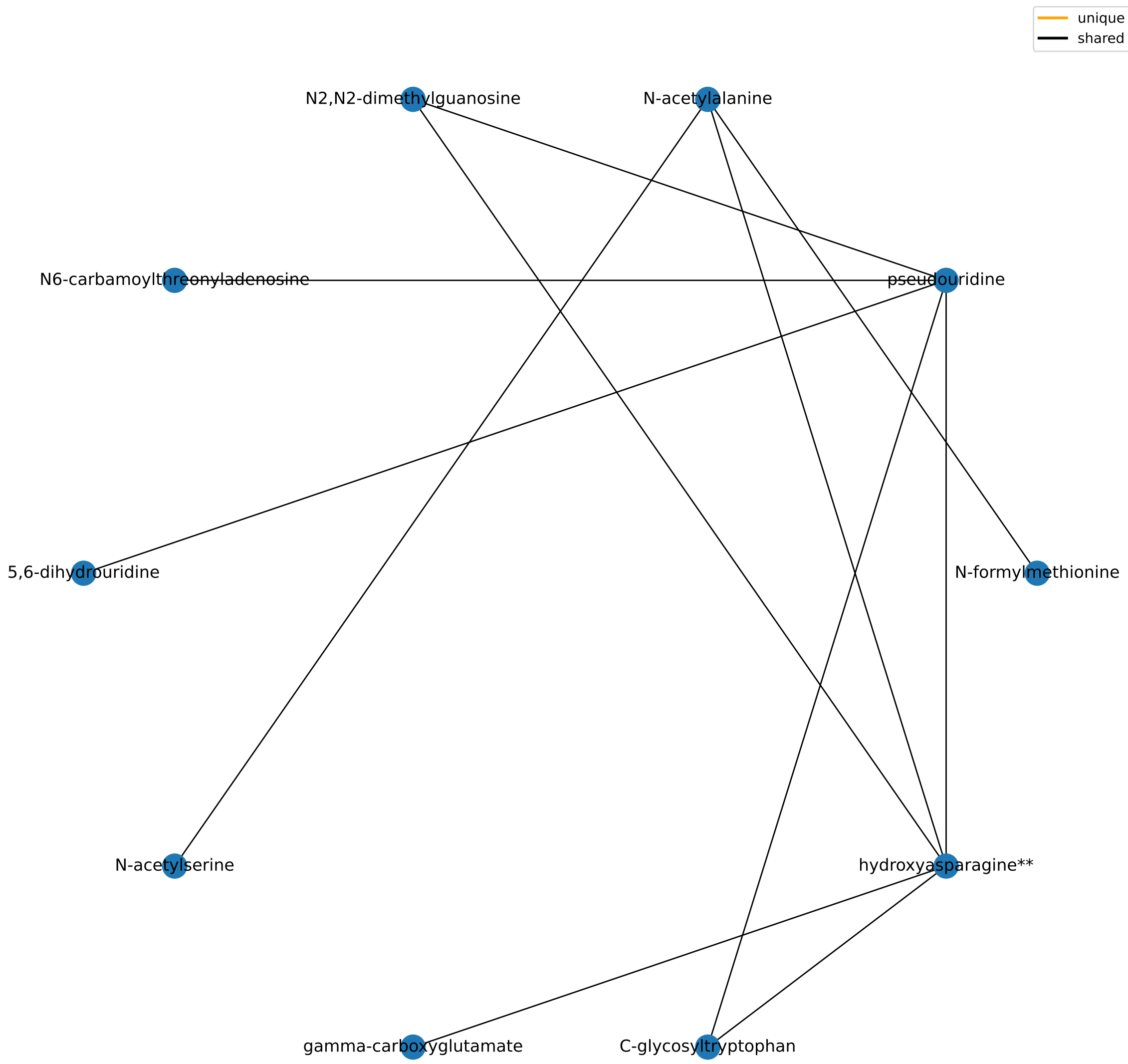
Component 4



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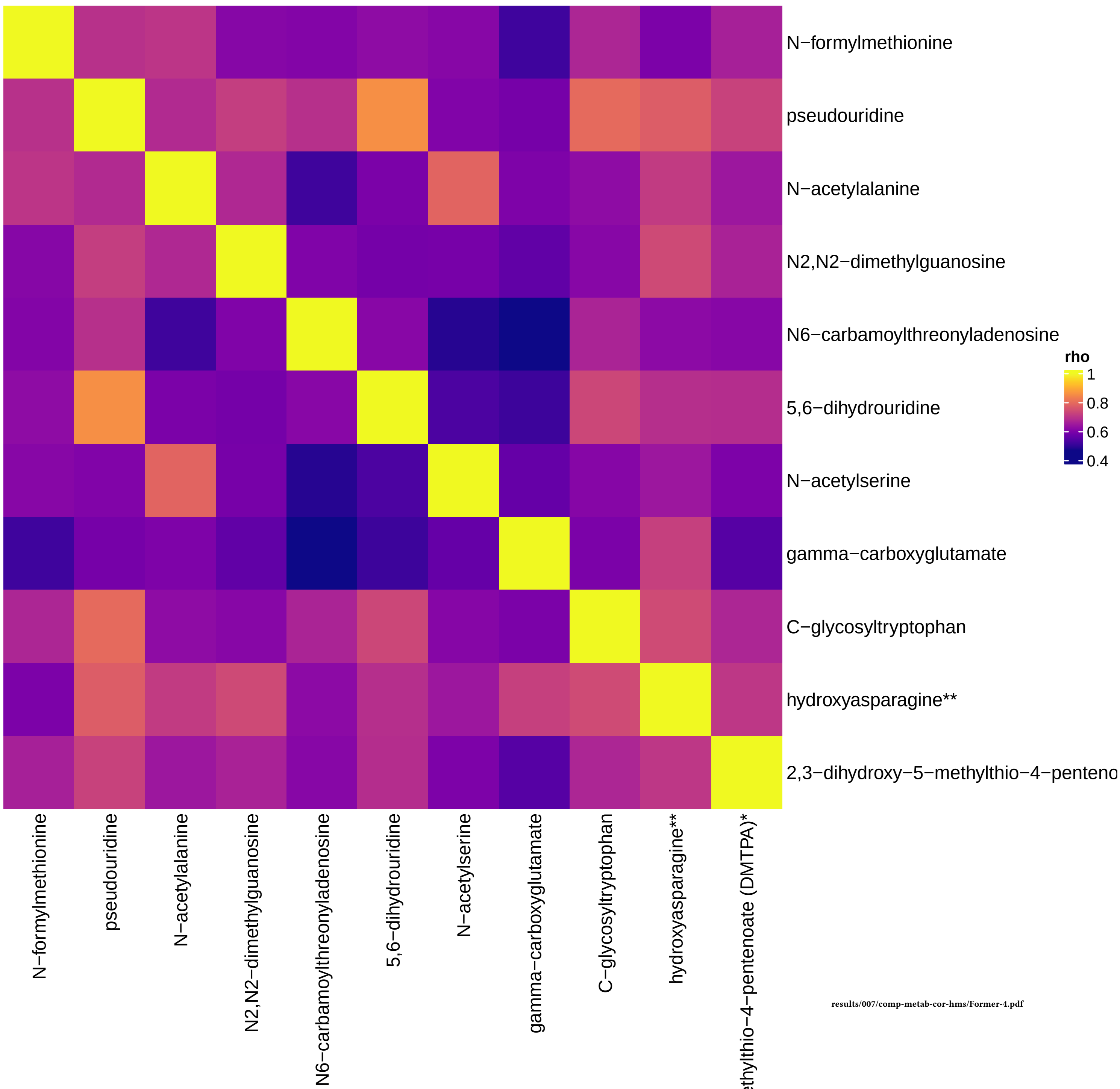
Former smokers, comp 4



results/007/comp-metab-cor-hms/Former-4.pdf

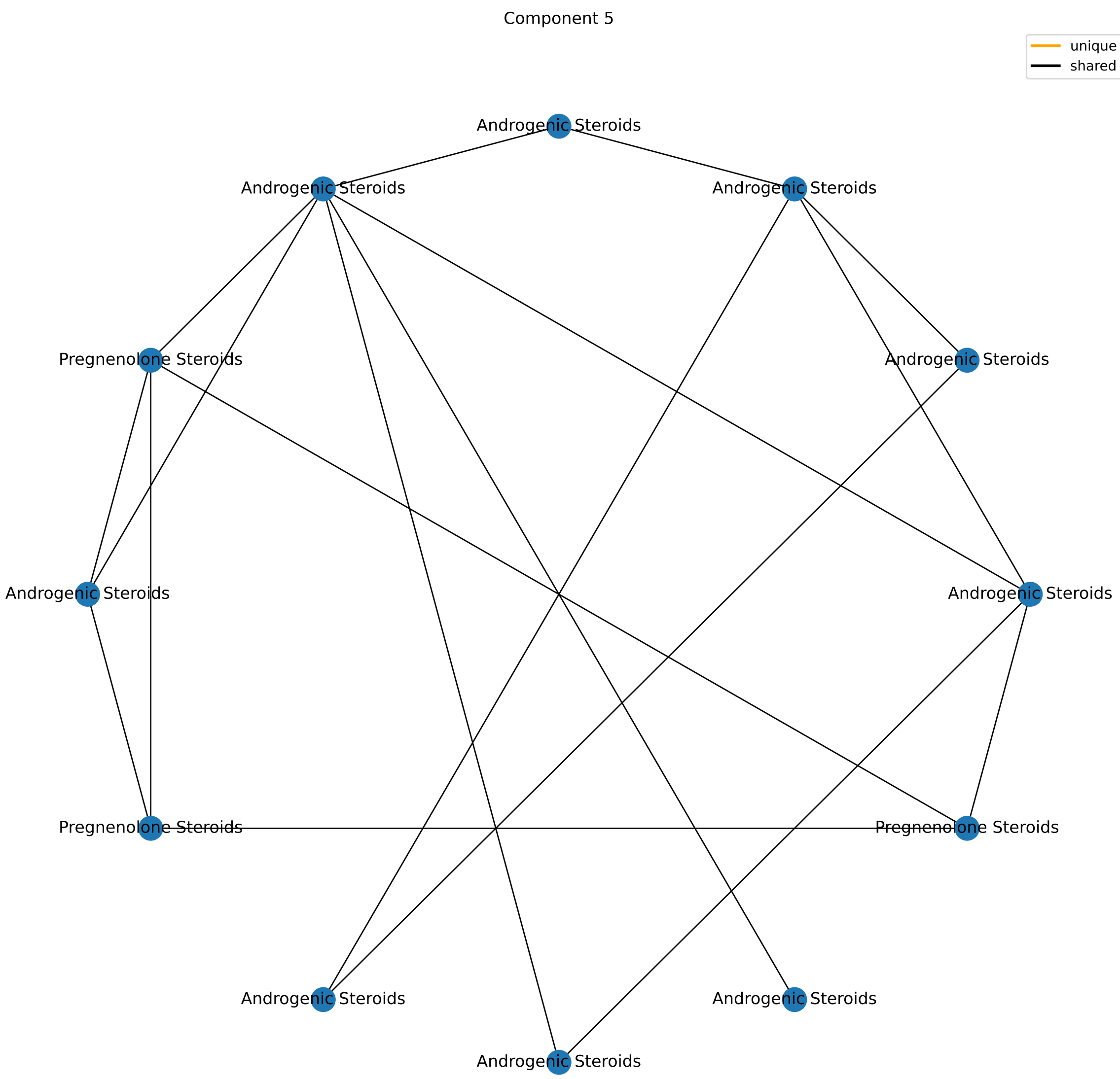
Component 5

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	13	6.21748359133489e-05	0.000559573522322014
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/form-components/5-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Androgenic Steroids	14	9	6.43764902868699e-15	6.69515498983447e-13
Pregnenolone Steroids	4	4	5.23942286754661e-08	2.72449989112424e-06
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1
Benzoate Metabolism	21	0	1	1

results/006/enrichment/form-components/5-subpathway.csv



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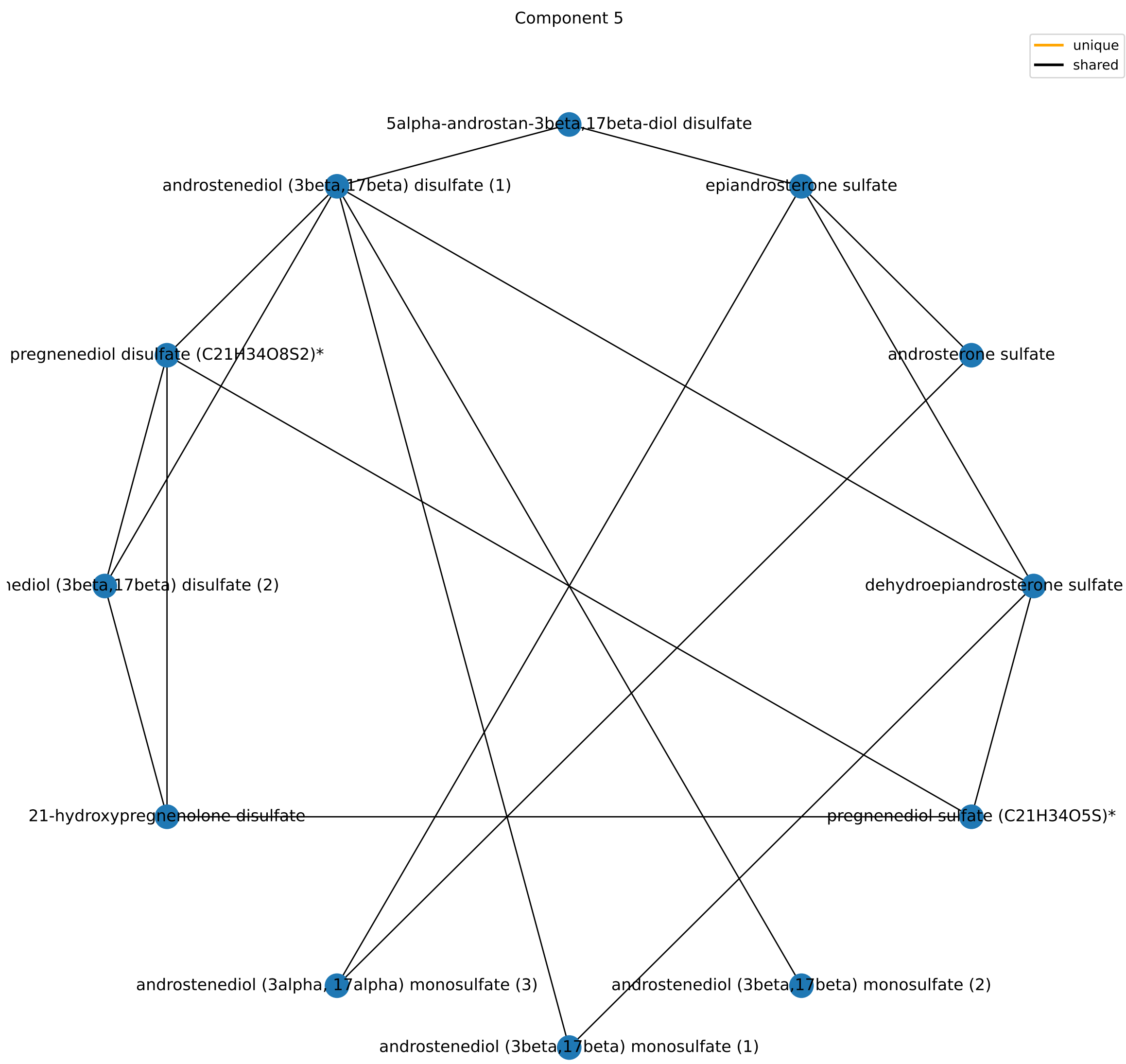

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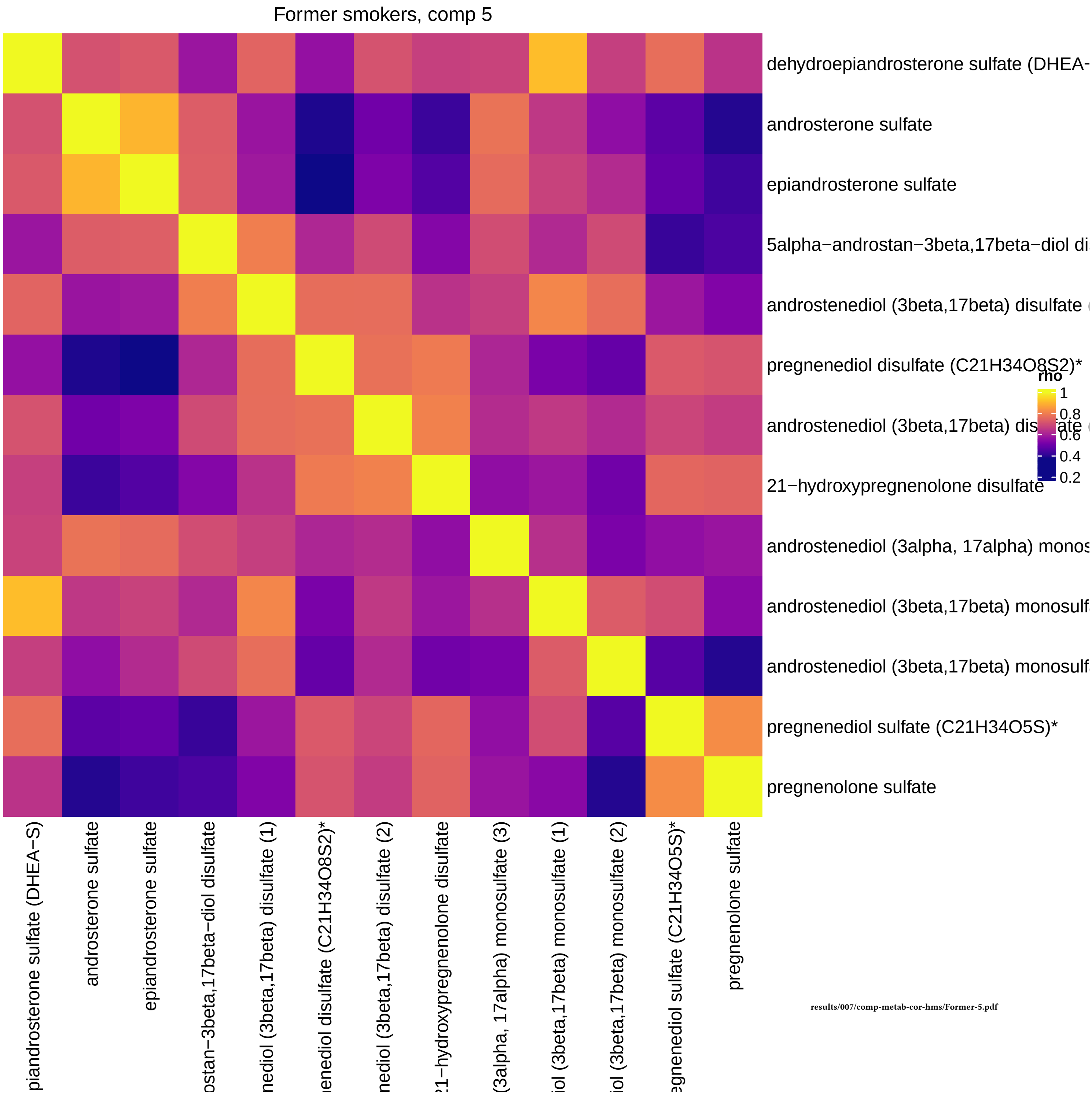


results/007/comp-metab-cor-hms/Former-5.pdf

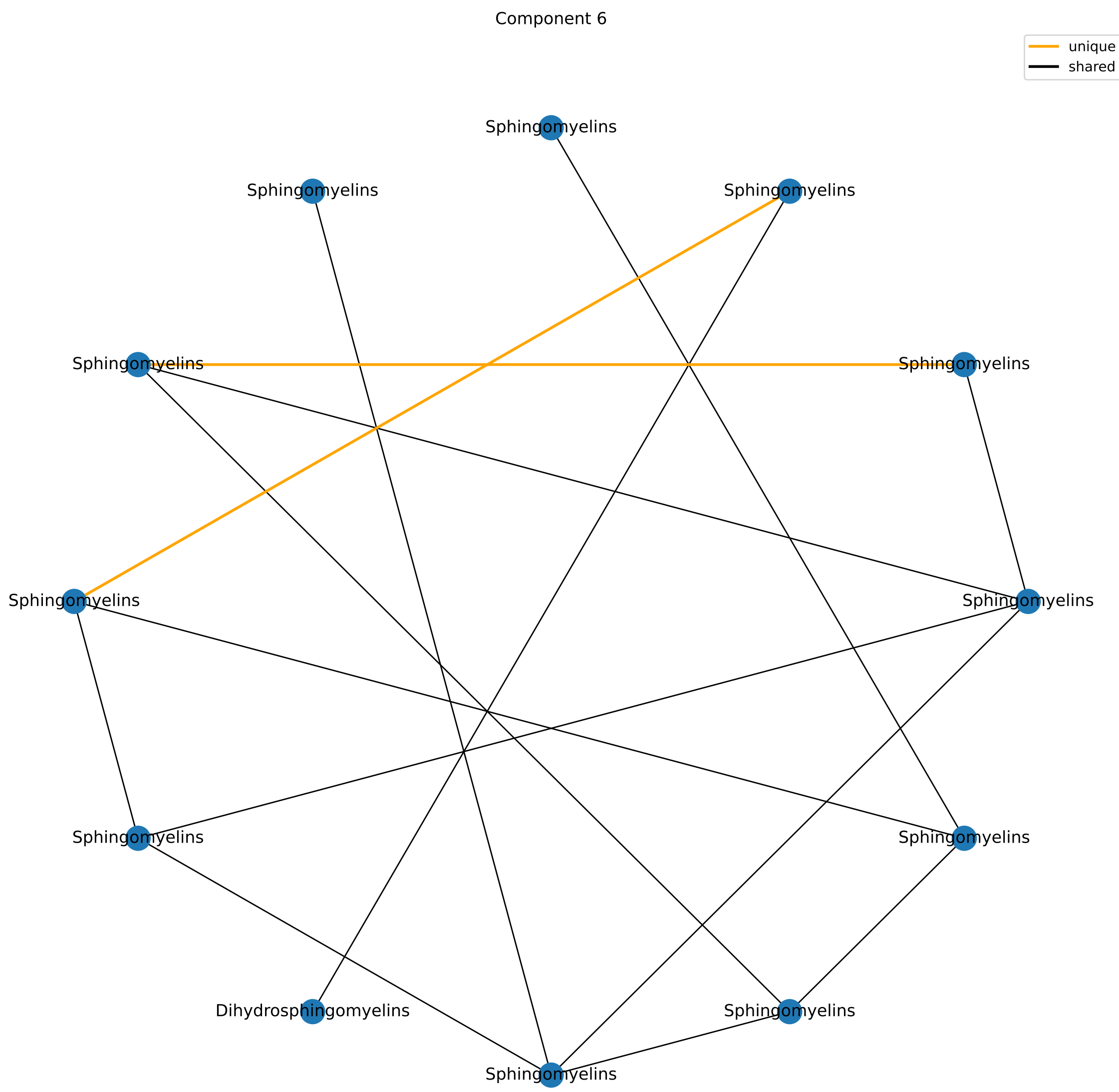
Component 6

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	13	6.21748359133489e-05	0.00055957352322014
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/form-components/6-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Sphingomyelins	28	12	5.63322370701747e-18	5.85855265529817e-16
Dihydrosphingomyelins	5	1	0.0830725862938696	1
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1

results/006/enrichment/form-components/6-subpathway.csv



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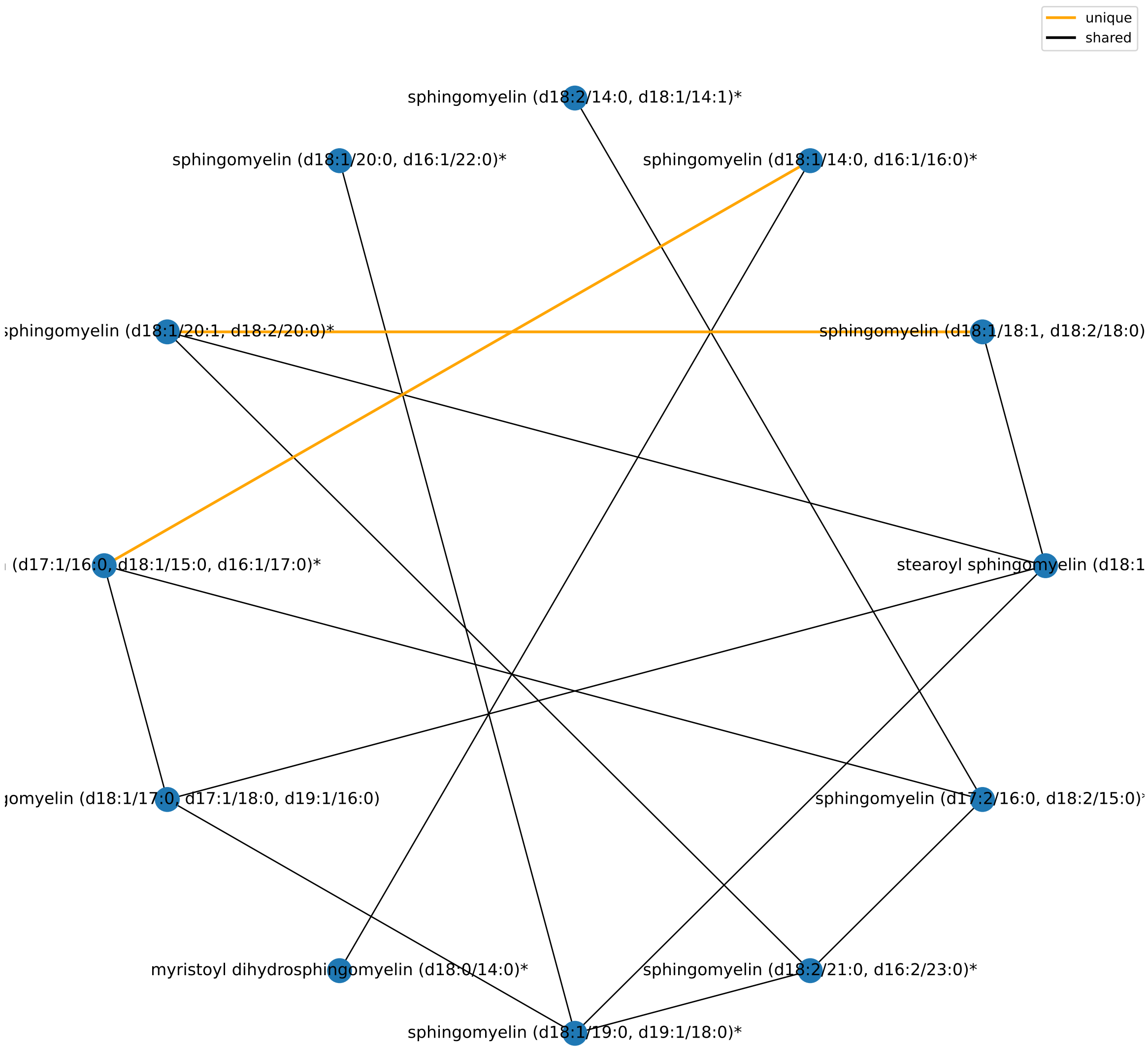
Component 6



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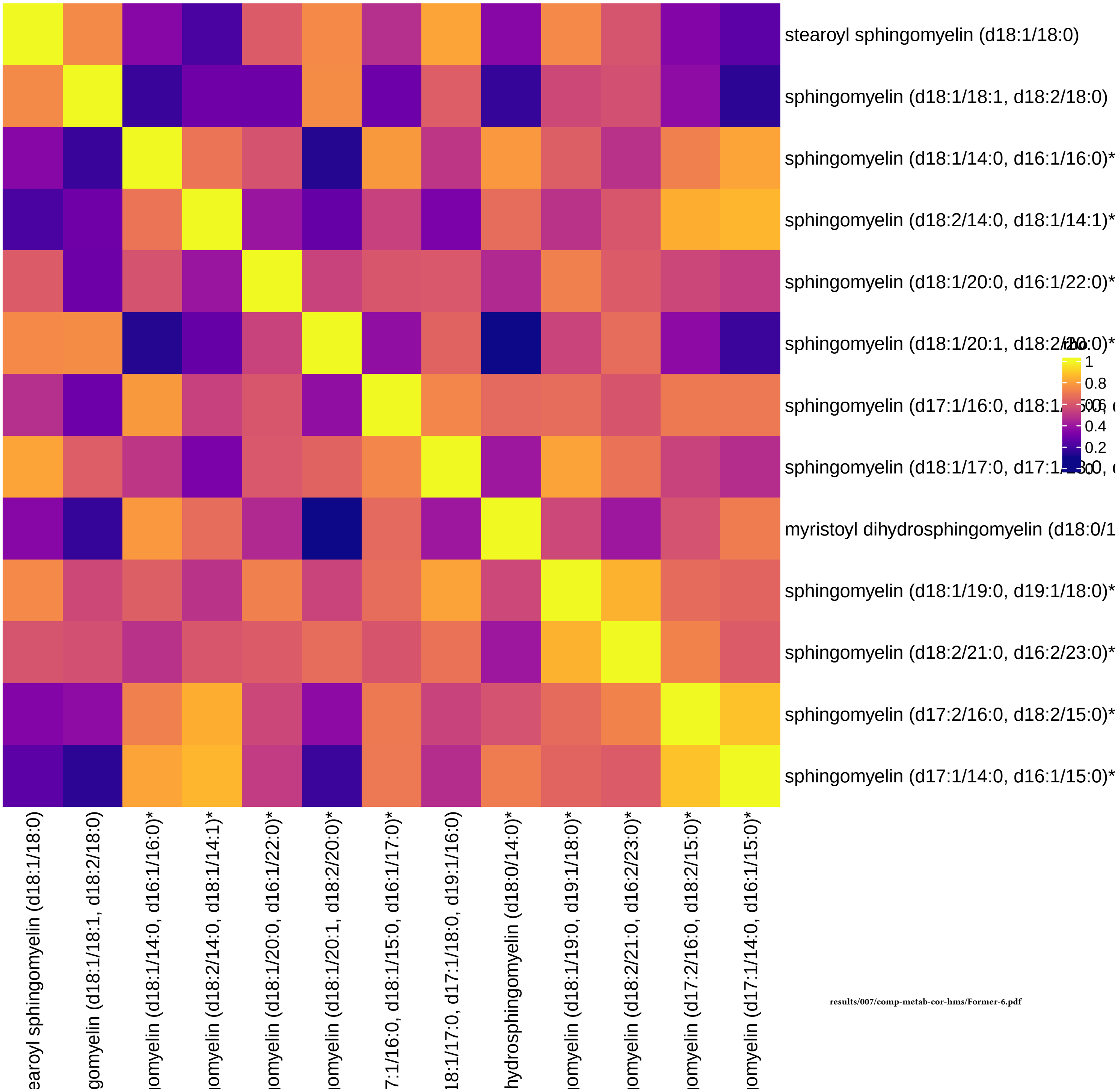
Former smokers, comp 6



results/007/comp-metab-cor-hms/Former-6.pdf

Component 7

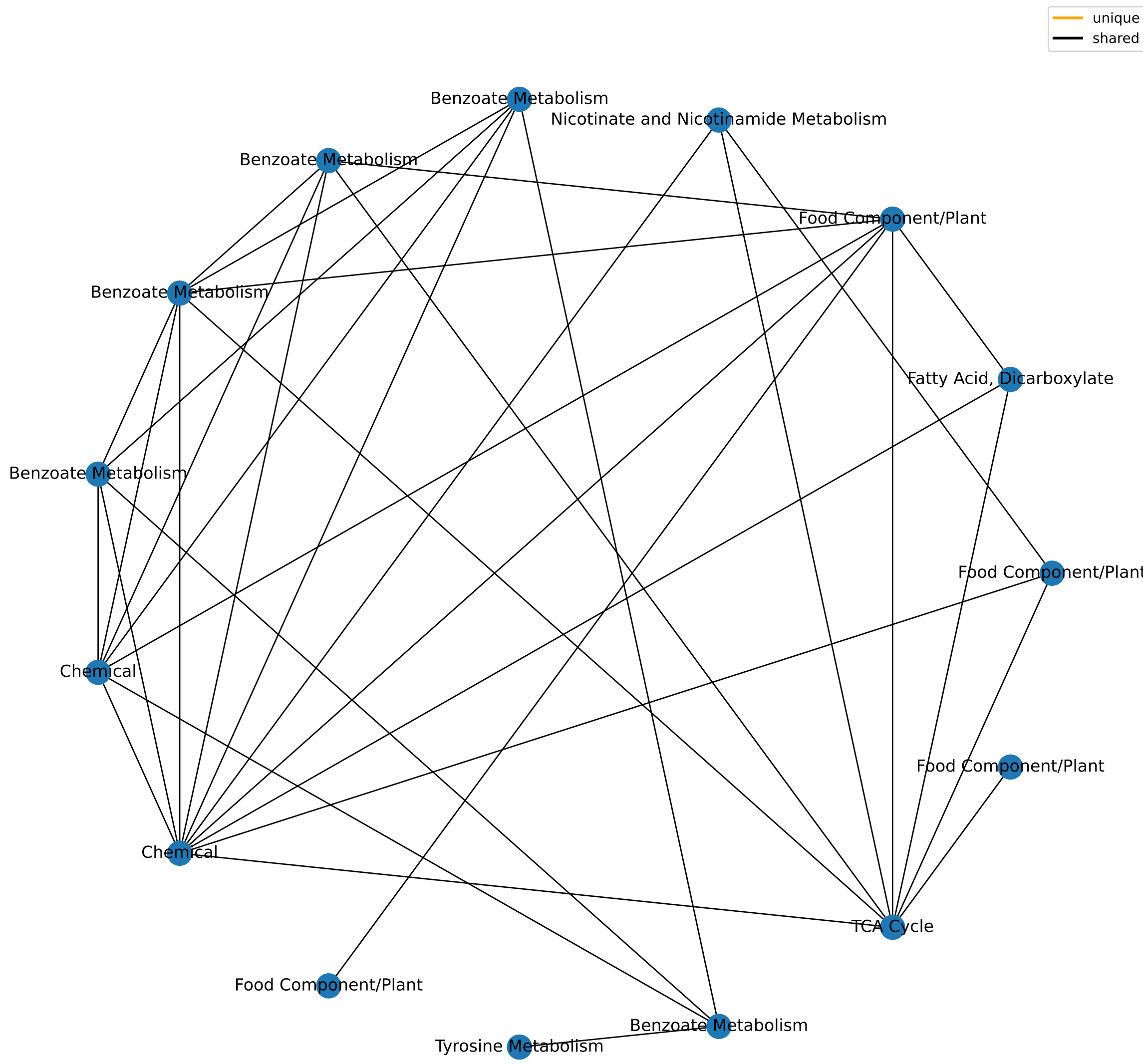
Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Xenobiotics	96	12	1.06469012466951e-08	9.5822111220256e-08
Energy	10	1	0.193162089459707	0.869229402568682
Cofactors and Vitamins	25	1	0.418461685816973	1
Amino Acid	180	1	0.987572508825868	1
Lipid	363	1	0.999974509667724	1
Carbohydrate	24	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1

results/006/enrichment/form-components/7-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Benzoate Metabolism	21	5	3.54641374718115e-05	0.00368827029706839
Food Component/Plant	38	5	0.000707569928652132	0.0367936362899109
Chemical	18	2	0.0525073793369186	1
Nicotinate and Nicotinamide Metabolism	5	1	0.101435089608838	1
TCA Cycle	9	1	0.175550347892662	1
Tyrosine Metabolism	17	1	0.306918033651192	1
Fatty Acid, Dicarboxylate	23	1	0.392273187538788	1
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1

results/006/enrichment/form-components/7-subpathway.csv

Component 7



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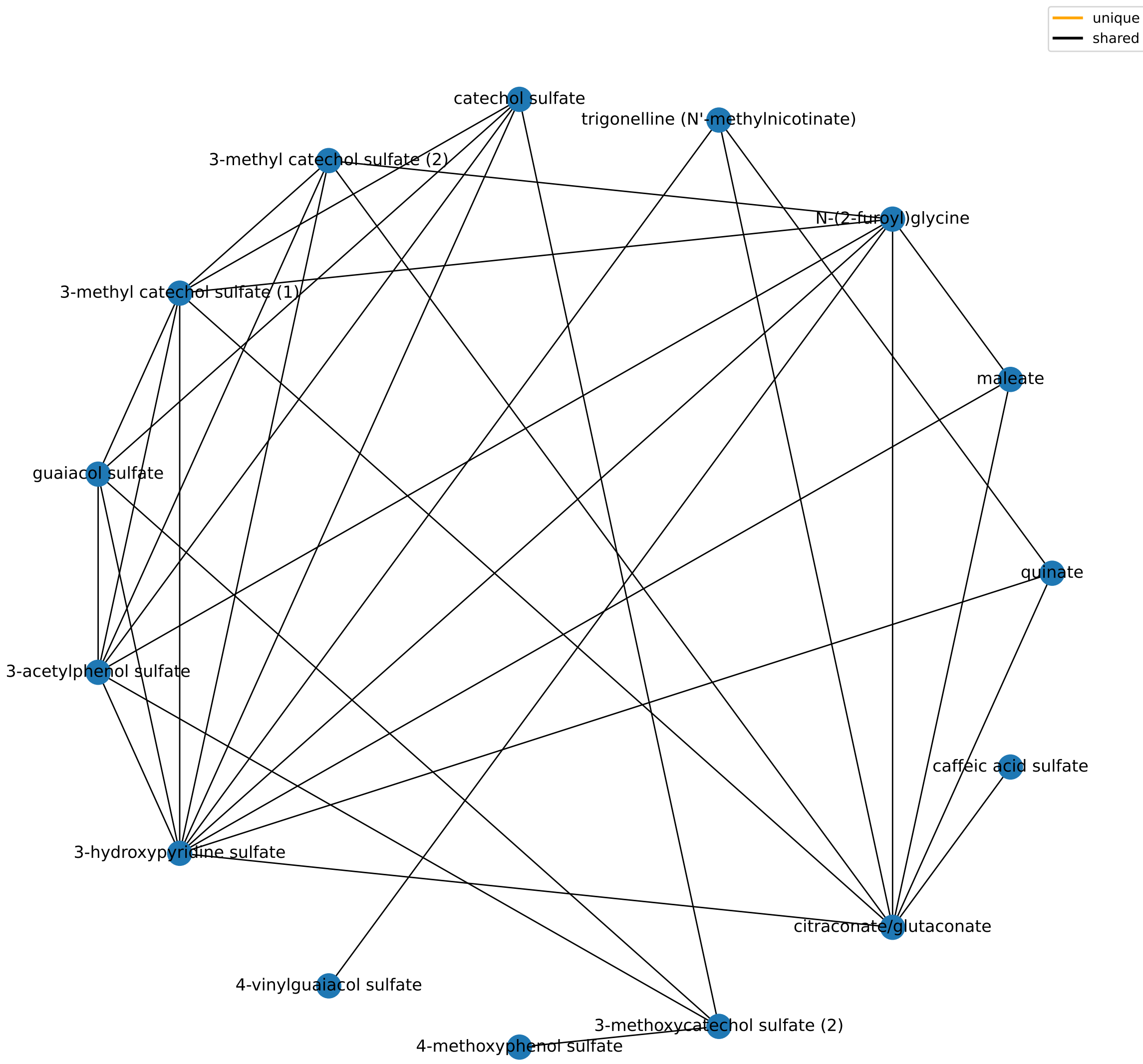
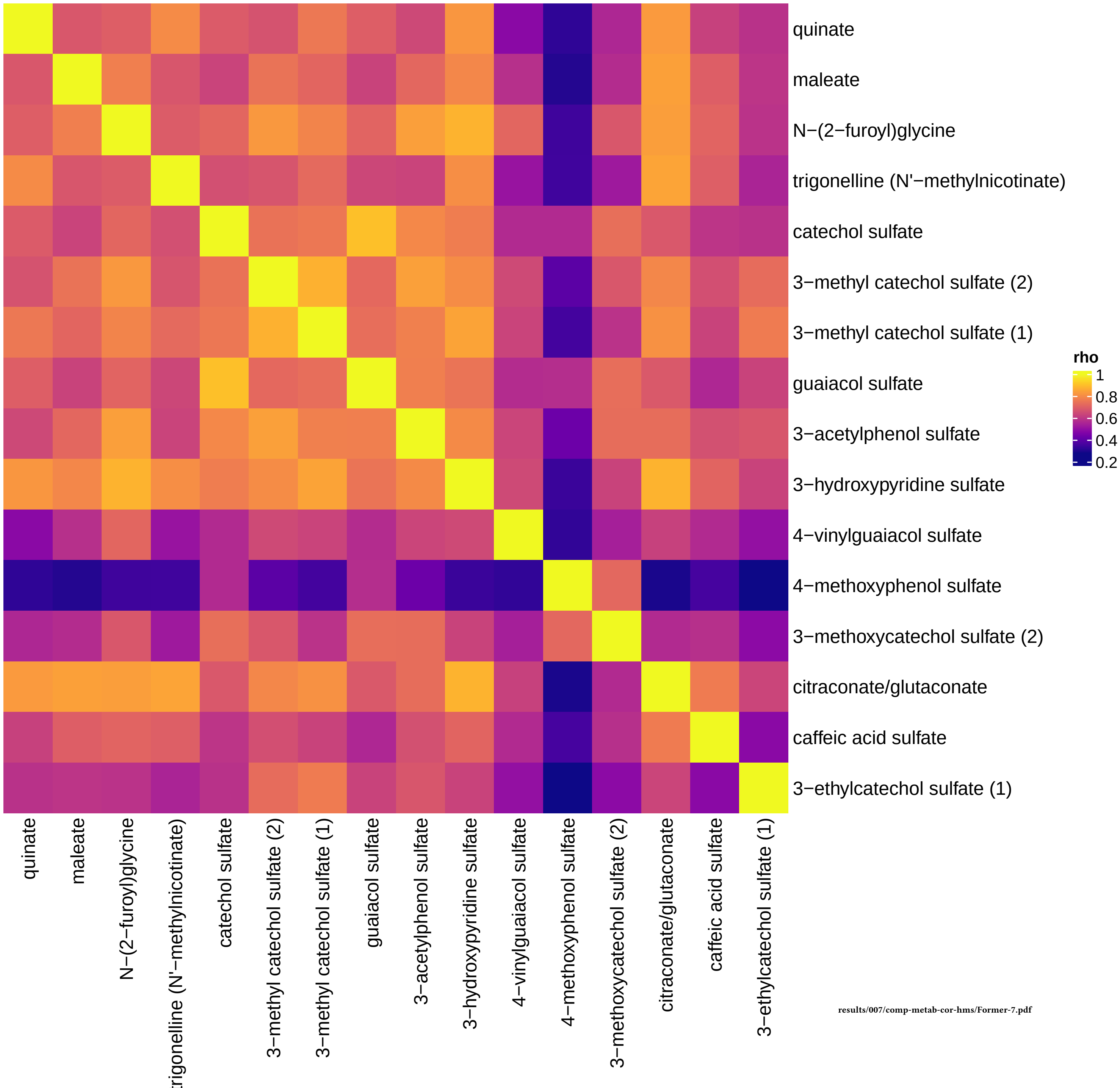
Component 7



results/005_1/component-graphs/former/chemical_name-6.pdf

Former smokers, comp 7



results/007/comp-metab-cor-hms/Former-7.pdf

Component 8

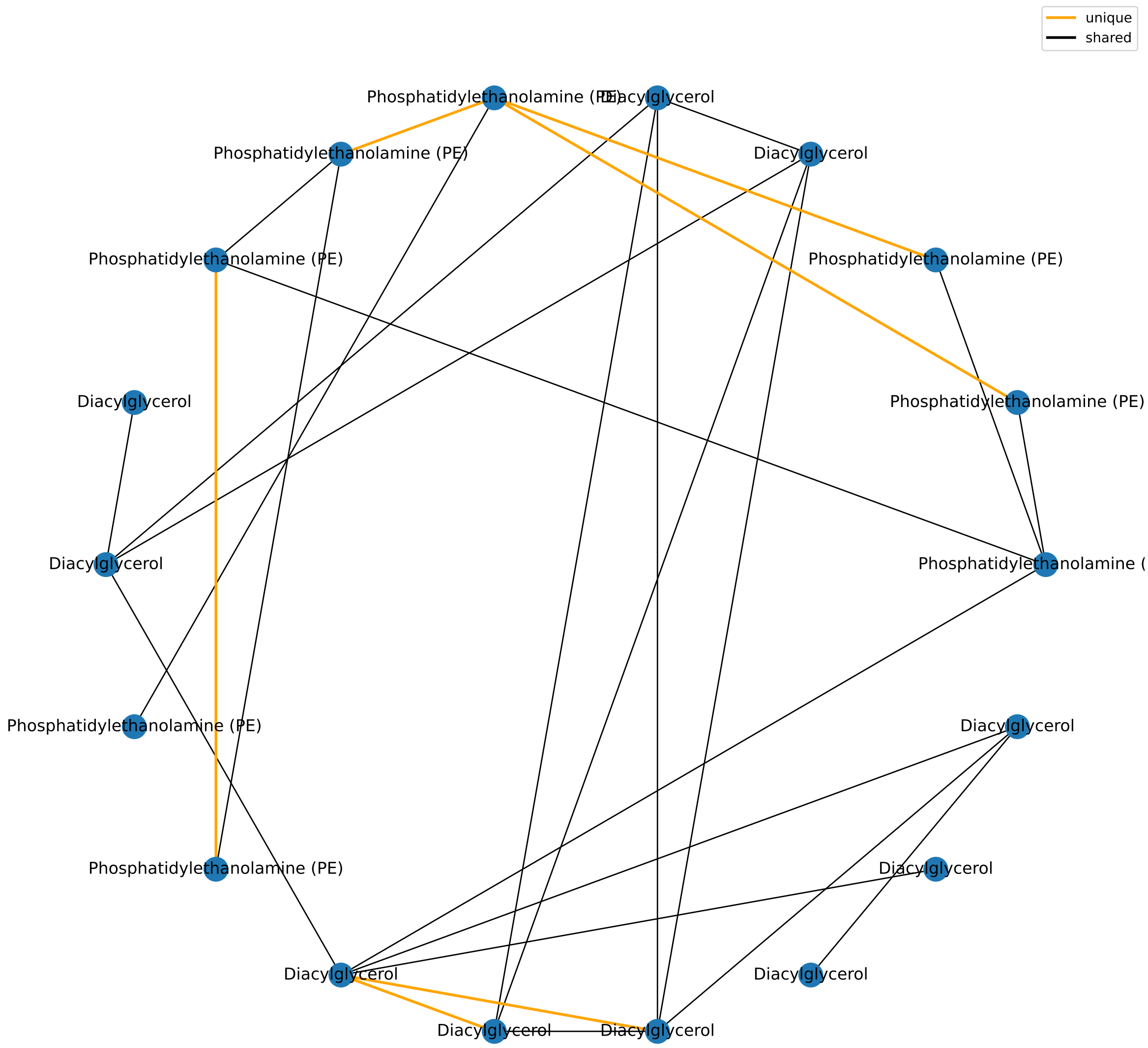
Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	19	6.5334139676174e-07	5.88007257085566e-06
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/form-components/8-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Diacylglycerol	19	11	4.77388701520288e-15	4.964842495811e-13
Phosphatidylethanolamine (PE)	11	8	4.60309215338075e-12	2.39360791975799e-10
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1
Ascorbate and Aldarate Metabolism	3	0	1	1
Bacterial/Fungal	2	0	1	1

results/006/enrichment/form-components/8-subpathway.csv

Component 8



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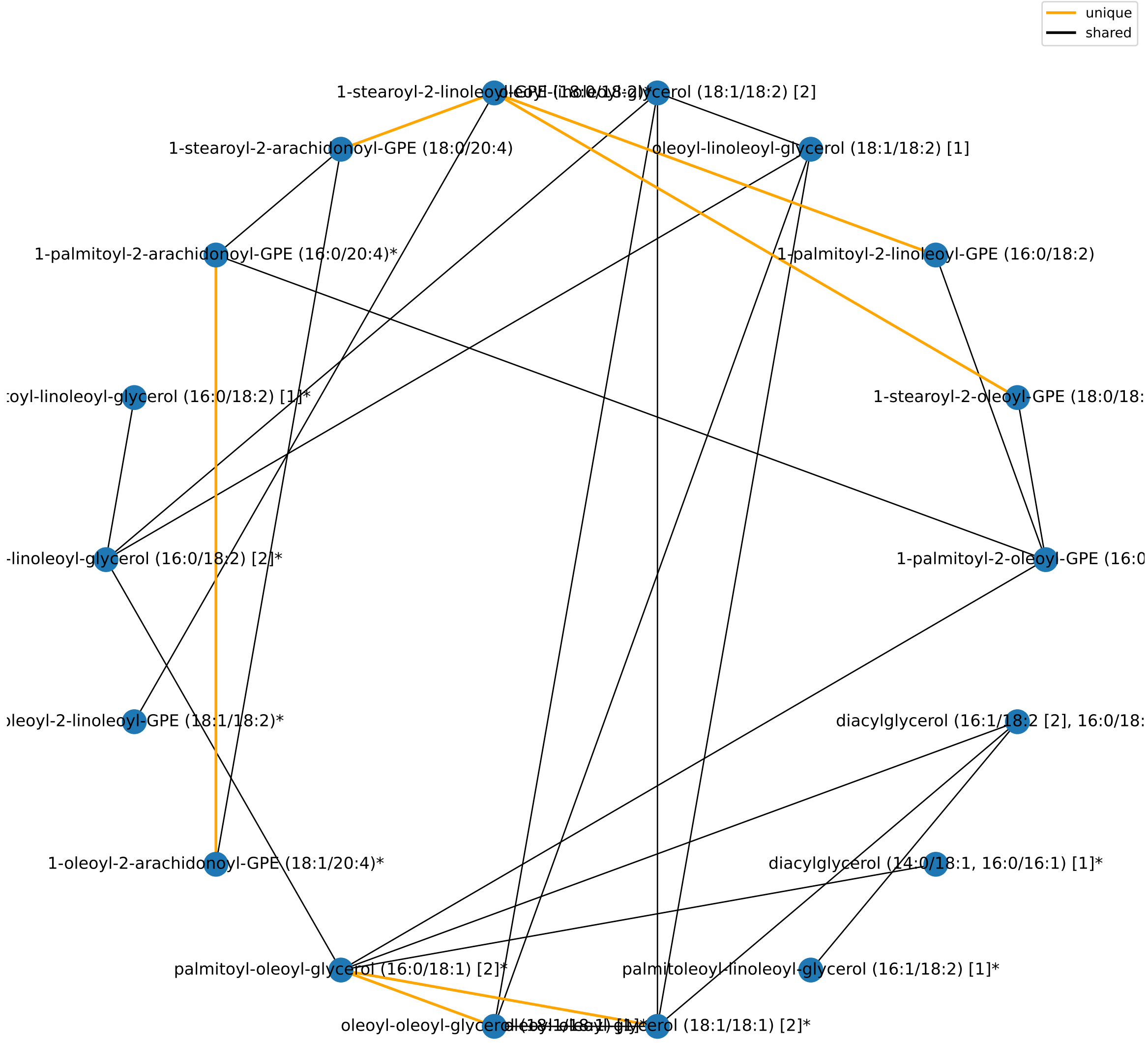
Component 8



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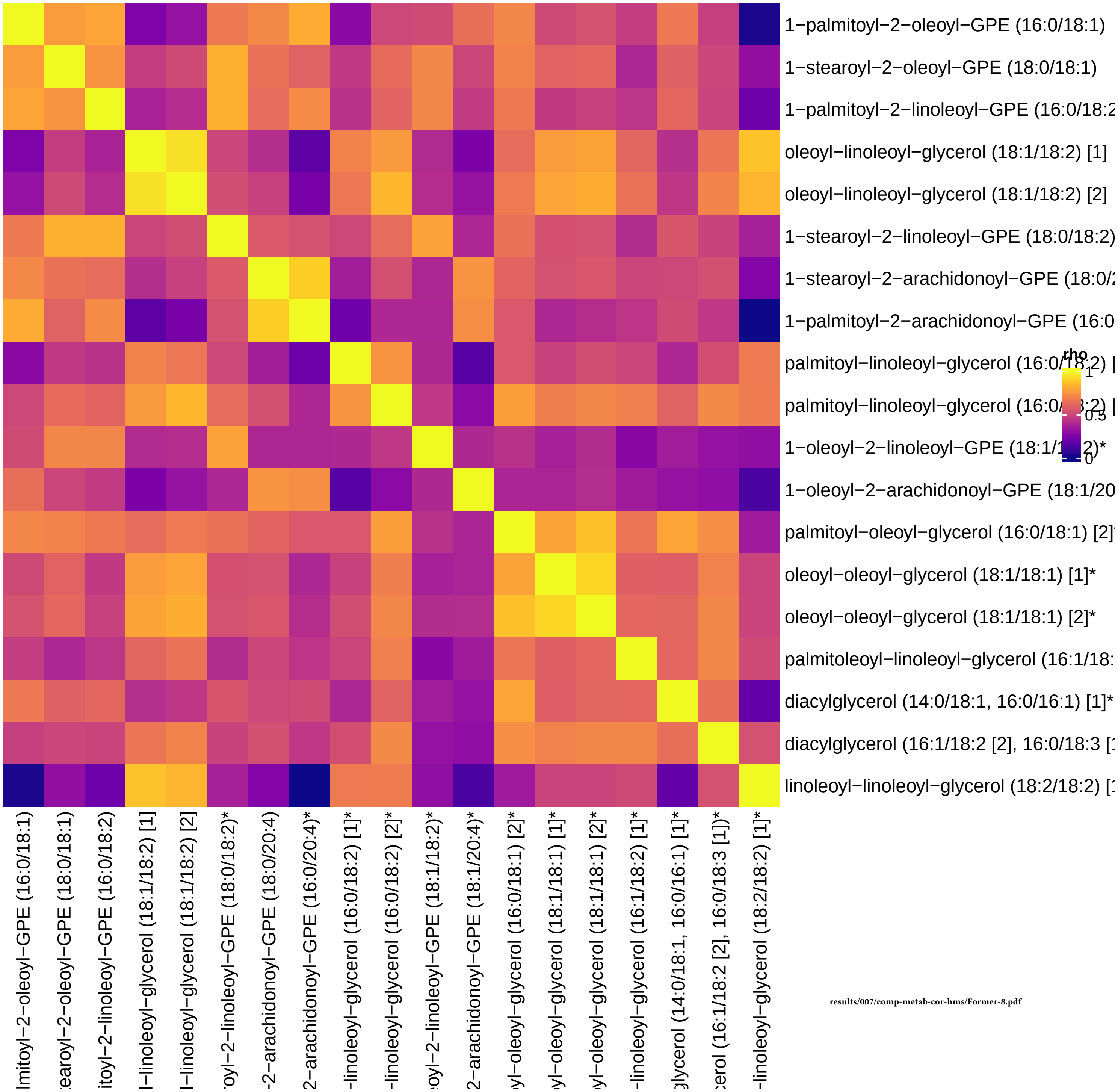
Former smokers, comp 8



results/007/comp-metab-cor-hms/Former-8.pdf

Component 9

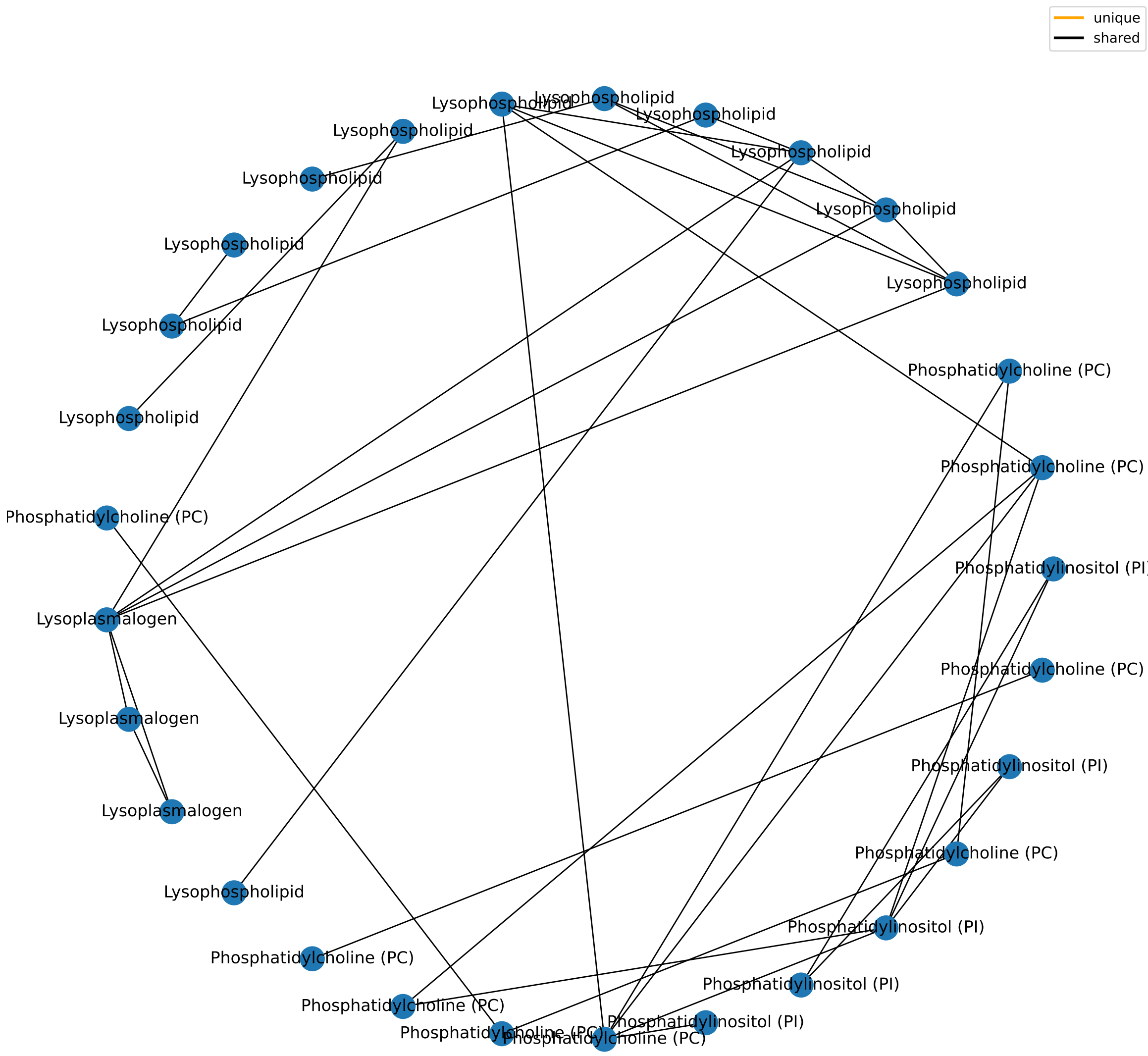
Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	30	1.33203816870287e-10	1.19883435183258e-09
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/form-components/9-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lysophospholipid	24	12	2.58948355074026e-12	2.69306289276987e-10
Phosphatidylcholine (PC)	19	10	1.36685992579985e-10	7.10767161415921e-09
Phosphatidylinositol (PI)	6	5	4.03994492909295e-07	1.40051424208556e-05
Lysoplasmalogen	3	3	5.61552549506666e-05	0.00146003662871733
Acetylated Peptides	3	0	1	1
Advanced Glycation End-product	1	0	1	1
Alanine and Aspartate Metabolism	8	0	1	1
Aminosugar Metabolism	5	0	1	1
Androgenic Steroids	14	0	1	1

results/006/enrichment/form-components/9-subpathway.csv

Component 9



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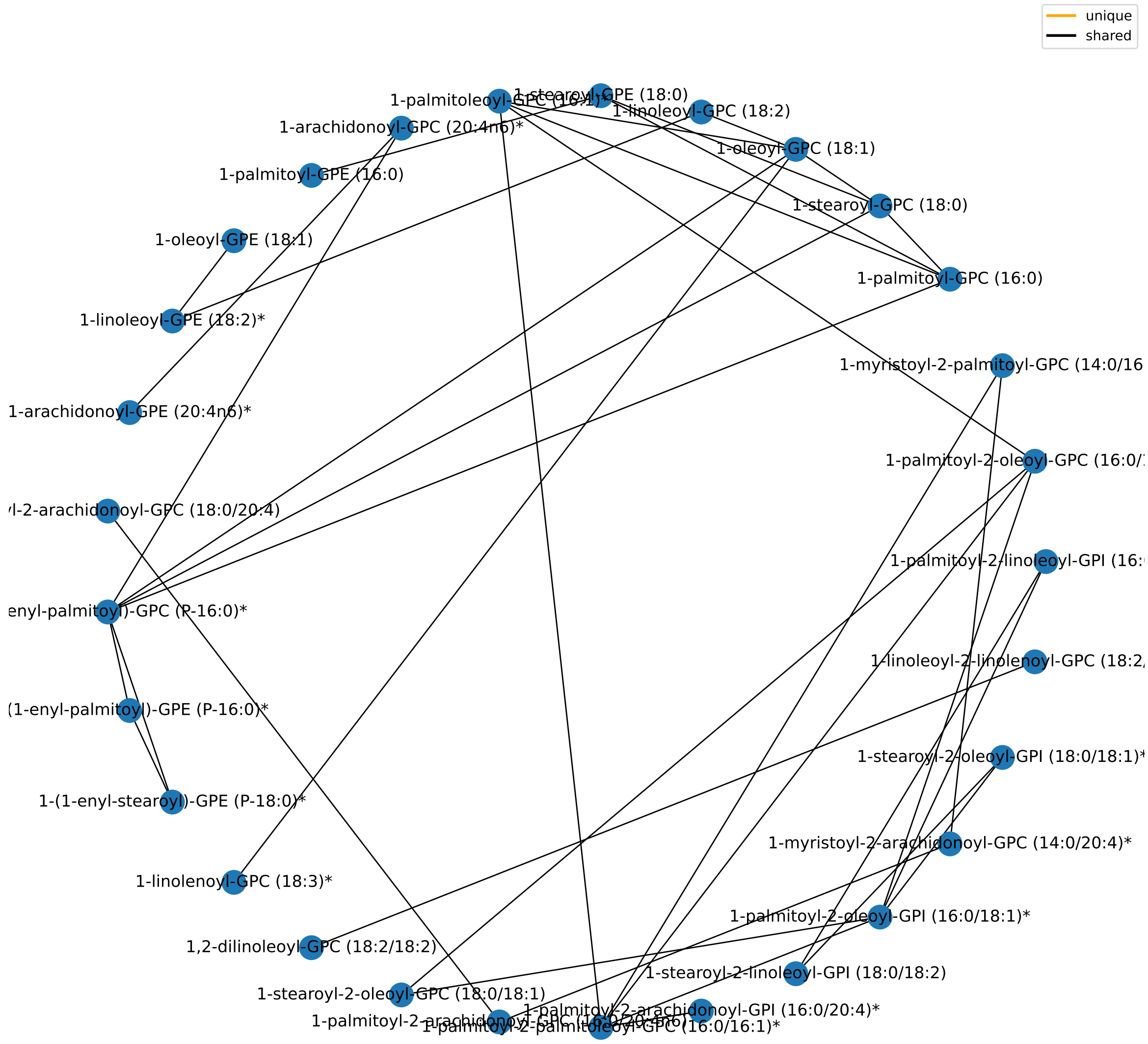
Component 9



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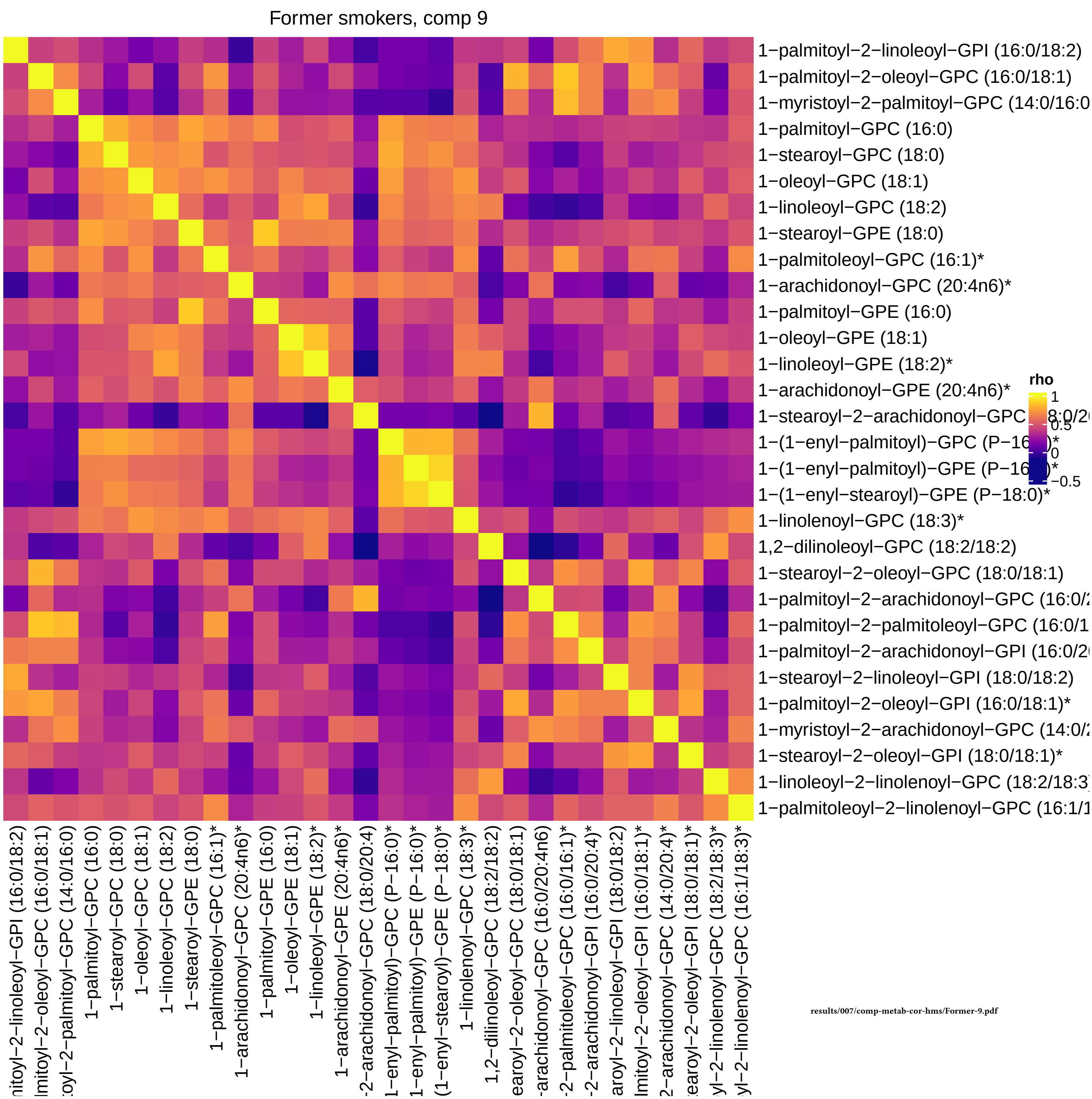


results/007/comp-metab-cor-hms/Former-9.pdf

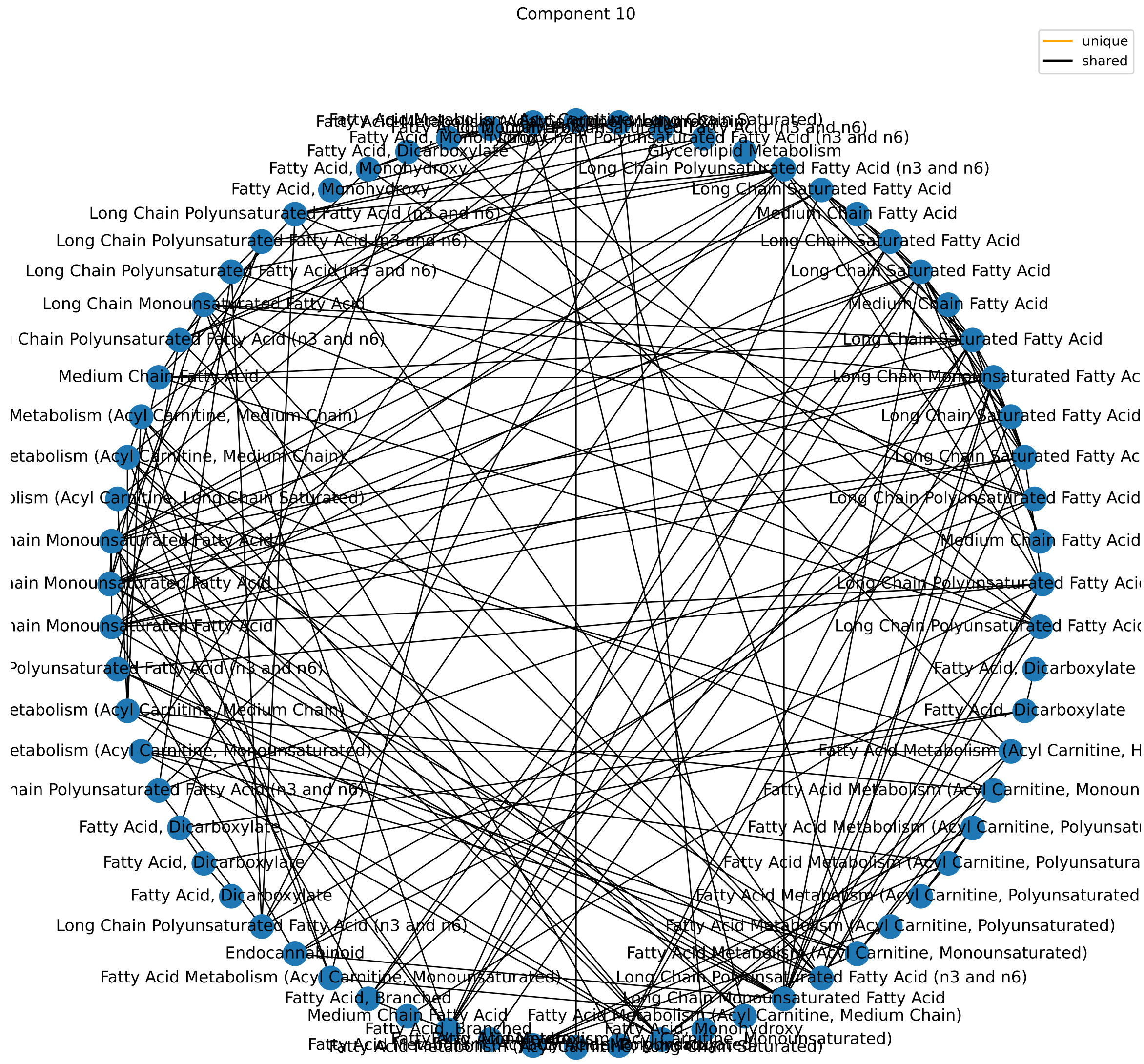
Component 10

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	69	2.0932052248421e-24	1.88388470235789e-23
Amino Acid	180	0	1	1
Carbohydrate	24	0	1	1
Cofactors and Vitamins	25	0	1	1
Energy	10	0	1	1
Nucleotide	32	0	1	1
Partially Characterized Molecules	3	0	1	1
Peptide	25	0	1	1
Xenobiotics	96	0	1	1

results/006/enrichment/form-components/10-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	7.35712223379008e-16	7.65140712314168e-14
Long Chain Monounsaturated Fatty Acid	7	6	3.0158208543665e-06	0.000104548456284705
Long Chain Saturated Fatty Acid	7	6	3.0158208543665e-06	0.000104548456284705
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	5.46116024646361e-06	0.000113592133126443
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	5.46116024646361e-06	0.000113592133126443
Fatty Acid, Monohydroxy	20	8	0.000164196711287389	0.00284607632898141
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	5	0.0002453961965353	0.0031901505549589
Medium Chain Fatty Acid	8	5	0.0002453961965353	0.0031901505549589
Fatty Acid, Dicarboxylate	23	7	0.00287693912993314	0.033244629945894

results/006/enrichment/form-components/10-subpathway.csv



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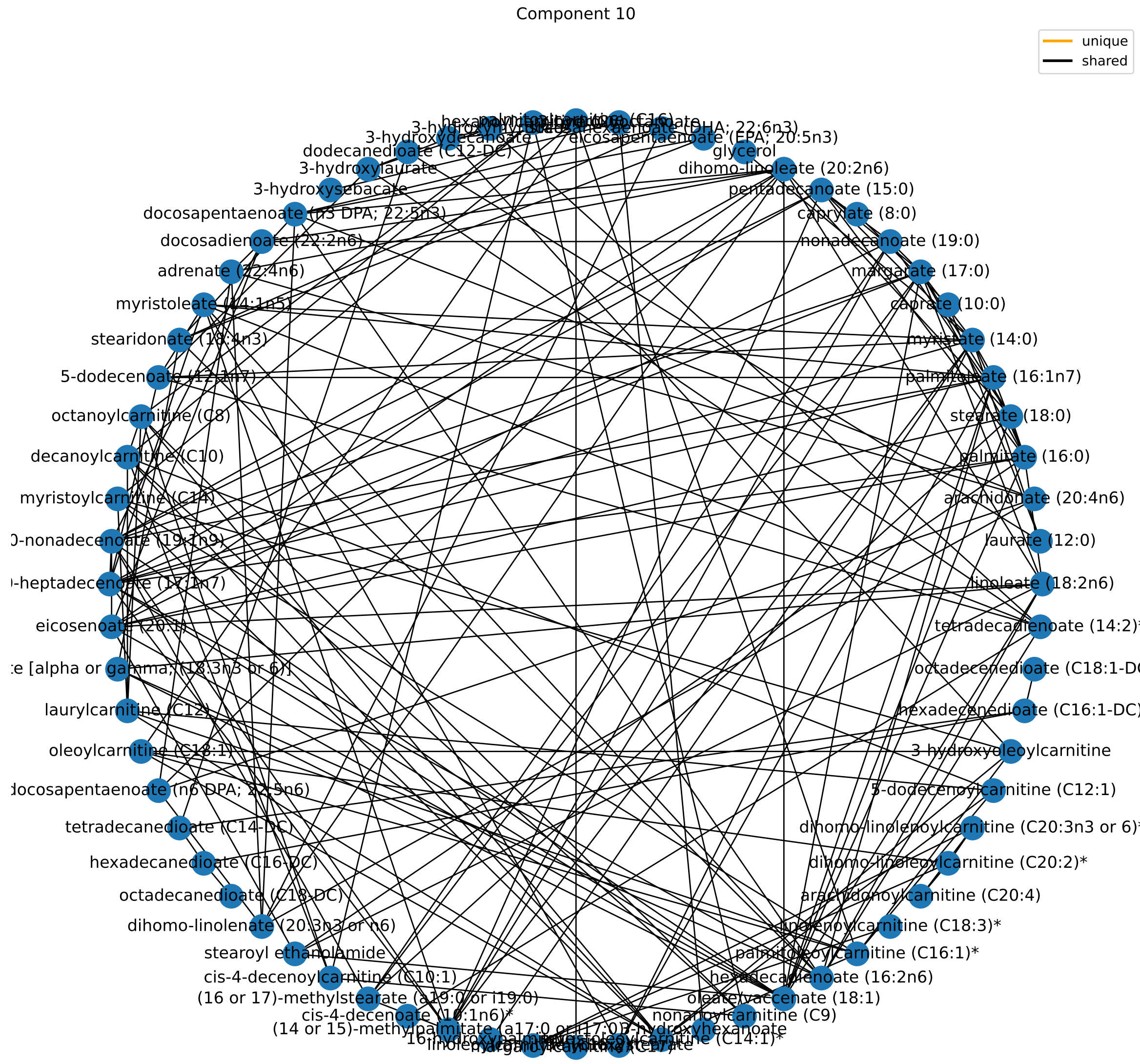

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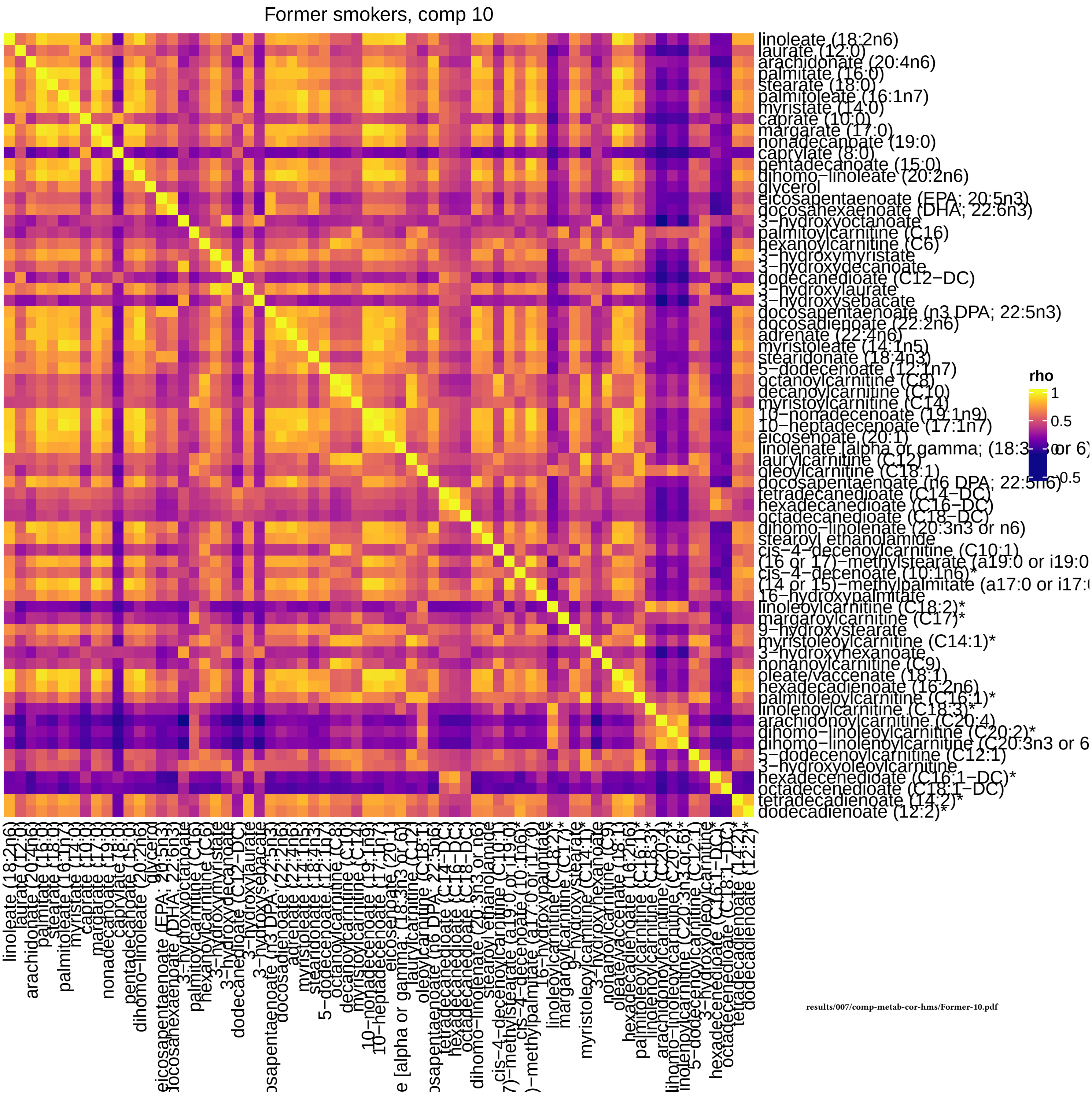


results/007/comp-metab-cor-hms/Former-10.pdf

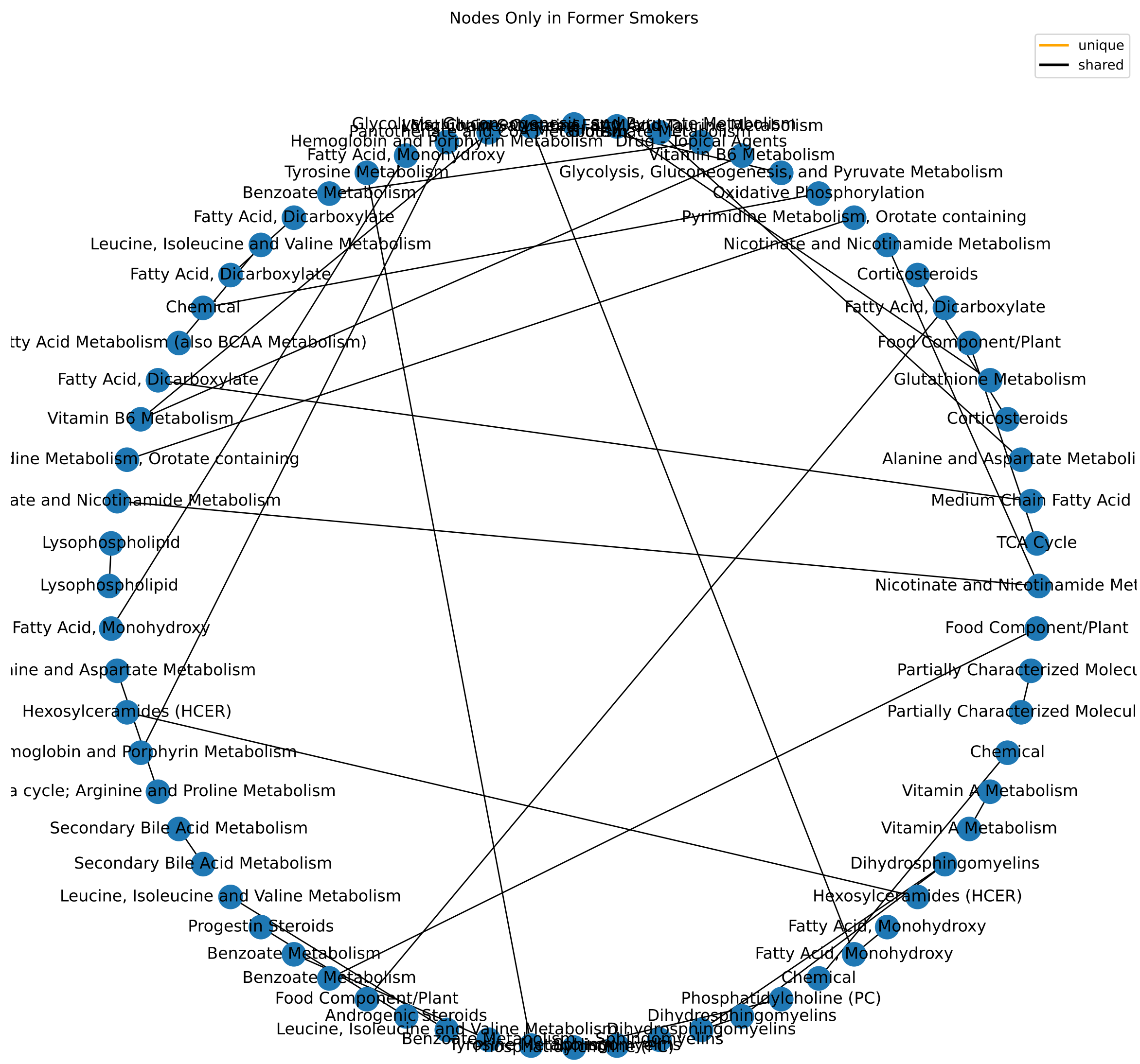
Unique To Former Smokers

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Cofactors and Vitamins	25	10	0.000149526718672046	0.00134574046804841
Partially Characterized Molecules	3	2	0.0345974462139159	0.155688507962622
Xenobiotics	96	13	0.267098312136676	0.700065558179091
Energy	10	2	0.311140248079596	0.700065558179091
Nucleotide	32	4	0.492074812112795	0.885734661803031
Lipid	363	38	0.769699163828548	0.993392162264282
Carbohydrate	24	2	0.772638348427775	0.993392162264282
Amino Acid	180	14	0.968554375337058	1
Peptide	25	0	1	1

results/006/enrichment/form-unique-superpathway.csv

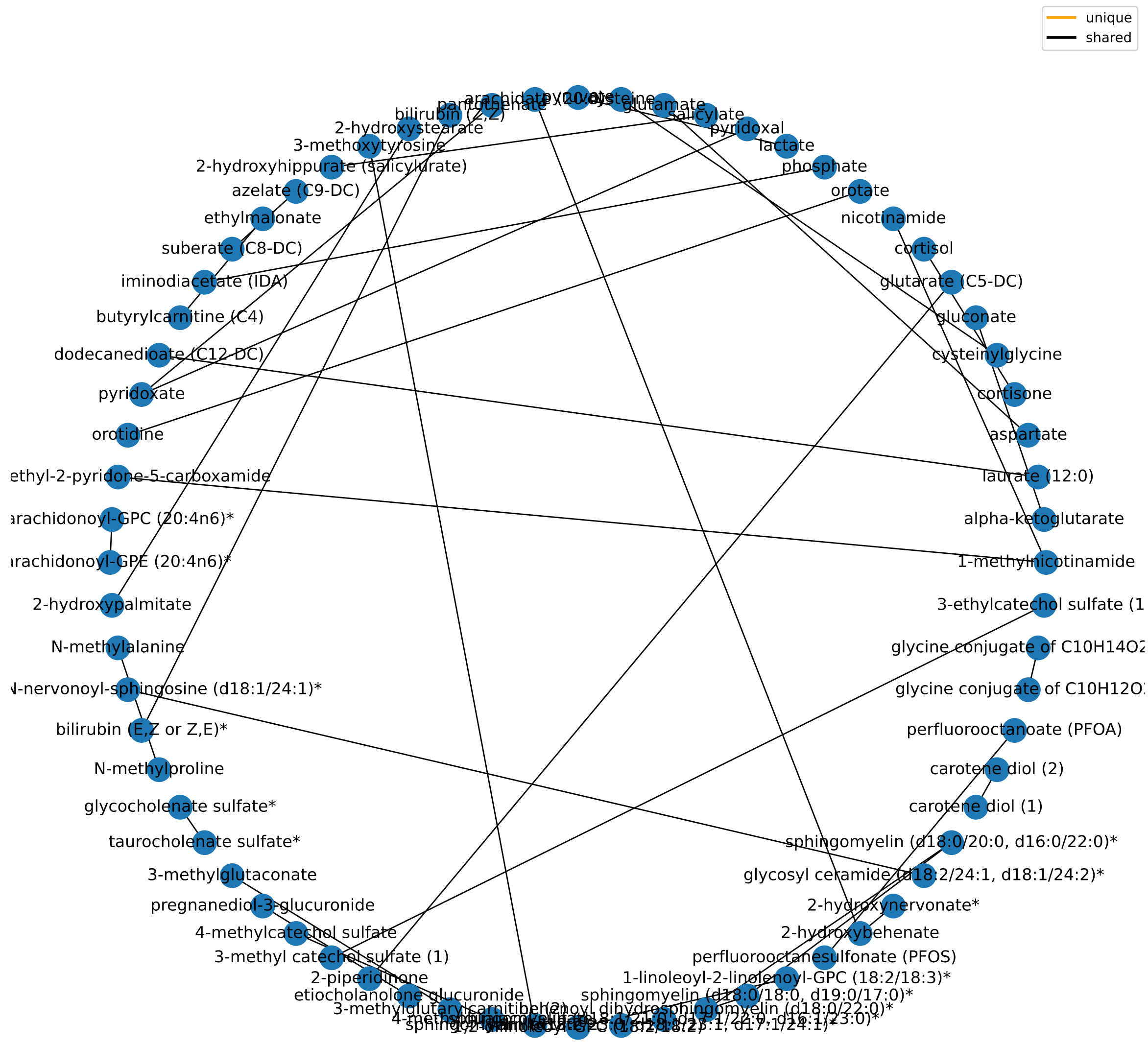
Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Dihydrosphingomyelins	5	3	0.0115312516114739	0.323523978487503
Nicotinate and Nicotinamide Metabolism	5	3	0.0115312516114739	0.323523978487503
Corticosteroids	2	2	0.012443229941827	0.323523978487503
Vitamin B6 Metabolism	2	2	0.012443229941827	0.323523978487503
Partially Characterized Molecules	3	2	0.0345974462139159	0.599689067707875
Pyrimidine Metabolism, Orotate containing	3	2	0.0345974462139159	0.599689067707875
Fatty Acid, Monohydroxy	20	5	0.0627659842140751	0.8342818339325
Hemoglobin and Porphyrin Metabolism	4	2	0.0641755256871154	0.8342818339325
Vitamin A Metabolism	5	2	0.0992717084042096	0.971855760773965

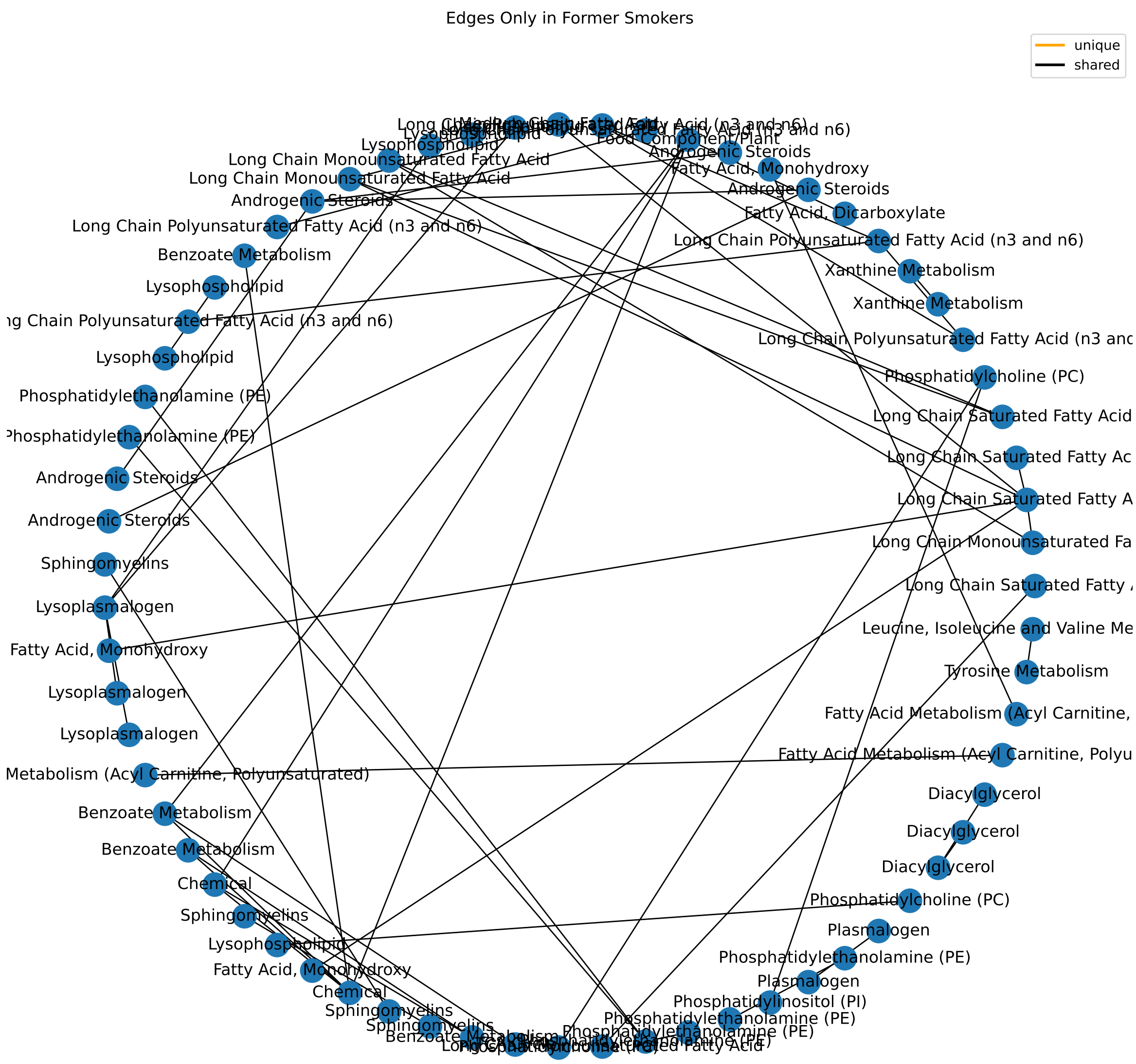
results/006/enrichment/form-unique-subpathway.csv



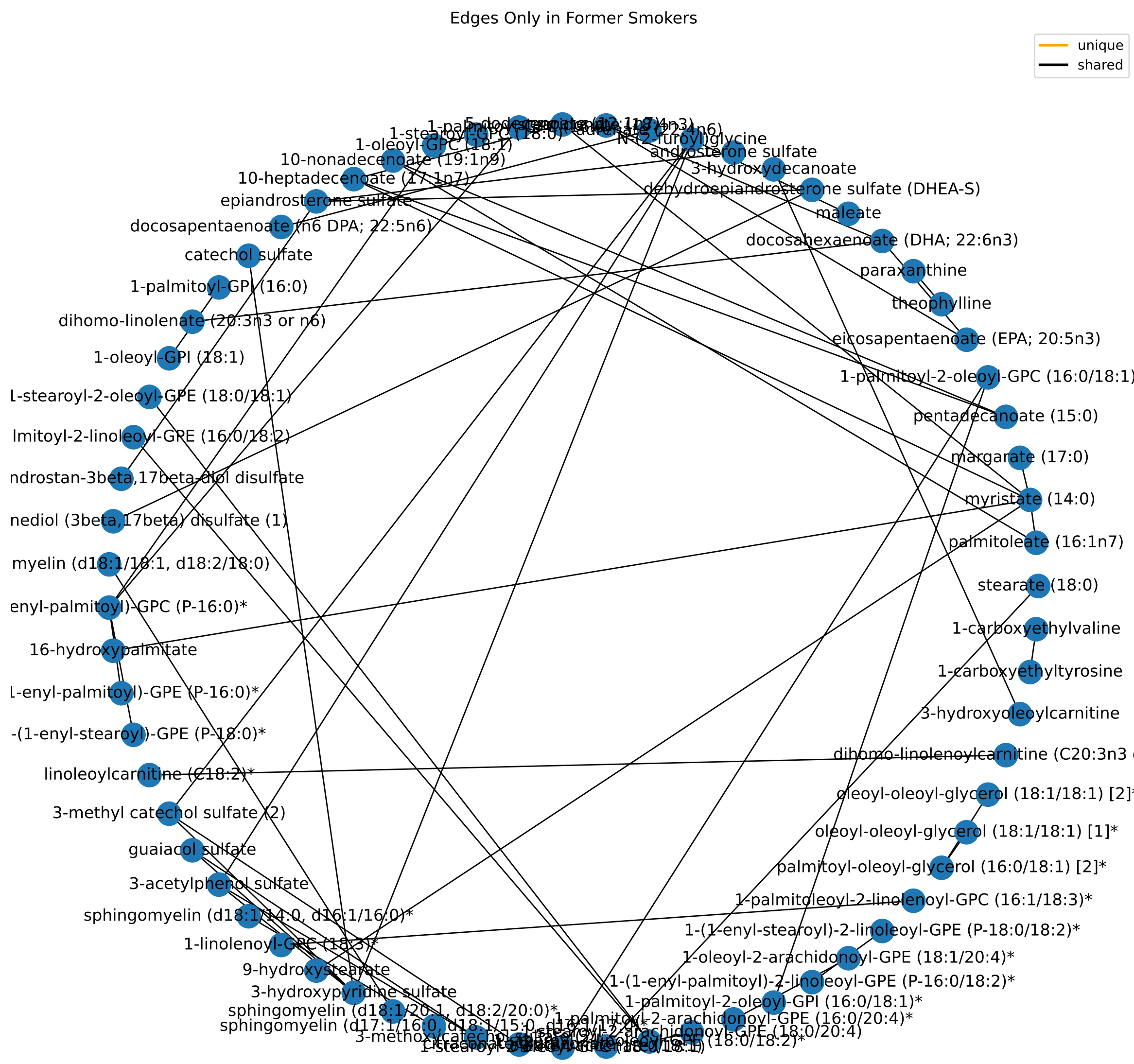
results/005_1/sub_pathway-node-only-form-graph.pdf

Nodes Only in Former Smokers





results/005_1/sub_pathway-edge-only-form-graph.pdf



results/005_1/chemical_name-edge-only-form-graph.pdf

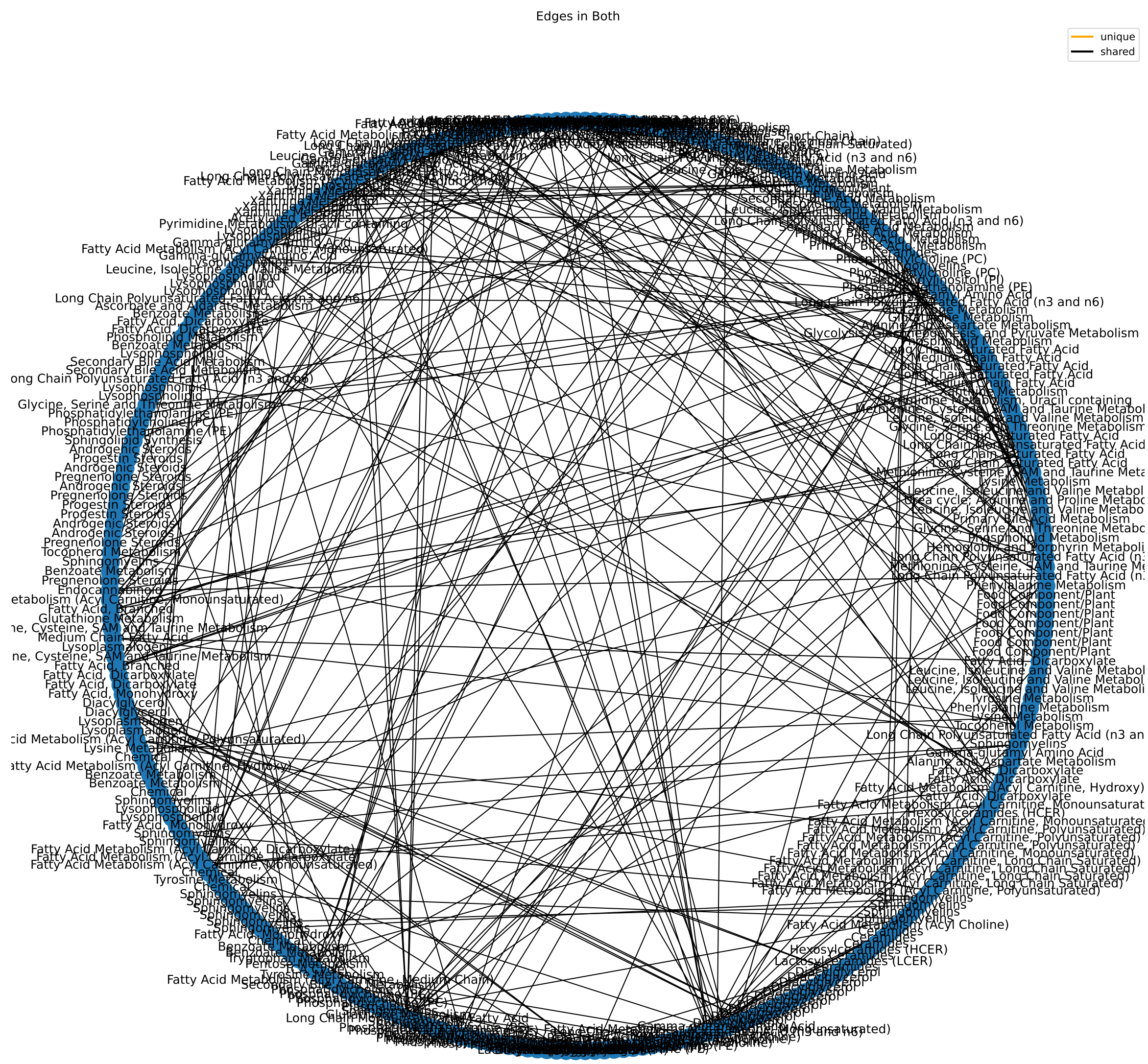
Intersection of Both

Super-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Lipid	363	215	2.8221021378912e-23	2.53989192410208e-22
Peptide	25	11	0.450079402996058	1
Xenobiotics	96	33	0.934447100476636	1
Cofactors and Vitamins	25	7	0.941250749535369	1
Energy	10	1	0.995010647987541	1
Carbohydrate	24	2	0.999952863918465	1
Nucleotide	32	2	0.999999238913724	1
Amino Acid	180	39	0.999999999832911	1
Partially Characterized Molecules	3	0	1	1

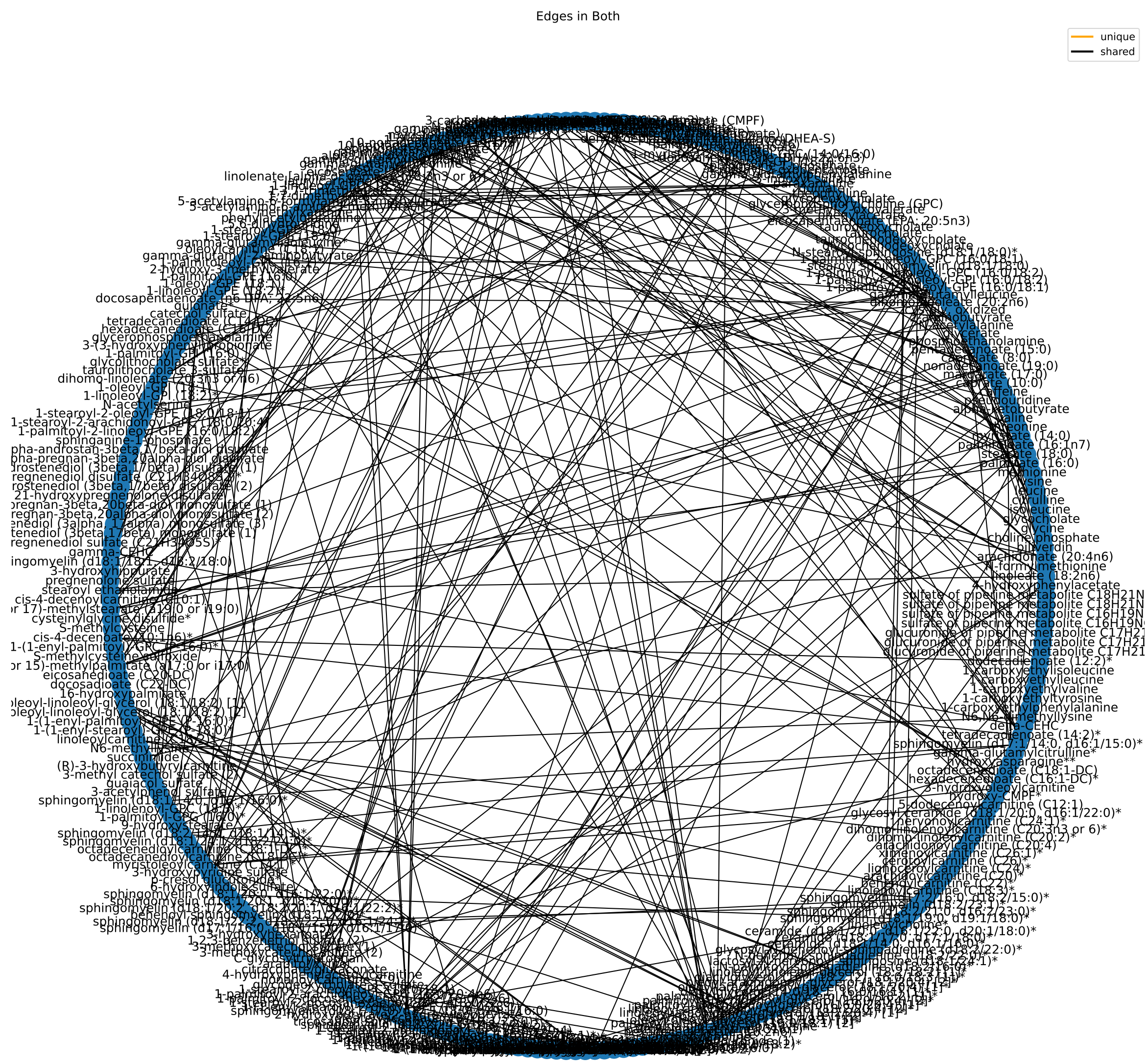
results/006/enrichment/intersection-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	3.06737607085555e-06	0.000319007111368977
Diacylglycerol	19	17	1.28924594069087e-05	0.000670407889159252
Phosphatidylethanolamine (PE)	11	10	0.000838596701336745	0.0225105767697082
Sphingomyelins	28	20	0.000865791414219547	0.0225105767697082
Plasmalogen	11	9	0.00670807230564668	0.129692942415898
Lysophospholipid	24	16	0.00868259050368756	0.129692942415898
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	7	0.00951175648944317	0.129692942415898
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	0.0112234277090681	0.129692942415898
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	0.0112234277090681	0.129692942415898

results/006/enrichment/intersection-subpathway.csv



results/005_1/sub_pathway-edge-shared-graph.pdf



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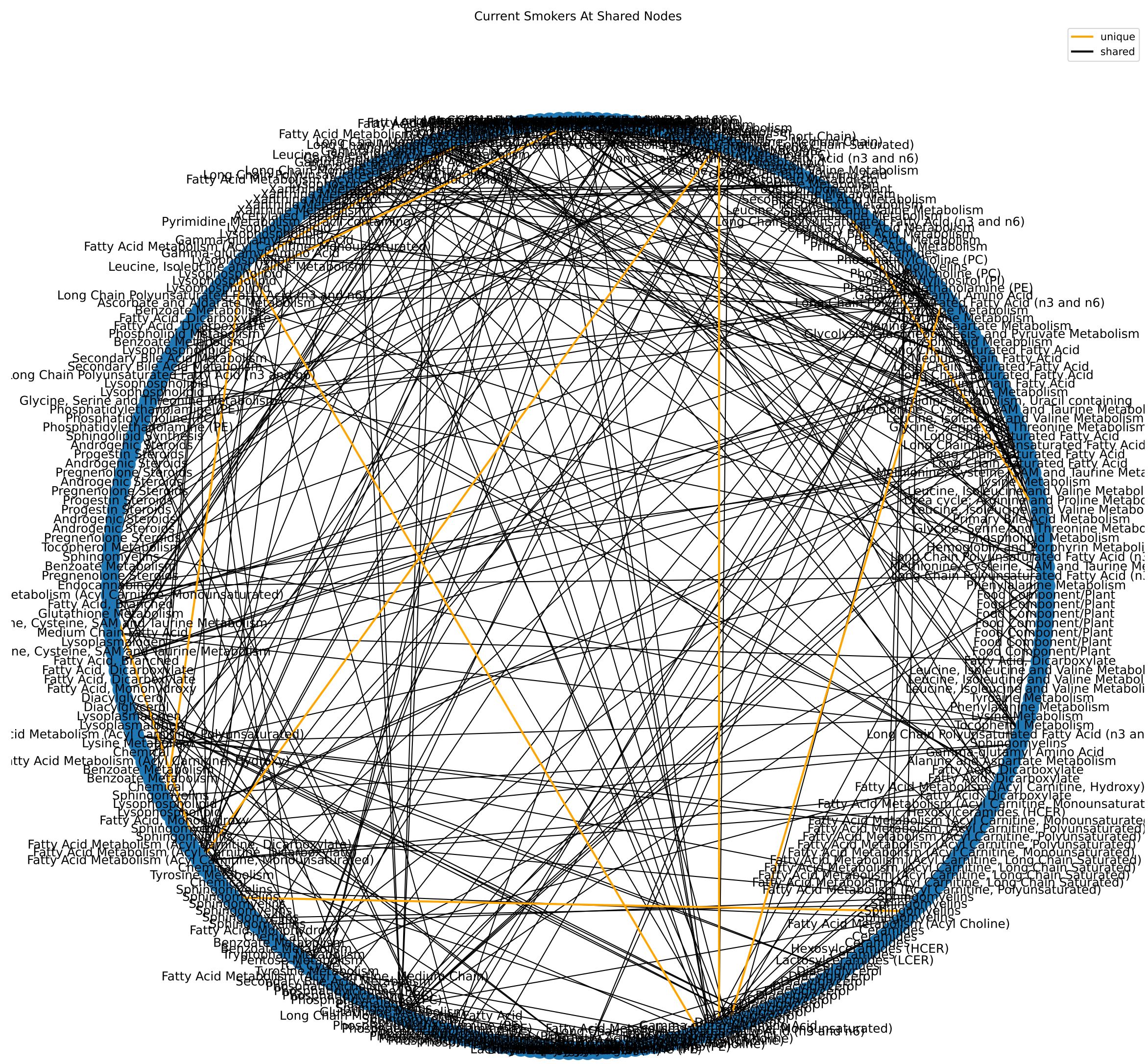

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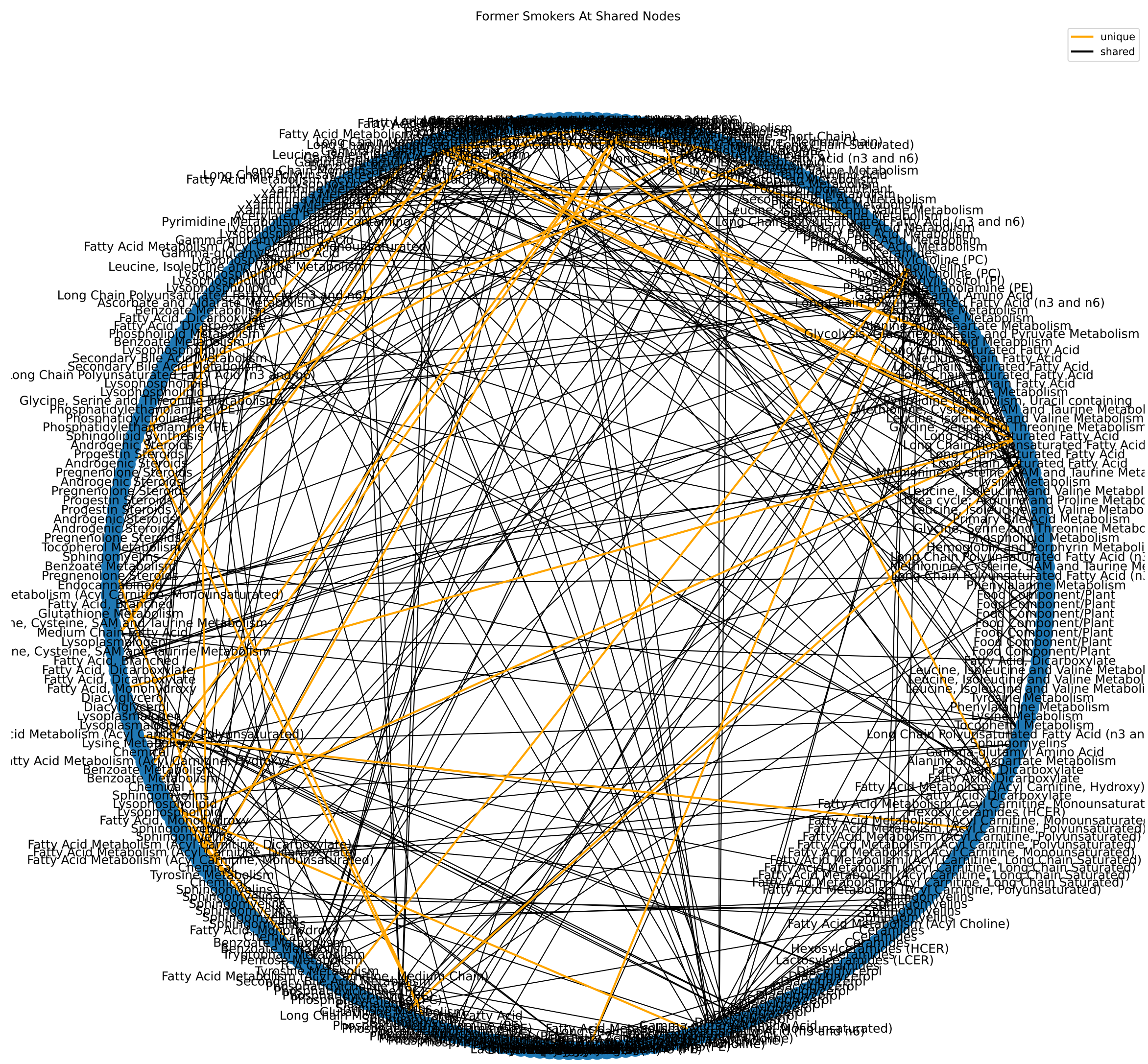

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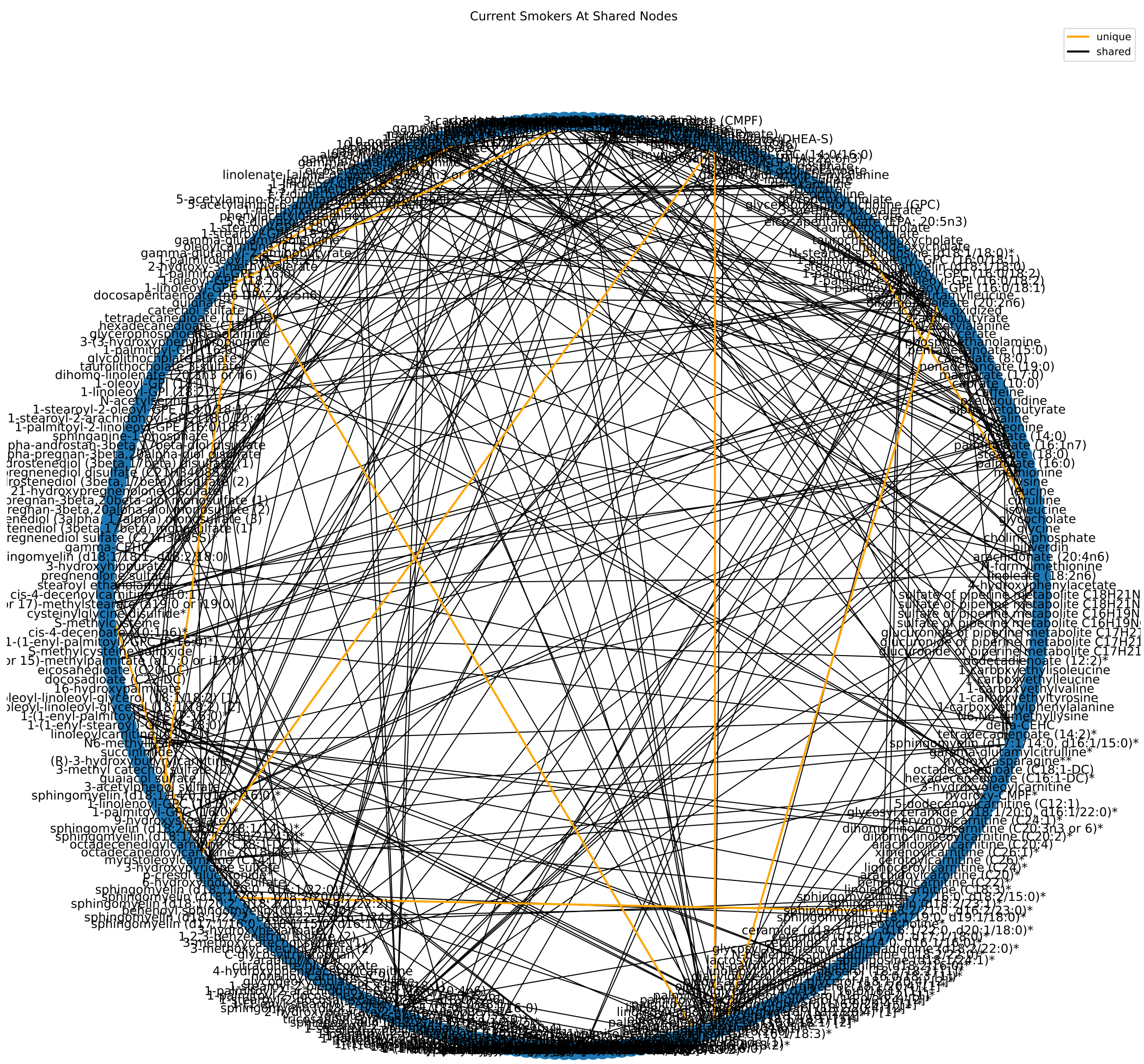


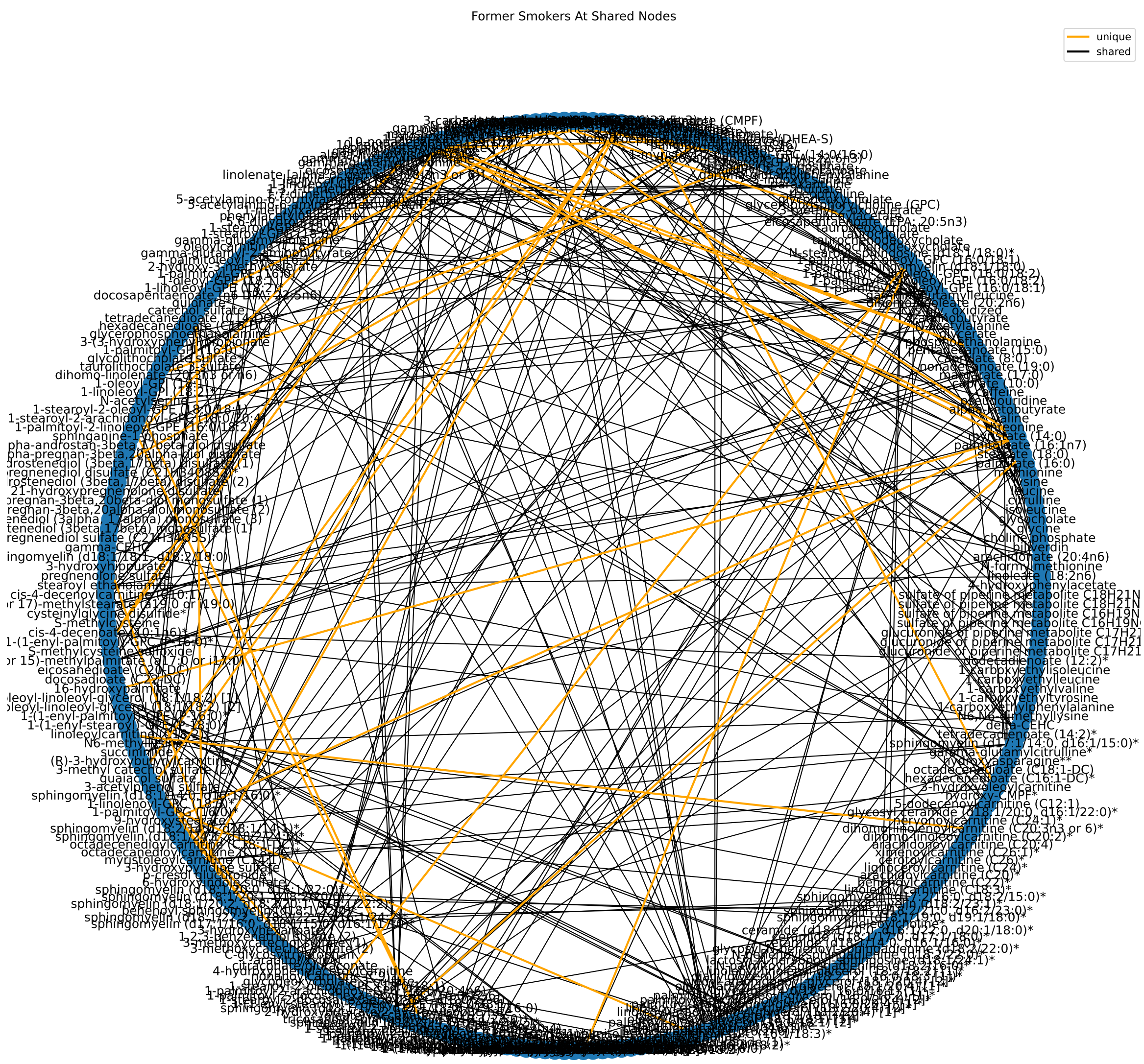
results/005_1/chemical_name-node-shared-curr-graph.pdf

Unique Edges (in orange)

Primary Bile Acid Metabolism glycocholate	Secondary Bile Acid Metabolism glycodeoxycholate
Gamma-glutamyl Amino Acid gamma-glutamylleucine	Gamma-glutamyl Amino Acid gamma-glutamyl-alpha-lysine

Primary Bile Acid Metabolism glycochenodeoxycholate	Secondary Bile Acid Metabolism glycodeoxycholate
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain) hexanoylcarnitine (C6)	Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) myristoleoylcarnitine (C14:1)*
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain) hexanoylcarnitine (C6)	Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) palmitoleoylcarnitine (C16:1)*
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosapentaenoate (n3 DPA; 22:5n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) adrenate (22:4n6)
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosadienoate (22:2n6)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosapentaenoate (n6 DPA; 22:5n6)
Lysophospholipid 1-palmitoyl-GPC (16:0)	Lysophospholipid 1-palmitoyl-GPE (16:0)
Xanthine Metabolism 1,7-dimethylurate	Xanthine Metabolism 5-acetylamino-6-amino-3-methyluracil
Lysophospholipid 1-palmitoleoyl-GPC (16:1)*	Lysophospholipid 1-linolenoyl-GPC (18:3)*
Lysophospholipid 1-palmitoleoyl-GPC (16:1)*	Phosphatidylcholine (PC) 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) cis-4-decenoylcarnitine (C10:1)	Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) myristoleoylcarnitine (C14:1)*
Sphingomyelins sphingomyelin (d18:1/20:1, d18:2/20:0)*	Sphingomyelins sphingomyelin (d18:2/21:0, d16:2/23:0)*
Diacylglycerol palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*	Diacylglycerol palmitoyl-arachidonoyl-glycerol (16:0/20:4) [2]*
Diacylglycerol oleoyl-oleoyl-glycerol (18:1/18:1) [2]*	Diacylglycerol diacylglycerol (16:1/18:2 [2], 16:0/18:3 [1])*
Sphingomyelins sphingomyelin (d18:2/21:0, d16:2/23:0)*	Sphingomyelins sphingomyelin (d18:2/23:1)*

results/005/curr-metabs/unique-edges-at-shared-nodes.txt



results/005_1/chemical_name-node-shared-form-graph.pdf

Unique Edges (in orange)

Long Chain Saturated Fatty Acid stearate (18:0)	Long Chain Monounsaturated Fatty Acid oleate/vaccenate (18:1)
Long Chain Monounsaturated Fatty Acid palmitoleate (16:1n7)	Long Chain Saturated Fatty Acid myristate (14:0)

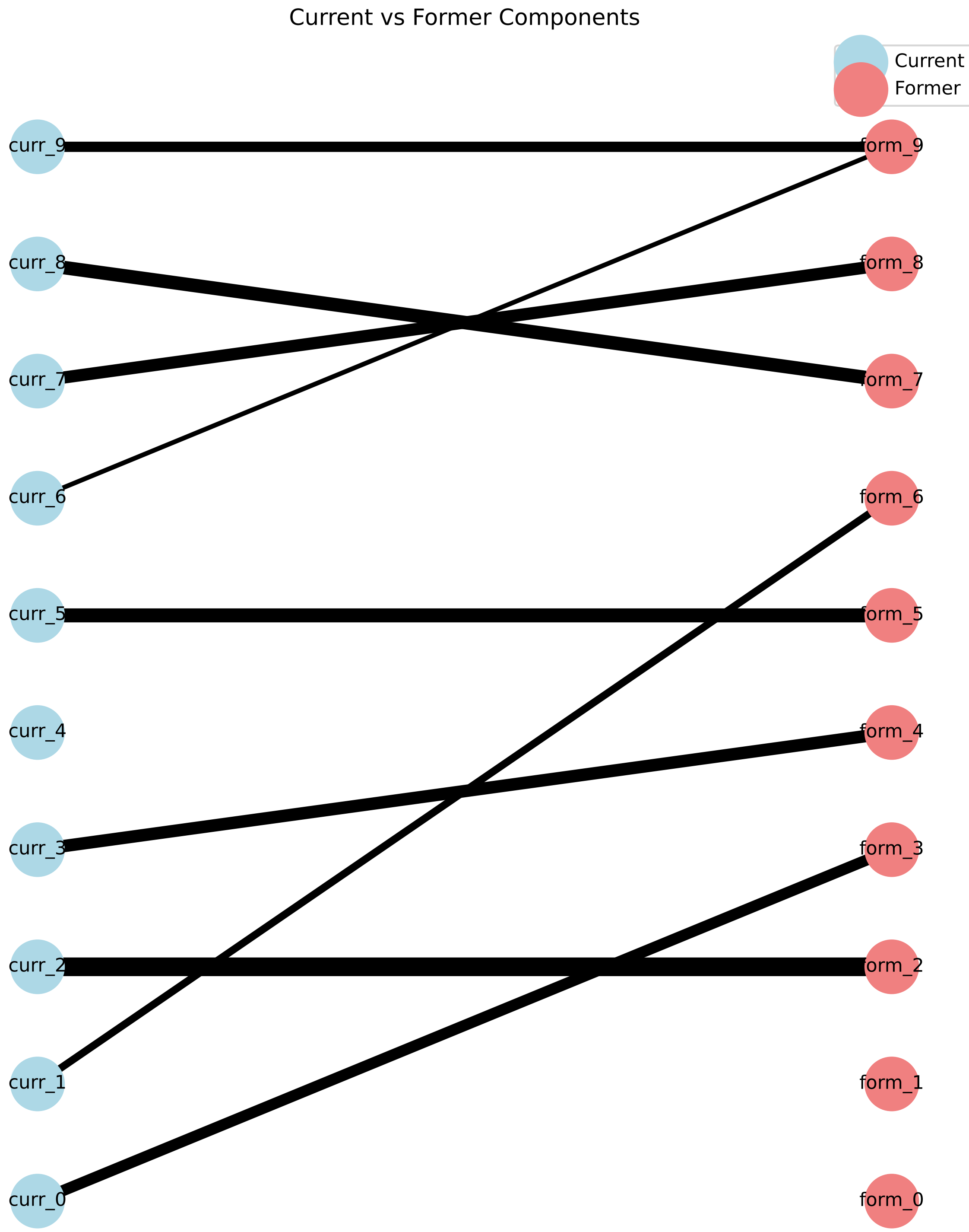
Long Chain Monounsaturated Fatty Acid palmitoleate (16:1n7)	Long Chain Monounsaturated Fatty Acid 10-nonadecenoate (19:1n9)
Long Chain Saturated Fatty Acid myristate (14:0)	Long Chain Saturated Fatty Acid margarate (17:0)
Long Chain Saturated Fatty Acid myristate (14:0)	Medium Chain Fatty Acid 5-dodecenoate (12:1n7)
Long Chain Saturated Fatty Acid myristate (14:0)	Long Chain Monounsaturated Fatty Acid 10-heptadecenoate (17:1n7)
Long Chain Saturated Fatty Acid myristate (14:0)	Fatty Acid, Monohydroxy 16-hydroxypalmitate
Long Chain Saturated Fatty Acid myristate (14:0)	Fatty Acid, Monohydroxy 9-hydroxystearate
Long Chain Saturated Fatty Acid pentadecanoate (15:0)	Long Chain Monounsaturated Fatty Acid 10-nonadecenoate (19:1n9)
Long Chain Saturated Fatty Acid pentadecanoate (15:0)	Long Chain Monounsaturated Fatty Acid 10-heptadecenoate (17:1n7)
Phosphatidylcholine (PC) 1-palmitoyl-2-oleoyl-GPC (16:0/18:1)	Phosphatidylcholine (PC) 1-stearoyl-2-oleoyl-GPC (18:0/18:1)
Phosphatidylcholine (PC) 1-palmitoyl-2-oleoyl-GPC (16:0/18:1)	Phosphatidylinositol (PI) 1-palmitoyl-2-oleoyl-GPI (16:0/18:1)*
Long Chain Polyunsaturated Fatty Acid (n3 and n6) eicosapentaenoate (EPA; 20:5n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosahexaenoate (DHA; 22:6n3)
Long Chain Polyunsaturated Fatty Acid (n3 and n6) eicosapentaenoate (EPA; 20:5n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) stearidonate (18:4n3)
Xanthine Metabolism theophylline	Xanthine Metabolism paraxanthine
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosahexaenoate (DHA; 22:6n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) stearidonate (18:4n3)
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosahexaenoate (DHA; 22:6n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) dihomo-linolenate (20:3n3 or n6)
Fatty Acid, Dicarboxylate maleate	Food Component/Plant N-(2-furoyl)glycine
Androgenic Steroids dehydroepiandrosterone sulfate (DHEA-S)	Androgenic Steroids epiandrosterone sulfate
Androgenic Steroids dehydroepiandrosterone sulfate (DHEA-S)	Androgenic Steroids androstenediol (3beta,17beta) disulfate (1)
Fatty Acid, Monohydroxy 3-hydroxydecanoate	Fatty Acid Metabolism (Acyl Carnitine, Hydroxy) 3-hydroxyoleoylcarnitine
Androgenic Steroids androsterone sulfate	Androgenic Steroids epiandrosterone sulfate
Food Component/Plant N-(2-furoyl)glycine	Benzoate Metabolism 3-methyl catechol sulfate (2)
Food Component/Plant N-(2-furoyl)glycine	Chemical 3-acetylphenol sulfate
Food Component/Plant N-(2-furoyl)glycine	Chemical 3-hydroxypyridine sulfate

Long Chain Polyunsaturated Fatty Acid (n3 and n6) adrenate (22:4n6)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosapentaenoate (n6 DPA; 22:5n6)
Medium Chain Fatty Acid 5-dodecenoate (12:1n7)	Long Chain Monounsaturated Fatty Acid 10-heptadecenoate (17:1n7)
Lysophospholipid 1-palmitoyl-GPC (16:0)	Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*
Lysophospholipid 1-stearoyl-GPC (18:0)	Lysophospholipid 1-oleoyl-GPC (18:1)
Lysophospholipid 1-oleoyl-GPC (18:1)	Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*
Androgenic Steroids epiandrosterone sulfate	Androgenic Steroids 5alpha-androstan-3beta,17beta-diol disulfate
Benzoate Metabolism catechol sulfate	Chemical 3-hydroxypyridine sulfate
Lysophospholipid 1-palmitoyl-GPI (16:0)	Lysophospholipid 1-oleoyl-GPI (18:1)
Phosphatidylethanolamine (PE) 1-stearoyl-2-oleoyl-GPE (18:0/18:1)	Phosphatidylethanolamine (PE) 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*
Phosphatidylethanolamine (PE) 1-palmitoyl-2-linoleoyl-GPE (16:0/18:2)	Phosphatidylethanolamine (PE) 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*
Sphingomyelins sphingomyelin (d18:1/18:1, d18:2/18:0)	Sphingomyelins sphingomyelin (d18:1/20:1, d18:2/20:0)*
Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*	Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPE (P-16:0)*
Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*	Lysoplasmalogen 1-(1-enyl-stearoyl)-GPE (P-18:0)*
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated) linoleoylcarnitine (C18:2)*	Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated) dihomo-linolenoylcarnitine (C20:3n3 or 6)*
Benzoate Metabolism 3-methyl catechol sulfate (2)	Chemical 3-hydroxypyridine sulfate
Benzoate Metabolism 3-methyl catechol sulfate (2)	TCA Cycle citraconate/glutaconate
Benzoate Metabolism guaiacol sulfate	Chemical 3-hydroxypyridine sulfate
Benzoate Metabolism guaiacol sulfate	Benzoate Metabolism 3-methoxycatechol sulfate (2)
Chemical 3-acetylphenol sulfate	Chemical 3-hydroxypyridine sulfate
Chemical 3-acetylphenol sulfate	Benzoate Metabolism 3-methoxycatechol sulfate (2)
Sphingomyelins sphingomyelin (d18:1/14:0, d16:1/16:0)*	Sphingomyelins sphingomyelin (d17:1/16:0, d18:1/15:0, d16:1/17:0)*
Lysophospholipid 1-linolenoyl-GPC (18:3)*	Phosphatidylcholine (PC) 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*
Phosphatidylethanolamine (PE) 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*	Phosphatidylethanolamine (PE) 1-stearoyl-2-arachidonoyl-GPE (18:0/20:4)

Phosphatidylethanolamine (PE) 1-palmitoyl-2-arachidonoyl-GPE (16:0/20:4)*	Phosphatidylethanolamine (PE) 1-oleoyl-2-arachidonoyl-GPE (18:1/20:4)*
Plasmalogen 1-(1-enyl-palmitoyl)-2-linoleoyl-GPE (P-16:0/18:2)*	Plasmalogen 1-(1-enyl-stearoyl)-2-linoleoyl-GPE (P-18:0/18:2)*
Diacylglycerol palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*	Diacylglycerol oleoyl-oleoyl-glycerol (18:1/18:1) [1]*
Diacylglycerol palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*	Diacylglycerol oleoyl-oleoyl-glycerol (18:1/18:1) [2]*
Tyrosine Metabolism 1-carboxyethyltyrosine	Leucine, Isoleucine and Valine Metabolism 1-carboxyethylvaline

results/005/form-metabs/unique-edges-at-shared-nodes.txt

Smoker Component Pairs



Current Smoker Component Metabolites

Component 1: pseudouridine, N-acetylserine, hydroxyasparagine**, C-glycosyltryptophan, N-acetylalanine, N-formylmethionine, 5,6-dihydrouridine,

Component 2: maleate, N-(2-furoyl)glycine, citraconate/glutaconate, quinate, trigonelline (N'-methylnicotinate), caffeic acid sulfate, 3-hydroxypyridine sulfate,

Component 3: 5-acetylamino-6-formylamino-3-methyluracil, caffeine, 1,3,7-trimethylurate, paraxanthine, 1-methylxanthine, 5-acetylamino-6-amino-3-methyluracil, 1,7-dimethylurate, theophylline,

Component 4: 21-hydroxypregnenolone disulfate, androstenediol (3beta,17beta) disulfate (2), 5alpha-androstan-3beta,17beta-diol disulfate, androstenediol (3beta,17beta) monosulfate (1), androstenediol (3beta,17beta) disulfate (1), dehydroepiandrosterone sulfate (DHEA-S), pregnenediol sulfate (C21H34O5S)*, pregnenolone sulfate, pregnenediol disulfate (C21H34O8S2)*,

Component 5: gamma-glutamylmethionine, isoleucine, gamma-glutamylphenylalanine, gamma-glutamylisoleucine*, gamma-glutamyl-alpha-lysine, gamma-glutamyl-epsilon-lysine, valine, leucine, lysine, gamma-glutamylleucine, methionine,

Component 6: sphingomyelin (d18:2/14:0, d18:1/14:1)*, sphingomyelin (d18:1/18:1, d18:2/18:0), sphingomyelin (d18:1/20:1, d18:2/20:0)*, sphingomyelin (d17:1/16:0, d18:1/15:0, d16:1/17:0)*, stearoyl sphingomyelin (d18:1/18:0), sphingomyelin (d18:1/17:0, d17:1/18:0, d19:1/16:0), sphingomyelin (d18:2/18:1)*, sphingomyelin (d18:2/21:0, d16:2/23:0)*, sphingomyelin (d18:2/23:1)*, sphingomyelin (d18:1/22:2, d18:2/22:1, d16:1/24:2)*, sphingomyelin (d18:1/20:2, d18:2/20:1, d16:1/22:2)*, sphingomyelin (d18:1/14:0, d16:1/16:0)*, sphingomyelin (d17:2/16:0, d18:2/15:0)*, sphingomyelin (d18:1/19:0, d19:1/18:0)*, myristoyl dihydrosphingomyelin (d18:0/14:0)*, sphingomyelin (d17:1/14:0, d16:1/15:0)*, sphingomyelin (d18:1/20:0, d16:1/22:0)*,

Component 7: linoleoylcarnitine (C18:2)*, decanoylcarnitine (C10), arachidonoylcarnitine (C20:4), palmitoleoylcarnitine (C16:1)*, cis-4-decenoylcarnitine (C10:1), laurylcarnitine (C12), linolenoylcarnitine (C18:3)*, myristoleoylcarnitine (C14:1)*, palmitoylcarnitine (C16), 5-dodecenoylcarnitine (C12:1), 3-hydroxyoleoylcarnitine, dihomo-linoleoylcarnitine (C20:2)*, dihomo-linolenoylcarnitine (C20:3n3 or 6)*, oleoylcarnitine (C18:1), hexanoylcarnitine (C6), myristoylcarnitine (C14), nonanoylcarnitine (C9), octanoylcarnitine (C8),

Component 8: 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*, 1-palmitoyl-2-oleoyl-GPC (16:0/18:1), 1-myristoyl-2-palmitoyl-GPC (14:0/16:0), 1-palmitoyl-2-oleoyl-GPI (16:0/18:1)*, 1-stearoyl-2-oleoyl-GPC (18:0/18:1), 1-(1-enyl-palmitoyl)-GPC (P-16:0)*, 1-oleoyl-GPE (18:1), 1-palmitoleoyl-GPC (16:1)*, 1-stearoyl-2-oleoyl-GPI (18:0/18:1)*, 1-oleoyl-GPC (18:1), 1-palmitoyl-2-linoleoyl-GPI (16:0/18:2), 1-linoleoyl-GPE (18:2)*, 1-linoleoyl-GPC (18:2), 1-palmitoyl-GPC (16:0), 1-linolenoyl-GPC (18:3)*, 1-stearoyl-2-linoleoyl-GPI (18:0/18:2), 1-palmitoyl-GPE (16:0), 1-palmitoyl-2-palmitoleoyl-GPC (16:0/16:1)*, 1-stearoyl-GPC (18:0), 1-stearoyl-GPE (18:0),

Component 9: oleoyl-linoleoyl-glycerol (18:1/18:2) [2], palmitoyl-linoleoyl-glycerol (16:0/18:2) [1]*, 1-stearoyl-2-arachidonoyl-GPE (18:0/20:4), oleoyl-arachidonoyl-glycerol (18:1/20:4) [2]*, 1-palmitoyl-2-arachidonoyl-GPE (16:0/20:4)*, 1-oleoyl-2-arachidonoyl-GPE (18:1/20:4)*, palmitoleoyl-linoleoyl-glycerol (16:1/18:2) [1]*, oleoyl-oleoyl-glycerol (18:1/18:1) [2]*, oleoyl-linoleoyl-glycerol (18:1/18:2) [1], palmitoyl-arachidonoyl-glycerol (16:0/20:4) [2]*, diacylglycerol (14:0/18:1, 16:0/16:1) [1]*, oleoyl-oleoyl-glycerol (18:1/18:1) [1]*, linoleoyl-linoleoyl-glycerol (18:2/18:2) [1]*, diacylglycerol (16:1/18:2 [2], 16:0/18:3 [1])*, palmitoyl-linoleoyl-glycerol (16:0/18:2) [2]*, 1-palmitoyl-2-linoleoyl-GPE (16:0/18:2), linoleoyl-arachidonoyl-glycerol (18:2/20:4) [2]*, palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*, 1-stearoyl-2-oleoyl-GPE (18:0/18:1), palmitoyl-arachidonoyl-glycerol (16:0/20:4) [1]*, 1-palmitoyl-2-oleoyl-GPE (16:0/18:1),

Component 10: 16-hydroxypalmitate, 9-hydroxystearate, linolenate [alpha or gamma; (18:3n3 or 6)], docosadienoate (22:2n6), 3-hydroxyhexanoate, 3-hydroxymyristate, cis-4-decenoate (10:1n6)*, linoleate (18:2n6), myristoleate (14:1n5), margarate (17:0), 3-hydroxylaurate, 3-hydroxydecanoate, 10-nonadecenoate (19:1n9), eicosenoate (20:1), arachidonate (20:4n6), (14 or 15)-methylpalmitate (a17:0 or i17:0), myristate (14:0), palmitate (16:0), stearate (18:0), dihomo-linolenate (20:3n3 or n6), docosahexaenoate (DHA; 22:6n3), 3-hydroxyoctanoate, dodecadienoate (12:2)*, 3-hydroxybutyrate (BHBA), stearidonate (18:4n3), oleate/vaccenate (18:1), nonadecanoate (19:0), tetradecadienoate (14:2)*, pentadecanoate (15:0), (16 or 17)-methylstearate (a19:0 or i19:0), 5-dodecenoate (12:1n7), docosapentaenoate (n3 DPA; 22:5n3), palmitoleate (16:1n7), dihomo-linoleate (20:2n6), eicosapentaenoate (EPA; 20:5n3), hexadecadienoate (16:2n6), stearoyl ethanolamide, docosapentaenoate (n6 DPA; 22:5n6), 10-heptadecenoate (17:1n7), adrenate (22:4n6),

Former Smoker Component Metabolites

Component 1: glucuronide of piperine metabolite C17H21NO3 (5)*, glucuronide of piperine metabolite C17H21NO3 (3)*, piperine, sulfate of piperine metabolite C16H19NO3 (3)*, sulfate of piperine metabolite C16H19NO3 (2)*, glucuronide of piperine metabolite C17H21NO3 (4)*,

Component 2: 1-(1-enyl-stearoyl)-2-arachidonoyl-GPE (P-18:0/20:4)*, 1-(1-enyl-stearoyl)-2-oleoyl-GPE (P-18:0/18:1), 1-(1-enyl-palmitoyl)-2-arachidonoyl-GPE (P-16:0/20:4)*, 1-(1-enyl-palmitoyl)-2-oleoyl-GPE (P-16:0/18:1)*, 1-(1-enyl-palmitoyl)-2-linoleoyl-GPE (P-16:0/18:2)*, 1-(1-enyl-stearoyl)-2-linoleoyl-GPE (P-18:0/18:2)*,

Component 3: 5-acetylamino-6-formylamino-3-methyluracil, caffeine, 1,3,7-trimethylurate, paraxanthine, 1-methylxanthine, 5-acetylamino-6-amino-3-methyluracil, 1,7-dimethylurate, theophylline,

Component 4: pseudouridine, N-acetylserine, N2,N2-dimethylguanosine, hydroxyasparagine**, C-glycosyltryptophan, N-acetylalanine, N6-carbamoylthreonyladenosine, N-formylmethionine, gamma-carboxyglutamate, 2,3-dihydroxy-5-methylthio-4-pentenoate (DMTPA)*, 5,6-dihydrouridine,

Component 5: epiandrosterone sulfate, androsterone sulfate, androstenediol (3beta,17beta) disulfate (2), 21-hydroxypregnenolone disulfate, androstenediol (3alpha, 17alpha) monosulfate (3), 5alpha-androstan-3beta,17beta-diol disulfate, pregnenediol disulfate (C21H34O8S2)*, androstenediol (3beta,17beta) monosulfate (1), androstenediol (3beta,17beta) disulfate (1), androstenediol (3beta,17beta) monosulfate (2), dehydroepiandrosterone sulfate (DHEA-S), pregnenolone sulfate, pregnenediol sulfate (C21H34O5S)*,

Component 6: sphingomyelin (d18:2/14:0, d18:1/14:1)*, sphingomyelin (d18:1/18:1, d18:2/18:0), sphingomyelin (d18:1/20:1, d18:2/20:0)*, sphingomyelin (d17:1/16:0, d18:1/15:0, d16:1/17:0)*, stearoyl sphingomyelin (d18:1/18:0), sphingomyelin (d18:1/17:0, d17:1/18:0, d19:1/16:0), sphingomyelin (d18:2/21:0, d16:2/23:0)*, sphingomyelin (d17:2/16:0, d18:2/15:0)*, myristoyl dihydrosphingomyelin (d18:0/14:0)*, sphingomyelin (d18:1/14:0, d16:1/16:0)*, sphingomyelin (d18:1/19:0, d19:1/18:0)*, sphingomyelin (d17:1/14:0, d16:1/15:0)*, sphingomyelin (d18:1/20:0, d16:1/22:0)*,

Component 7: 3-methyl catechol sulfate (2), guaiacol sulfate, 4-vinylguaiacol sulfate, 3-methyl catechol sulfate (1), 3-methoxycatechol sulfate (2), maleate, N-(2-furoyl)glycine, citraconate/glutaconate, 3-ethylcatechol sulfate (1), quinate, trigonelline (N'-methylnicotinate), caffeic acid sulfate, 4-methoxyphenol sulfate, 3-hydroxypyridine sulfate, catechol sulfate, 3-acetylphenol sulfate,

Component 8: oleoyl-linoleoyl-glycerol (18:1/18:2) [2], palmitoyl-linoleoyl-glycerol (16:0/18:2) [1]*, 1-stearoyl-2-arachidonoyl-GPE (18:0/20:4), 1-palmitoyl-2-arachidonoyl-GPE (16:0/20:4)*, 1-oleoyl-2-arachidonoyl-GPE (18:1/20:4)*, palmitoleoyl-linoleoyl-glycerol (16:1/18:2) [1]*, oleoyl-oleoyl-glycerol (18:1/18:1) [2]*, oleoyl-linoleoyl-glycerol (18:1/18:2) [1], 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*, diacylglycerol (14:0/18:1, 16:0/16:1) [1]*, oleoyl-oleoyl-glycerol (18:1/18:1) [1]*, linoleoyl-linoleoyl-glycerol (18:2/18:2) [1]*, diacylglycerol (16:1/18:2 [2], 16:0/18:3 [1])*, palmitoyl-linoleoyl-glycerol (16:0/18:2) [2]*, 1-palmitoyl-2-linoleoyl-GPE (16:0/18:2), palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*, 1-oleoyl-2-linoleoyl-GPE (18:1/18:2)*, 1-stearoyl-2-oleoyl-GPE (18:0/18:1), 1-palmitoyl-2-oleoyl-GPE (16:0/18:1),

Component 9: 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*, 1-palmitoyl-2-oleoyl-GPC (16:0/18:1), 1-(1-enyl-stearoyl)-GPE (P-18:0)*, 1-myristoyl-2-palmitoyl-GPC (14:0/16:0), 1-(1-enyl-palmitoyl)-GPE (P-16:0)*, 1-palmitoyl-2-oleoyl-GPI (16:0/18:1)*, 1-linoleoyl-2-linolenoyl-GPC (18:2/18:3)*, 1-stearoyl-2-oleoyl-GPC (18:0/18:1), 1-(1-enyl-palmitoyl)-GPC (P-16:0)*, 1-palmitoyl-2-arachidonoyl-GPI (16:0/20:4)*, 1-oleoyl-GPE (18:1), 1-palmitoleoyl-GPC (16:1)*, 1-stearoyl-2-oleoyl-GPI (18:0/18:1)*, 1-oleoyl-GPC (18:1), 1-palmitoyl-2-linoleoyl-GPI (16:0/18:2), 1-linoleoyl-GPE (18:2)*, 1-palmitoyl-2-arachidonoyl-GPC (16:0/20:4n6), 1-stearoyl-2-arachidonoyl-GPC (18:0/20:4), 1-linoleoyl-GPC (18:2), 1-palmitoyl-GPC (16:0), 1-linolenoyl-GPC (18:3)*, 1-arachidonoyl-GPE (20:4n6)*, 1-myristoyl-2-arachidonoyl-GPC (14:0/20:4)*, 1-stearoyl-2-linoleoyl-GPI (18:0/18:2), 1-palmitoyl-GPE (16:0), 1-palmitoyl-2-palmitoleoyl-GPC (16:0/16:1)*, 1-stearoyl-GPC (18:0), 1,2-dilinoleoyl-GPC (18:2/18:2), 1-stearoyl-GPE (18:0), 1-arachidonoyl-GPC (20:4n6)*,

Component 10: 16-hydroxypalmitate, decanoylcarnitine (C10), hexadecanedioate (C16-DC), 9-hydroxystearate, caprylate (8:0), linolenate [alpha or gamma; (18:3n3 or 6)], docosadienoate (22:2n6), cis-4-decenoylcarnitine (C10:1), 3-hydroxyhexanoate, 3-hydroxymyristate, linolenoylcarnitine (C18:3)*, palmitoylcarnitine (C16), cis-4-decenoate (10:1n6)*, linoleate (18:2n6), nonanoylcarnitine (C9), hexadecenedioate (C16:1-DC)*, myristoleate (14:1n5), tetradecanedioate (C14-DC), margarate (17:0), laurylcarnitine (C12), 3-hydroxysebacate, 3-hydroxylaurate, 3-hydroxydecanoate, 5-dodecenoylcarnitine (C12:1), 10-nonadecenoate (19:1n9), dihomo-linoleoylcarnitine (C20:2)*, eicosenoate (20:1), arachidonate (20:4n6), oleoylcarnitine (C18:1), (14 or 15)-methylpalmitate (a17:0 or i17:0), myristate (14:0), myristoylcarnitine (C14), hexanoylcarnitine (C6), linoleoylcarnitine (C18:2)*, arachidonoylcarnitine (C20:4), palmitate (16:0), stearate (18:0), dihomo-linolenate (20:3n3 or n6), docosahexaenoate (DHA; 22:6n3), 3-hydroxyoctanoate, dodecadienoate (12:2)*, laurate (12:0), stearidonate (18:4n3), caprate (10:0), myristoleoylcarnitine (C14:1)*, margaroylcarnitine (C17)*, oleate/vaccenate (18:1), nonadecanoate (19:0), tetradecadienoate (14:2)*, octanoylcarnitine (C8), pentadecanoate (15:0), (16 or 17)-methylstearate (a19:0 or i19:0), dodecanedioate (C12-DC), 5-dodecenoate (12:1n7), docosapentaenoate (n3 DPA; 22:5n3), palmitoleate (16:1n7), dihomo-linoleate (20:2n6), eicosapentaenoate (EPA; 20:5n3), hexadecadienoate (16:2n6), stearoyl ethanolamide, palmitoleoylcarnitine (C16:1)*, octadecanedioate (C18-DC), docosapentaenoate (n6 DPA; 22:5n6), 10-heptadecenoate (17:1n7), 3-hydroxyoleoylcarnitine, dihomo-linolenoylcarnitine (C20:3n3 or 6)*, glycerol, adrenate (22:4n6), octadecenedioate (C18:1-DC),